TASK 79-21-02-400-802-G00

5. Engine Surface Air Cooling Oil Cooler 2 Installation

(Figure 403 and Figure 404)

A. General

(1) This task gives the instructions to install the engine Surface Air Cooled Oil Cooler (SACOC) 2.

B. References

Reference	Title
12-13-11-600-801	Engine Servicing (Oil Replenishing) (P/B 301)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
A50474 [CP2256]	Locking Agent - Liquid	
D00599 [CP2442]	Oil - Engine	
D00601 [CP2101]	Vaseline - Graphite Mineral	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
426	Right Thrust Reverser, Engine 2

G. Engine Surface Air Cooling Oil Cooler 2 Installation

SUBTASK 79-21-02-420-002-G00

WARNING

DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION

DO NOT LET OIL GET ON THE ENGINE, OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE AREAS THAT OIL FALLS ON. OIL CAN CAUSE DAMAGE TO PAINT AND RUBBER.

(1) Install the engine SACOC 2 [2]:

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(a) Make sure that the mating faces are clean and clear of unwanted materials.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0181

(b) Lubricate the threads of the 14 bolts with plates [20] and two bolts with integrated washers [15] with graphite mineral vaseline, D00601 [CP2101].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0181

(c) Lubricate the threads of the 14 bolts with integrated washers [20] and two bolts with integrated washers [15] with graphite mineral vaseline, D00601 [CP2101].

PTF ALL

CAUTION MAKE SURE THAT YOU ALIGN THE LUG OF THE BRACKET CORRECTLY WITH THE NOTCH OF THE ENGINE SURFACE AIR COOLED OIL COOLER (SACOC) 2. IF YOU DO NOT OBEY, DAMAGE TO THE ENGINE SACOC 2 CAN OCCUR.

(d) Install the engine SACOC 2 [2] on the engine.

(e) Remove the protective covers from the engine SACOC 2 [2], oil tube [17], and oil tube [18].

(f) Put the engine SACOC 2 [2] and bracket [21] in position at the same time.

1) Slide the bracket [21] between the engine SACOC 2 [2] and the fan case.

2) Make sure that the assembly direction is correct.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0181

CAUTION BE CAREFUL WHEN YOU INSTALL THE BOLTS ON THE AIR/OIL EX-CHANGER. IF YOU DO NOT OBEY, DAMAGE TO THE LIP SEAL OF THE AIR/OIL EX-CHANGER CAN OCCUR.

(g) Install the 14 bolts with plates [20] and two bolts with integrated washers [15] that attach the engine SACOC 2 [2] to the engine.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0181

CAUTION BE CAREFUL WHEN YOU INSTALL THE BOLTS ON THE AIR/OIL EX-CHANGER. IF YOU DO NOT OBEY, DAMAGE TO THE LIP SEAL OF THE AIR/OIL EX-CHANGER CAN OCCUR.

(h) Install the 14 bolts with integrated washers [20] and two bolts with integrated washers [15] that attach the engine SACOC 2 [2] to the engine.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0181

(i) Tighten the bolts with plates [20] from the top to the bottom, following the sequence 1 to 16 then re-tighten the bolts with plates [20] a second time following the same sequence order as follows (Figure 404):

1) Tighten the two bolts with plates [20], following the specific sequence (start by 1 and finish by 2), to 110.6 in-lb (12.5 N·m) - 119.5 in-lb (13.5 N·m).

2) Tighten the two bolts with integrated washers [15], following the specific sequence (start by 3 and finish by 4), to 110.6 in-lb (12.5 N·m) - 119.5 in-lb (13.5 N·m).

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PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0181 (Continued)

3) Tighten the 12 remaining bolts with plates [20], from the top to the bottom following the specific sequence (start by 5 until 16), to 110.6 in-lb (12.5 N·m) - 119.5 in-lb (13.5 N·m).

4) Tighten the bolts with plates [20] and two bolts with integrated washers [15] again in accordance with the requested torque value, follow the sequence order from the top to the bottom.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0181

(j) Tighten the bolts with integrated washers [20] from the top to the bottom, following the sequence 1 to 16 then re-tighten the bolts with plates [20] a second time following the same sequence order as follows (Figure 404):

1) Tighten the two bolts with integrated washers [20], following the specific sequence nj,nj(start by 1 and finish by 2), to 110.6 in-lb (12.5 N·m) - 119.5 in-lb (13.5 N·m).

2) Tighten the two bolts with integrated washers [15], following the specific sequence (start by 3 and finish by 4), to 110.6 in-lb (12.5 N·m) - 119.5 in-lb (13.5 N·m).

3) Tighten the 12 remaining bolts with integrated washers [20], from the top to the bottom following the specific sequence (start by 5 until 16), to 110.6 in-lb (12.5 N·m) - 119.5 in-lb (13.5 N·m).

4) Tighten the bolts with integrated washers [20] and two bolts with integrated washers [15] again in accordance with the requested torque value, follow the sequence order from the top to the bottom.

PTF ALL

(k) Install the attachment clamp [22] and attachment clamp [25] to secure the engine SACOC 2 [2] as follows:

1) Apply Loctite 241, A50474 [CP2256], on the threads and under the head of the two screws [23].

2) Lubricate the threads of the two screws [24] with graphite mineral vaseline, D00601 [CP2101].

3) Install the attachment clamp [22] and two screws [23] that attach the engine SACOC 2 [2] to the bracket [21].

4) Install the attachment clamp [25] and two screws [24] that attach the engine SACOC 2 [2] to the bracket [26].

5) Tighten the two screws [23] to 84.1 in-lb (9.5 N·m) - 92.9 in-lb (10.5 N·m).

6) Tighten the screws [24] to 84.1 in-lb (9.5 N·m) - 92.9 in-lb (10.5 N·m).

(l) Connect the oil tube [18] to the engine SACOC 2 [2] as follows:

1) Lubricate the threads of the coupling nut [19] with oil, D00599 [CP2442].

2) Connect the oil tube [18] with the coupling nut [19] to the engine SACOC 2 [2].

3) Tighten the coupling nut [19] to 442.5 in-lb (50 N·m) - 531 in-lb (60 N·m).

(m) Connect the oil tube [17] to the engine SACOC 2 [2] as follows:

1) Lubricate the threads of the coupling nut [16] with oil, D00599 [CP2442].

2) Connect the oil tube [17] with the coupling nut [16] to the engine SACOC 2 [2].

3) Tighten the coupling nut [16] to 442.5 in-lb (50 N·m) - 531 in-lb (60 N·m).

(n) Clean the work area and clear the tools used to do the task.

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H. Put the Airplane Back to Its Usual Condition

SUBTASK 79-21-02-610-001-G00

(1) Do this task: Engine Servicing (Oil Replenishing), TASK 12-13-11-600-801.

SUBTASK 79-21-02-410-002-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do these tasks in sequence to safely close the right thrust reverser:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
416	Right Thrust Reverser, Engine 1
426	Right Thrust Reverser, Engine 2

(b) Close the fan cowl panels (TASK 71-11-04-410-801-G00).

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 79-21-02-860-004-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

I. Engine Surface Air Cooling Oil Cooler 2 Installation Test

SUBTASK 79-21-02-730-002-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ENGINE SURFACE AIR COOLING OIL COOLER - INSPECTION/CHECK

1. General

A. This procedure has one task:

(1) Engine Surface Air Cooling Oil Cooler Inspection.

TASK 79-21-02-211-801-G00

2. Engine Surface Air Cooling Oil Cooler Inspection

(Figure 601)

A. General

(1) This task gives the instructions to do the inspection of the engine Surface Air Cooling Oil Cooler (SACOC).

B. References

Reference	Title
71-00-00-800-802-G00	Engine Operation Limits (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
79-21-02 P/B 401	ENGINE SURFACE AIR COOLING OIL COOLER - REMOVAL/INSTALLATION
79-21-02-000-801-G00	Engine Surface Air Cooling Oil Cooler 1 Removal (P/B 401)
79-21-02-000-802-G00	Engine Surface Air Cooling Oil Cooler 2 Removal (P/B 401)
79-21-02-400-801-G00	Engine Surface Air Cooling Oil Cooler 1 Installation (P/B 401)
79-21-02-400-802-G00	Engine Surface Air Cooling Oil Cooler 2 Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-21826	Stone - Oil, Aluminum Oxide, Fine Grit (+400) Part #: 61463686770 Supplier: 44197
STD-858	Tag - DO NOT OPERATE
STD-1081	Flashlight - Explosion Proof
STD-3907	Mirror - Dental
STD-21746	Set - Carbide Bur

D. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	
G51791	Cloth - Abrasive, Silicon Carbide, 500 Grit	

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E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Inspection

SUBTASK 79-21-02-860-005-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 79-21-02-010-006-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

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WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Engine Surface Air Cooling Oil Cooler Inspection

SUBTASK 79-21-02-211-003-G00

(1) Use the explosion proof flashlight, STD-1081, to shine the light at the forward face of the heat exchanger, and look for these conditions:

(a) Make sure that no bird debris is present before you continue (TASK 71-00-00-800-802-G00).

(b) Debris in the heat exchanger fins.

NOTE: The most likely area of damage is the forward facing surface of the heat exchanger. You will need a small hand held dental mirror, STD-3907, to see the forward face of the heat exchanger.

1) If you find debris, clean to remove debris.

NOTE: If you see little to no light through the heat exchanger fins during the check with a flashlight, this indicates that airflow is significantly obstructed and the cooling capability could be diminished. You can do the cleaning procedure to remove debris and improve airflow.

(c) Inspection of the fins.

1) Broken, teared or cracked fins are not permitted.

a) If broken, teared or cracked fins are present, do the applicable tasks:

- Engine Surface Air Cooling Oil Cooler 1 Removal, TASK 79-21-02-000-801-G00 and Engine Surface Air Cooling Oil Cooler 1 Installation, TASK 79-21-02-400-801-G00
- Engine Surface Air Cooling Oil Cooler 2 Removal, TASK 79-21-02-000-802-G00 and Engine Surface Air Cooling Oil Cooler 2 Installation, TASK 79-21-02-400-802-G00.

2) Missing fins

a) Not permitted.

b) Not repairable.

c) If it is necessary, replace or send to manufacturer.

3) Nick, scratch, gouge or dent on the fin.

a) Permitted up to 0.01 in. (0.254 mm) in depth for the nick or scratch and gouge without loss of the protective layer.

b) Permitted up to 0.1 in. (2.54 mm) in depth for dent or blending depth without loss of the protective layer.

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c) Raised metal is not permitted.

d) If raised metal is present, do these steps:

NOTE: Dents indication types on fins are considered as bent fins indication type, therefore the maximum number of cumulated finding to be considered on fins is 42 findings on both engine surface air cooling oil cooler.

<1> If 43 fins are found with findings, replace the engine surface air cooling oil coolers (PAGEBLOCK 79-21-02/401).

<2> Use a carbide bur set, STD-21746, and hand grind out findings to remove heavy burrs or raised metal.

<3> Use an oil stone, COM-21826, to remove light burrs.

<4> Use a 500 grit abrasive cloth, G51791, to remove edges of repaired surfaces smoothly into adjacent surfaces.

<5> Do not remove more material than necessary.

NOTE: No high metal must remain. The transition with the non-blended area must be smooth.

NOTE: After blending, the maximum depth on the blended area must be not over than 0.1 in. (2.54 mm).

4) The first and last four rows of fins are removed.

a) If the surface cooler is found with missing fins in any other area, replace it with a new surface cooler assembly.

NOTE: A maximum of first four rows and last four rows of fins are removed for manufacturing purposes and will not adversely affect the function of the coolers.

5) Bent fins

a) A total of 42 bent fins, without loss of the protective layer, is acceptable on both air-oil exchanger. If cracks are visible, remove and replace the part.

b) If there is more than 21 bent fins on 1 air-oil exchanger, proceed to the following repair steps.

<1> Insert a locally manufactured straightening tool between the bent fins and gently apply force to straighten the fins back to approximately 90 degrees to the cooler body.

NOTE: The fins should have a gap of 0.085 in. (2.159 mm) (reference) between them.

<a> If you damage any fins when you straight it, remove the surface cooler.

<2> You may use needle nose pliers or similar type tool to straighten small localized areas of fin deformation.

<3> Continue the same process to repair all the bent fins.

NOTE: Fins can become loose or dislodged from repeated or over bending. After straightening all fins, check to assure the fins are secure. You should not be able to move the individual fins with your fingers. If the fins move easily with just finger pressure, the surface cooler should be replaced.

(d) Examine the sealing surfaces in zone A, B, and C.

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1) Examine all the sealing surfaces of the cooler for evidence of scratches, raised metal, nicks, gouges, cracks, and dents.
2) The requirements of inspection criteria apply to all sealing surfaces.
   a) Any number up to 0.001 in. (0.025 mm) is permitted.
   b) Do a check for scratches, raised metal, nicks, gouges, and cracks.
      <1> Permitted up to 0.010 in. (0.254 mm) in depth of scratches/gouges.
      <2> Raised metal or cracks are not permitted.
      <3> Not repairable.
      <4> If it is necessary, replace or send to manufacturer.
   c) Do a check for dents in zone A.
      <1> Permitted up to 0.01 in. (0.254 mm) in depth without loss of the protective layer.
      <2> Raised metal or cracks are not permitted.
      <3> If raised metal is present, do these steps:
         <a> Use a carbide bur set, STD-21746, and hand grind out findings to remove heavy burrs or raised metal.
         <b> Use an oil stone, COM-21826, to remove light burrs.
         <c> Use a 500 grit abrasive cloth, G51791, to remove edges of repaired surfaces smoothly into adjacent surfaces.
         <d> Do not remove more material than necessary.

            NOTE: No high metal must remain. The transition with the non-blended area must be smooth.

            NOTE: After blending, the maximum depth on the blended area must be less than 0.032 in. (0.81 mm).
      <4> If it is necessary, replace or send to manufacturer.

(e) Oil stains or leakage.
   1) Oil stains or leakage are not permitted.
      a) If you find them, do the applicable tasks:
         • Engine Surface Air Cooling Oil Cooler 1 Removal, TASK 79-21-02-000-801-G00 and Engine Surface Air Cooling Oil Cooler 1 Installation, TASK 79-21-02-400-801-G00
         • Engine Surface Air Cooling Oil Cooler 2 Removal, TASK 79-21-02-000-802-G00 and Engine Surface Air Cooling Oil Cooler 2 Installation, TASK 79-21-02-400-802-G00.

(f) Inspection of the fireseal.
   1) Remove the engine surface air cooling oil coolers, do these tasks:
      • Engine Surface Air Cooling Oil Cooler 1 Removal, TASK 79-21-02-000-801-G00
      • Engine Surface Air Cooling Oil Cooler 2 Removal, TASK 79-21-02-000-802-G00.
   2) The fit of the fireseal on the manifold must be sufficient for it to not fall off under its own weight.
   3) Missing fireseal, finger or spacer, torn or missing bulb seal, or spacer separated from seal is not permitted.

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a) If you find them, replace the fireseal as follows:

<1> Remove and discard the unserviceable fireseal from cooler.

<2> Clean the area with alcohol, B00676 [CP1041].

<3> Make sure that the surface is dry.

<4> Install the new fireseal.

4) Install the engine surface air cooling oil coolers, do these tasks:

- Engine Surface Air Cooling Oil Cooler 1 Installation, TASK 79-21-02-400-801-G00
- Engine Surface Air Cooling Oil Cooler 2 Installation, TASK 79-21-02-400-802-G00.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 79-21-02-410-004-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 79-21-02-860-006-G00

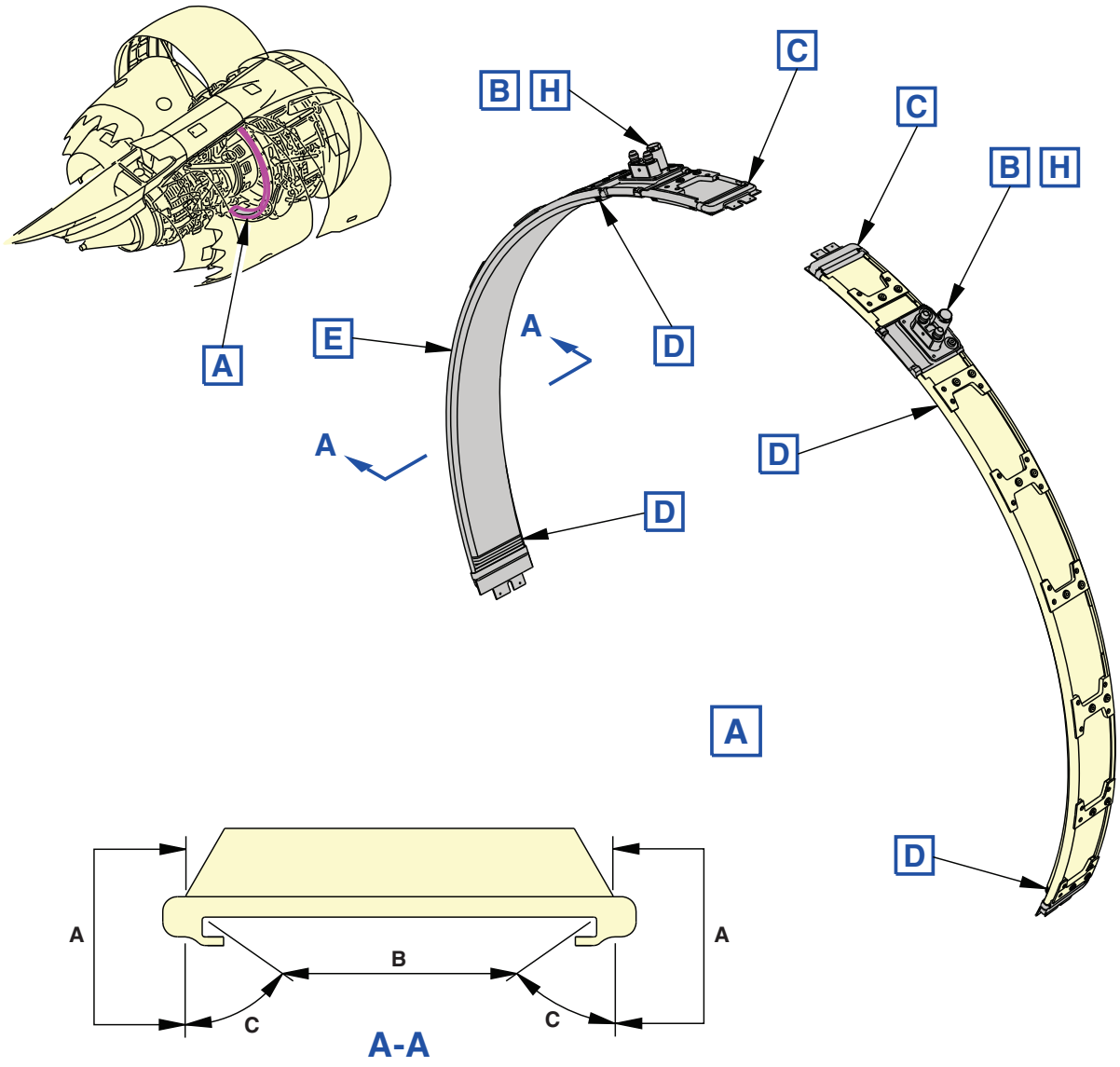
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

END OF TASK

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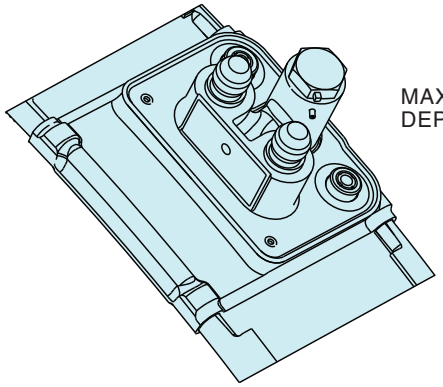
ZONE ALLOWABLE SCRATCH/GOUGE:

ZONE A: 0.010 in (0.254 mm) SCRATCH/GOUGE DEPTH
ZONE B: 0.010 in (0.254 mm) SCRATCH/GOUGE DEPTH
ZONE C: 0.010 in (0.254 mm) SCRATCH/GOUGE DEPTH
NO HIGH METAL IN ZONES A, B, C
NO MISSING FIN ALLOWED 0.010 in (0.254 mm) SCRATCH/GOUGE DEPTH ALLOWABLE ON FIN

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Engine Surface Air Cooling Oil Cooler Inspection
Figure 601/79-21-02-990-805-G00 (Sheet 1 of 4)

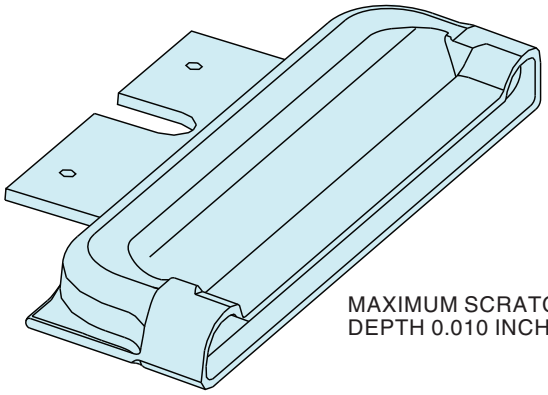
ECCN 9E991 BOEING PROPRIETARY - See title page for details



MAXIMUM SCRATCH / GOUGE
DEPTH 0.010 INCH (0.254 mm)

**ALLOWABLE SCRATCH / GOUGE DEPTH
ON THE INLET / OUTLET MANIFOLDS**

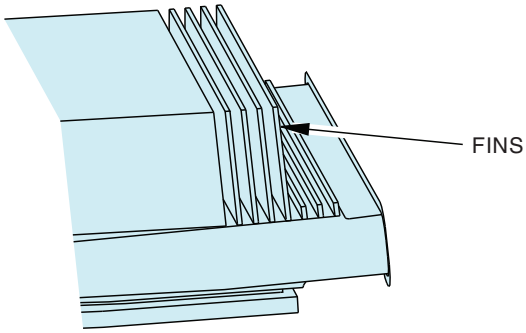
B



MAXIMUM SCRATCH / GOUGE
DEPTH 0.010 INCH (0.254 mm)

**ALLOWABLE SCRATCH / GOUGE DEPTH
ON THE RETURN MANIFOLDS**

C



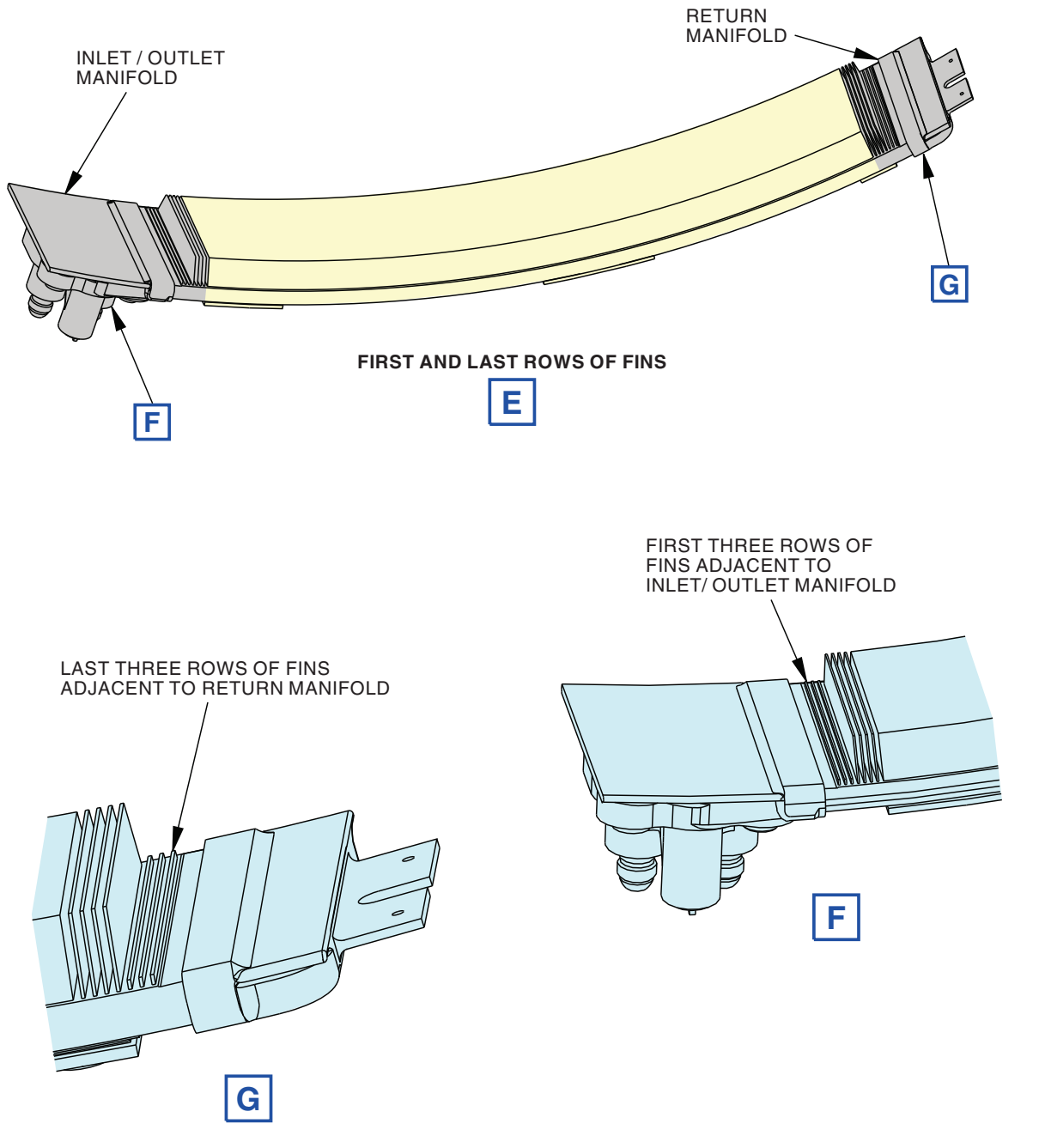
FINS

D

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**Engine Surface Air Cooling Oil Cooler Inspection
Figure 601/79-21-02-990-805-G00 (Sheet 2 of 4)**

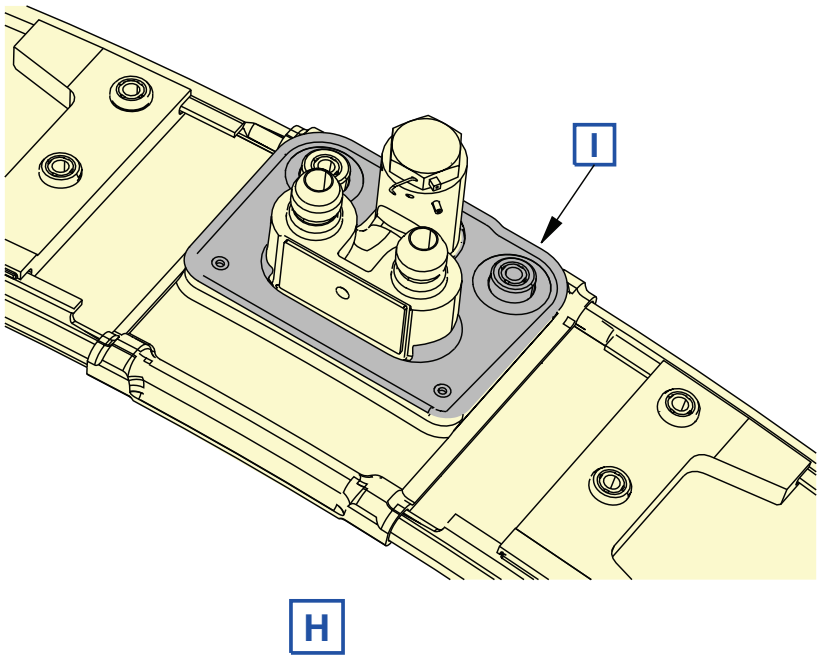
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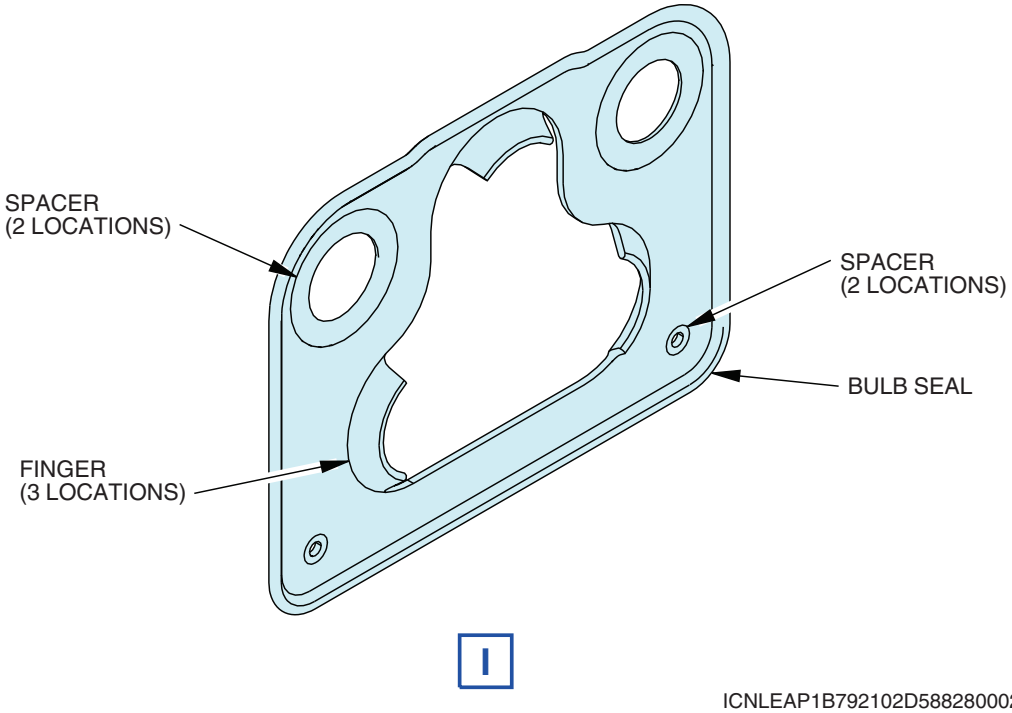


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Engine Surface Air Cooling Oil Cooler Inspection
Figure 601/79-21-02-990-805-G00 (Sheet 3 of 4)

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Engine Surface Air Cooling Oil Cooler Inspection
Figure 601/79-21-02-990-805-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LUBRICATION UNIT - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Lubrication Unit Removal

(2) Lubrication Unit Installation.

TASK 79-21-03-000-801-G00

2. Lubrication Unit Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the lubrication unit.

B. References

Reference	Title
12-13-11-600-803	Engine Oil Change Servicing (P/B 301)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
79-35-01-000-801-G00	Oil Filter Differential Pressure Sensor Removal (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14893	Guide Pins - Installation, Lubrication Unit Part #: 956A8673G02 Supplier: 58828
STD-858	Tag - DO NOT OPERATE
STD-1055	Container - Oil Resistant, 5 Gallon (19 Liter)

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
G02345 [CP8001]	Wire - Safety, 0.032 Inch (0.8 mm) Diameter	AMS 5687

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
4	Preformed packing	79-21-07-01-020	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1

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(Continued)

Number	Name/Location
415	Left Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2

H. Prepare for the Removal

SUBTASK 79-21-03-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 79-21-03-010-001-G00

(2) Do these tasks in sequence to safely open the left thrust reverser on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left fan cowl panel (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left thrust reverser (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

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I. Lubrication Unit Removal

SUBTASK 79-21-03-680-001-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Drain the oil from the lubrication unit [1]:

CAUTION DO NOT SPILL THE OIL. USE A MINIMUM OF TWO CONTAINERS TO DRAIN THE OIL. A LARGE VOLUME OF OIL WILL DRAIN FROM THE LUBE PUMP.

(a) Put the oil resistant container (5 gal)(19 Liter), STD-1055, below the lubrication unit [1].

(b) Drain the oil tank (TASK 12-13-11-600-803).

(c) Remove the drain plug [3] from the lubrication unit [1].

1) Cut the safety wire [5] from the drain plug [3].
2) Remove the drain plug [3] from the lubrication unit [1].
3) Remove and discard the preformed packing [4] from the drain plug [3].
4) Let the oil drain into the oil resistant container (5 gal)(19 Liter), STD-1055.

(d) Install the drain plug [3] on the lubrication unit [1].

1) Make sure that the mating faces are clean and clear of unwanted materials.
2) Lubricate a new preformed packing [4] with the oil, D00599 [CP2442].
3) Install the new preformed packing [4] on the drain plug [3].
4) Install the drain plug [3] on the lubrication unit [1].
5) Tighten the drain plug [3] to 67.45 in-lb (7.6 N·m) - 74.55 in-lb (8.4 N·m).
6) Install a safety wire [5] (safety wire, G02345 [CP8001]) on the drain plug [3].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 79-21-03-020-001-G00

(2) Do this task: Oil Filter Differential Pressure Sensor Removal, TASK 79-35-01-000-801-G00.

SUBTASK 79-21-03-030-001-G00

(3) Disconnect the oil tube [6] from the lubrication unit [1]:

(a) Remove the four bolts [7] from the oil tube [6].

(b) Disconnect the oil tube [6] from the lubrication unit [1].

(c) Remove and discard the gasket seal [8] from the oil tube [6].

(d) Install the protective covers on the openings.

SUBTASK 79-21-03-030-002-G00

(4) Disconnect the oil tube [9] from the lubrication unit [1]:

(a) Remove the four bolts [10] from the oil tube [9].

(b) Disconnect the oil tube [9] from the lubrication unit [1].

(c) Remove and discard the gasket seal [11] from the oil tube [9].

(d) Install the protective covers on the openings.

SUBTASK 79-21-03-030-003-G00

(5) Disconnect the oil tube [22] and oil tube [29]:

(a) Remove the nut [12] and bolt [13] that attach the clamp [14] and clamp [15] to the bracket [16].

(b) Remove the clamp [14] and oil tube [22] from the bracket [16].

(c) Remove the clamp [15] and oil tube [29] from the bracket [16].

SUBTASK 79-21-03-010-002-G00

(6) Disconnect the oil tube [22] and oil tube [29] from the bracket [21]:

(a) Remove the nut [17] and bolt [18] that attach the clamp [19] and clamp [20] to the bracket [21].

(b) Remove the clamp [19] and oil tube [22] from the bracket [21].

(c) Remove the clamp [20] and oil tube [29] from the bracket [21].

SUBTASK 79-21-03-010-003-G00

(7) Disconnect the oil tube [22] from the lubrication unit [1]:

(a) Remove the electrical harness HJ3A [23] from the clip [24].

(b) Remove the two bolts [25] that attach the bracket [28] to the oil tube [22].

(c) Remove the bracket [28] from the oil tube [22].

(d) Remove the bolt [26] from the oil tube [22].

(e) Disconnect the oil tube [22] from the lubrication unit [1].

(f) Remove and discard the gasket seal [27] from the oil tube [22].

(g) Install the protective covers on the openings.

SUBTASK 79-21-03-010-004-G00

(8) Disconnect the oil tube [29] from the lubrication unit [1]:

(a) Remove the three bolts [30] from the oil tube [29].

(b) Disconnect the oil tube [29] from the lubrication unit [1].

(c) Remove and discard the gasket seal [31] from the oil tube [29].

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(d) Install the protective covers on the openings.

SUBTASK 79-21-03-010-005-G00

(9) Disconnect the oil tube [32] from the AGB [35]:

(a) Remove the four bolts [33] from the oil tube [32].

(b) Disconnect the oil tube [32] from the AGB [35].

(c) Remove and discard the gasket C seal [34] from the AGB [35].

(d) Install the protective covers on the openings.

SUBTASK 79-21-03-010-006-G00

(10) Disconnect the oil tube [32] from the lubrication unit [1]:

(a) Remove the three bolts [37] from the oil tube [32].

(b) Disconnect the oil tube [32] from the lubrication unit [1].

(c) Remove and discard the gasket seal [38] from the oil tube [32].

(d) Install the protective covers on the openings.

SUBTASK 79-21-03-010-007-G00

(11) Disconnect the oil tube [39] from the lubrication unit [1]:

(a) Remove the three bolts [40] from the oil tube [39].

(b) Disconnect the oil tube [39] from the lubrication unit [1].

(c) Remove and discard the gasket seal [41] from the oil tube [39].

(d) Install the protective covers on the openings.

SUBTASK 79-21-03-010-008-G00

(12) Disconnect the oil tube [42] from the lubrication unit [1]:

(a) Remove the three bolts [43] from the oil tube [42].

(b) Disconnect the oil tube [42] from the lubrication unit [1].

(c) Remove and discard the gasket seal [44] from the oil tube [42].

(d) Install the protective covers on the openings.

SUBTASK 79-21-03-010-009-G00

(13) Disconnect the oil tube [45] from the lubrication unit [1]:

(a) Remove the three bolts [46] from the oil tube [45].

(b) Disconnect the oil tube [45] from the lubrication unit [1].

(c) Remove and discard the gasket seal [47] from the oil tube [45].

(d) Install the protective covers on the openings.

SUBTASK 79-21-03-020-002-G00

WARNING

BE CAREFUL WHEN YOU MOVE THE COMPONENT. THE COMPONENT IS HEAVY. INJURIES TO PERSONS CAN OCCUR.

(14) Remove the lubrication unit [1] from the AGB input pad [54]:

NOTE: The lubrication unit weighs 28.7 lb (13.0 kg).

(a) Remove any three bolts [48] and three washers [49] not adjacent to each other that attach the lubrication unit [1] to the AGB input pad [54].

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(b) Install the guide pins, SPL-14893, through the three bolt holes of the lubrication unit [1] to the AGB input pad [54].

(c) Remove the remaining six bolts [48] and six washers [49] that attach the lubrication unit [1] to the AGB input pad [54].

CAUTION

THE AGB SHAFT SLEEVE CAN STAY ATTACHED TO THE LUBE PUMP DRIVE-SHAFT WHEN YOU REMOVE THE LUBE PUMP FROM THE AGB. MAKE SURE THAT THE AGB SHAFT SLEEVE STAYS ON THE AGB. IF YOU DO NOT INSTALL THE AGB SHAFT SLEEVE, THE ENGINE WILL NOT HAVE SUFFICIENT LUBRICATION WHEN YOU OPERATE AND DAMAGE TO THE ENGINE CAN OCCUR.

(d) Carefully pull the lubrication unit [1] rearward until the splines of the drive shaft [50] disengage from the AGB input pad [54].

(e) Make sure that the AGB shaft sleeve [55] remains installed in the AGB drive gear and that it is not pulled out together with the lubrication unit drive shaft.

(f) Remove and discard the preformed packing [51] from the groove of the lubrication unit [1].

(g) Remove and discard the preformed packing [52] from the groove of the drive shaft [50].

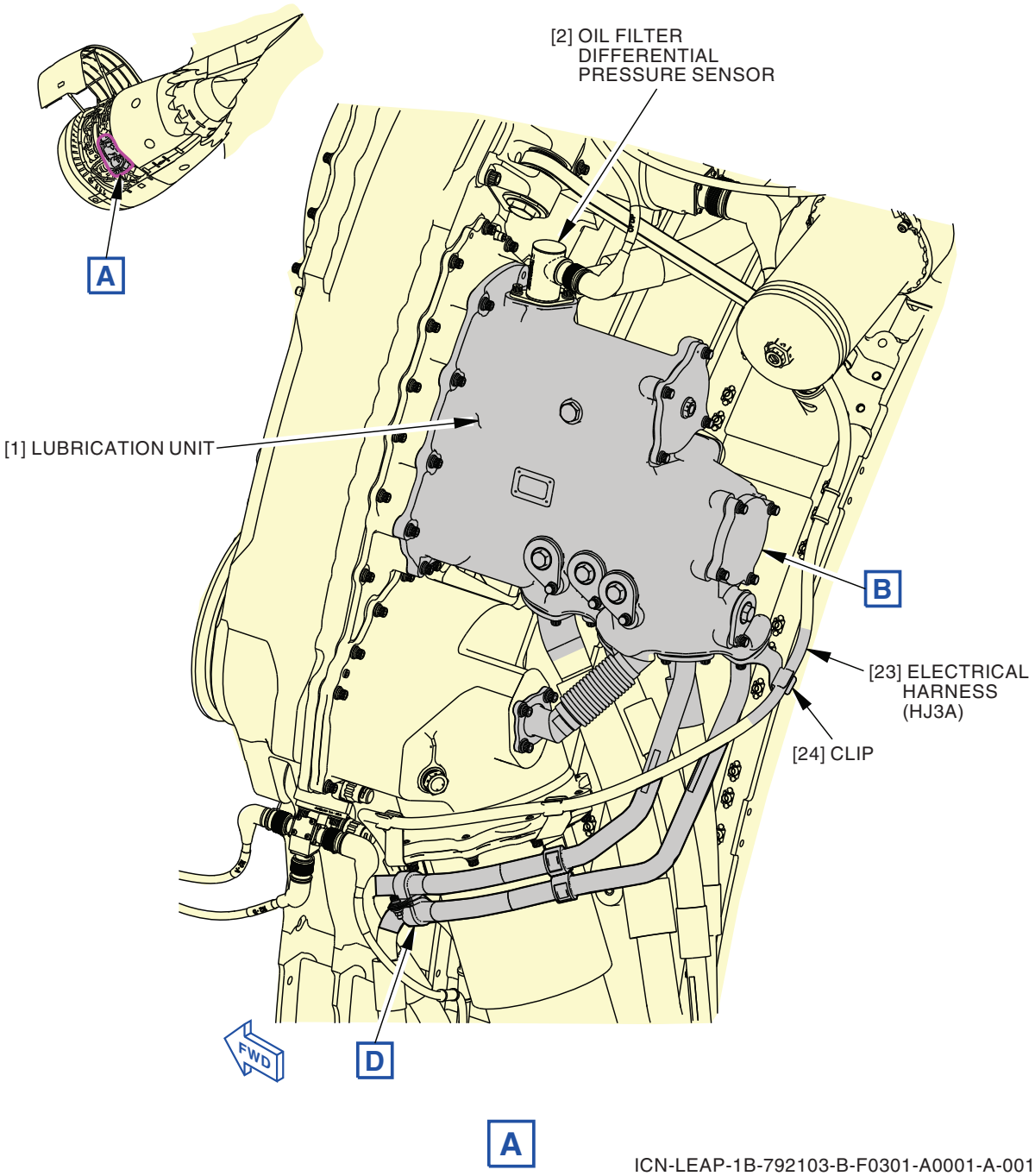
(h) Remove and discard the preformed packing [53] from the groove of the lubrication unit [1].

(i) Remove the guide pins.

(j) Install the protective covers on the openings.

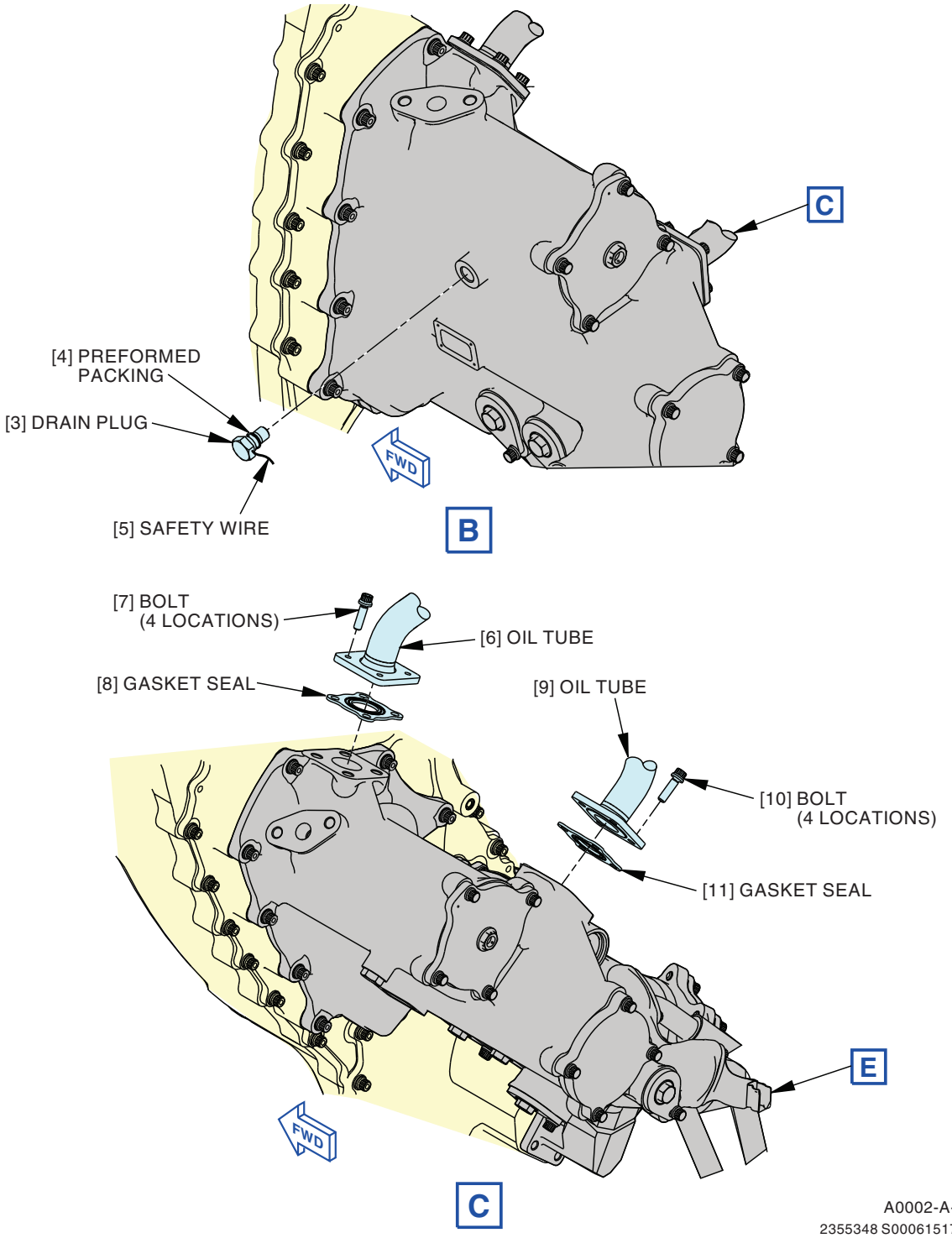
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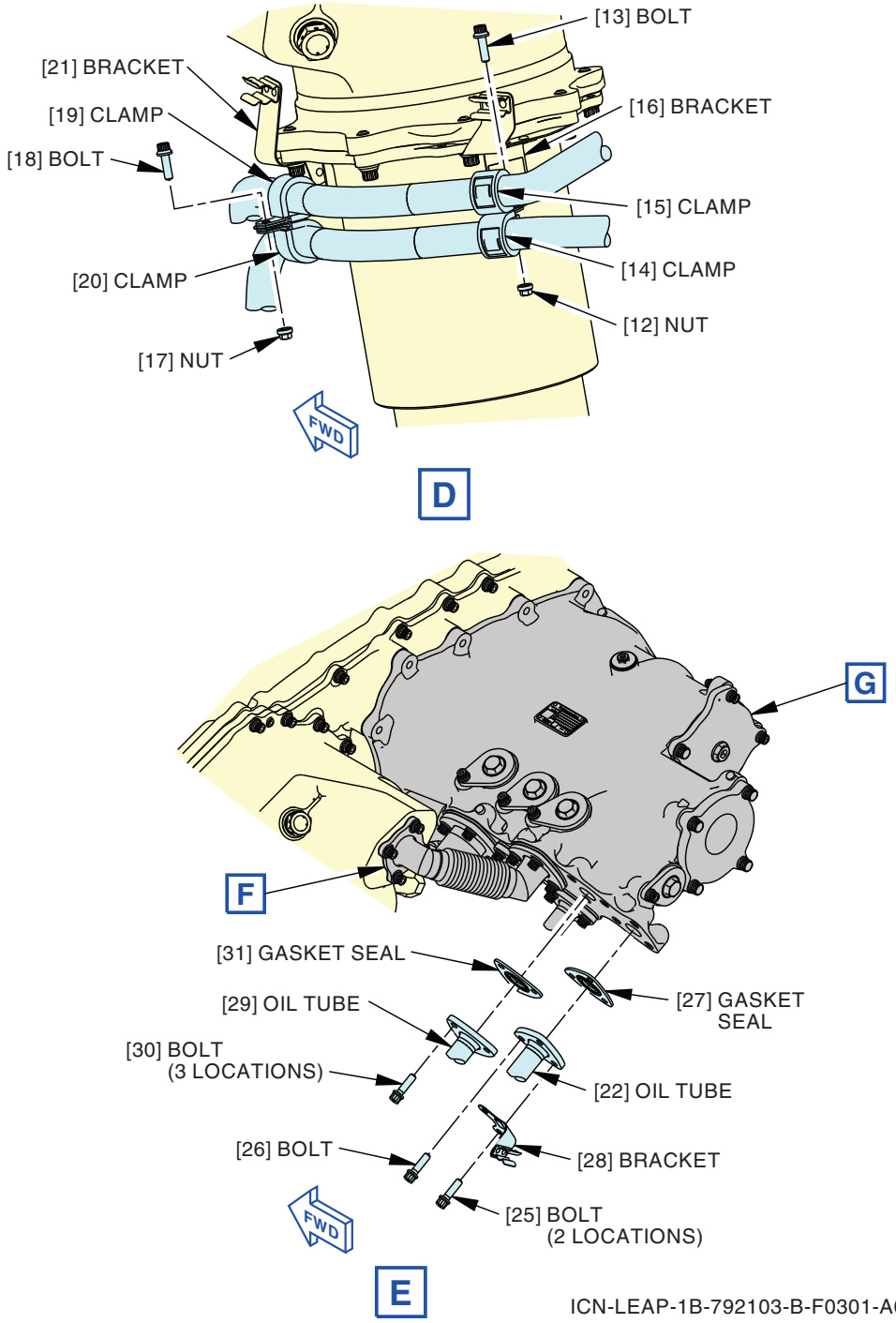
Lubrication Unit Installation
Figure 401/79-21-03-990-802-G00 (Sheet 1 of 7)

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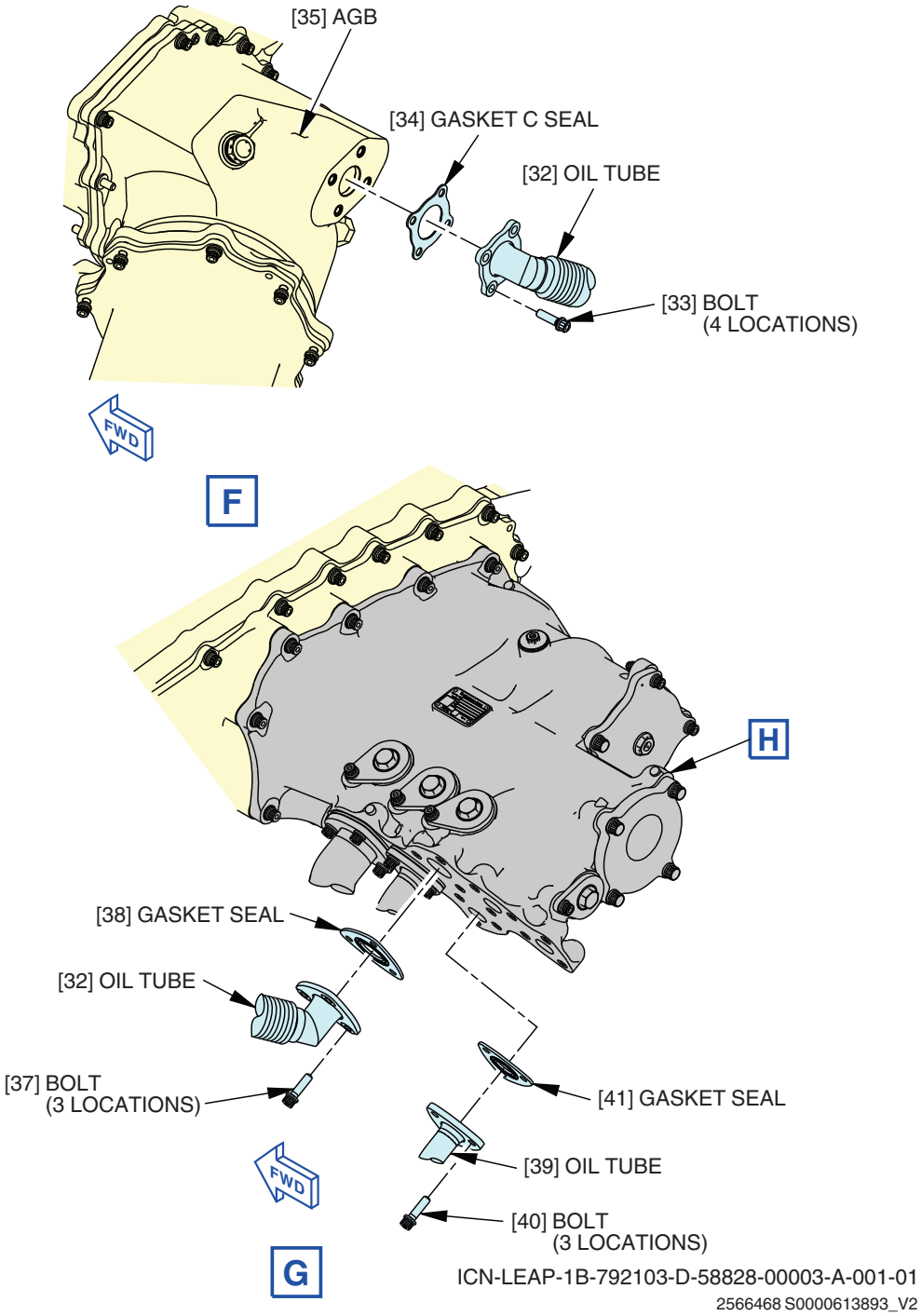
Lubrication Unit Installation
Figure 401/79-21-03-990-802-G00 (Sheet 2 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



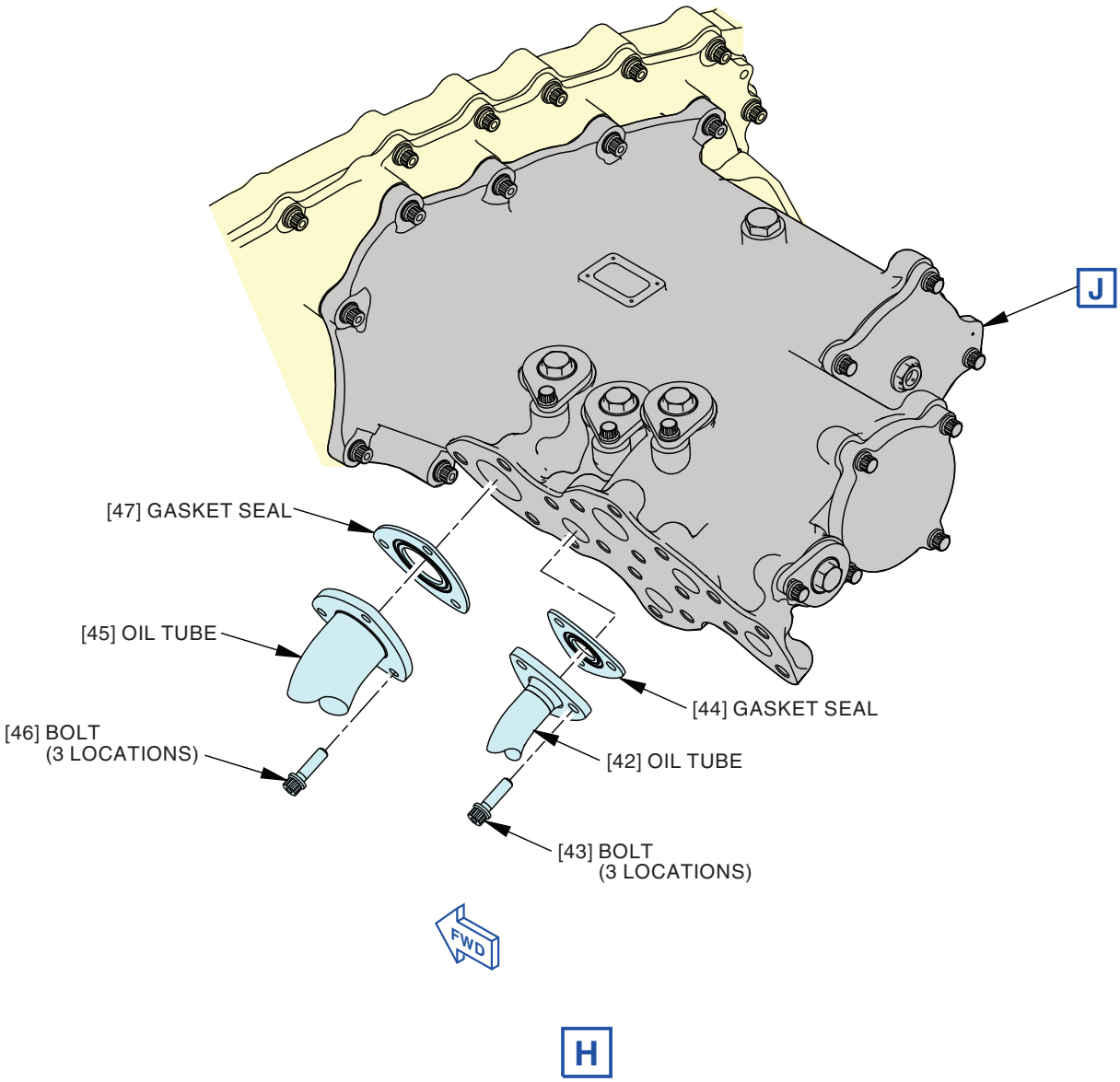
Lubrication Unit Installation
Figure 401/79-21-03-990-802-G00 (Sheet 3 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Lubrication Unit Installation
Figure 401/79-21-03-990-802-G00 (Sheet 4 of 7)

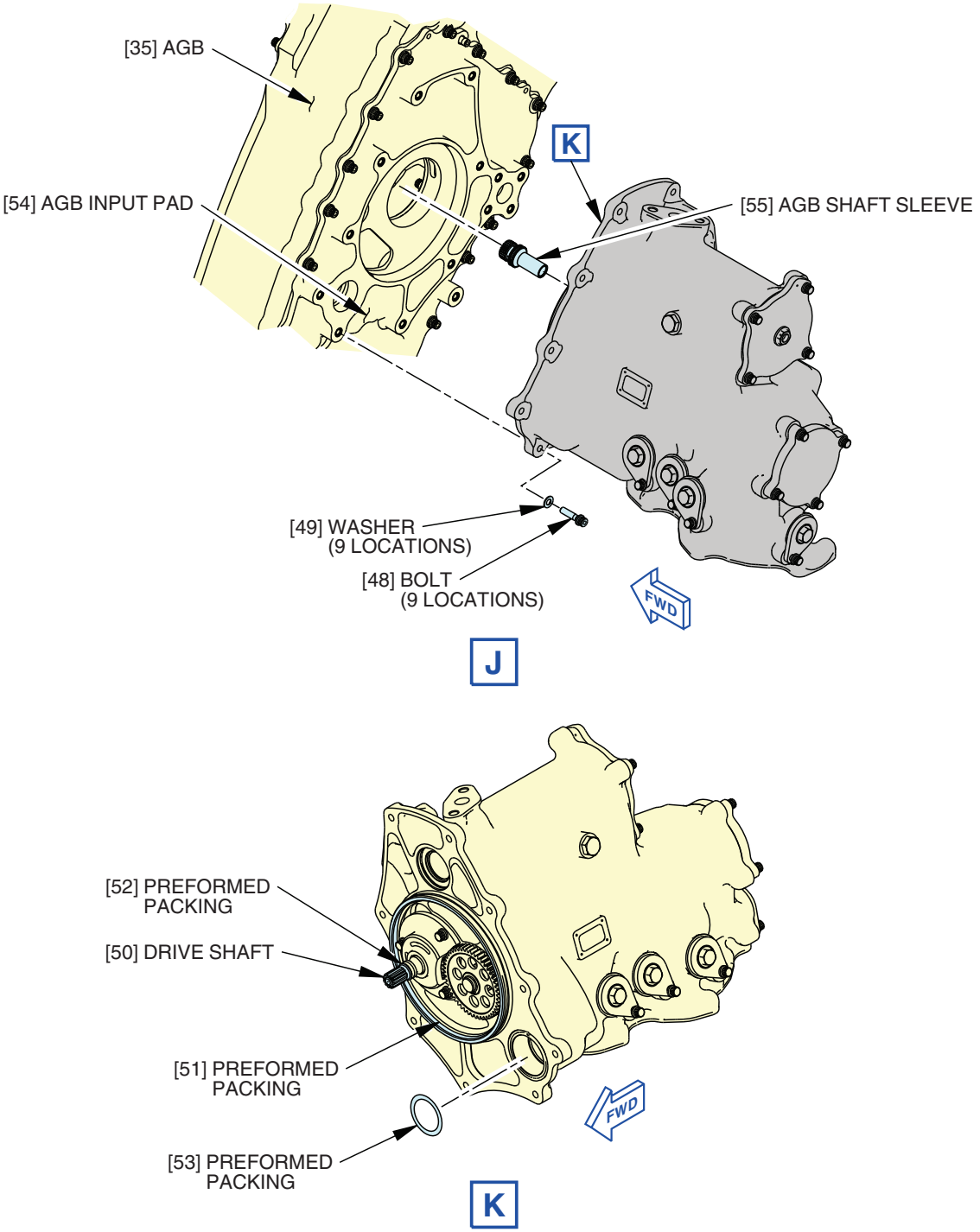
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Lubrication Unit Installation
Figure 401/79-21-03-990-802-G00 (Sheet 5 of 7)

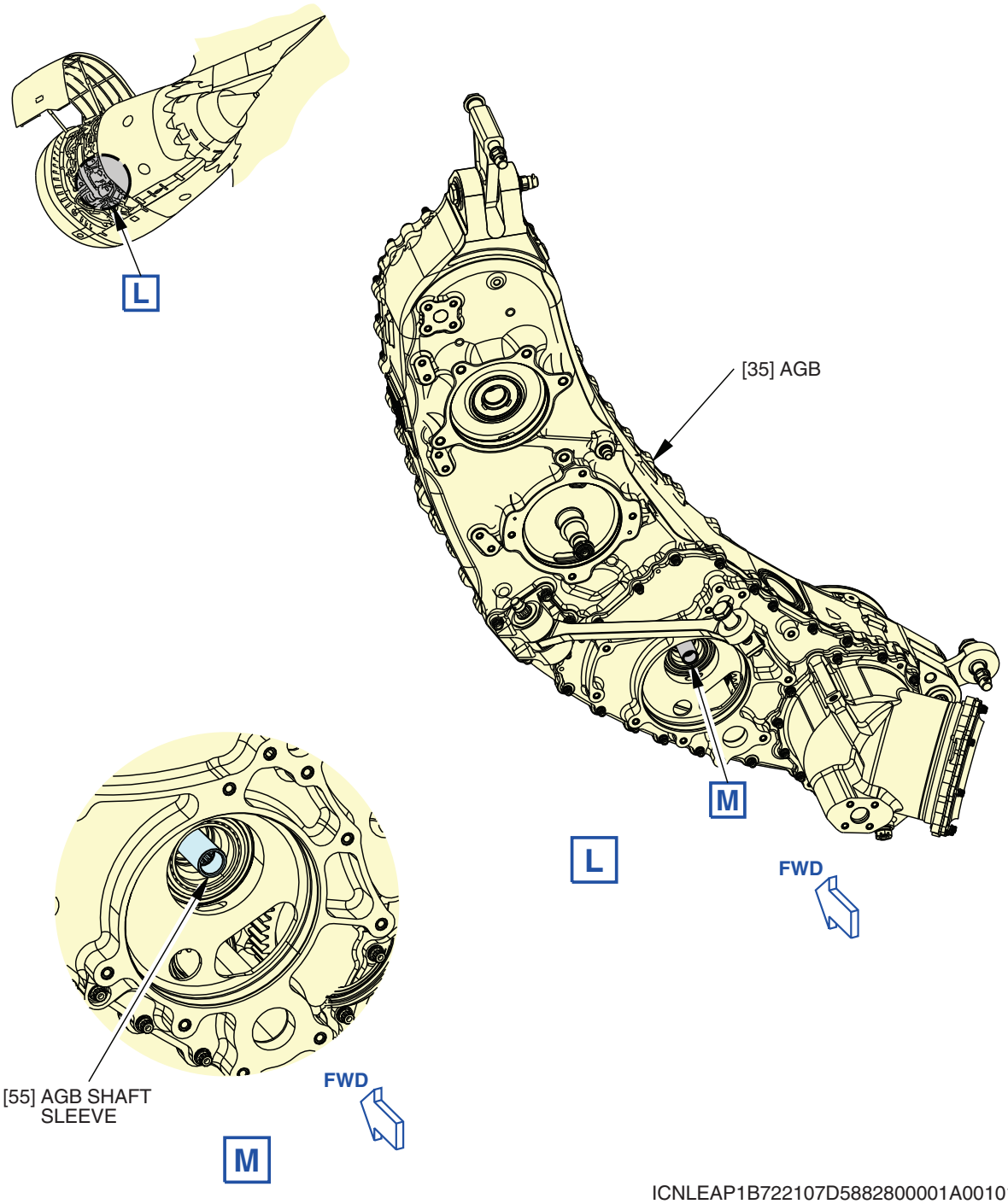
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Lubrication Unit Installation
Figure 401/79-21-03-990-802-G00 (Sheet 6 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Lubrication Unit Installation
Figure 401/79-21-03-990-802-G00 (Sheet 7 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 79-21-03-400-801-G00

3. Lubrication Unit Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the lubrication unit.

B. References

Reference	Title
12-13-11-600-801	Engine Servicing (Oil Replenishing) (P/B 301)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-00-00-910-801-G00	High Pressure Rotor (N2) Turning for Borescope (P/B 201)
72-63-01-610-802-G00	Accessory Gearbox Assembly Oil Replenishing (P/B 201)
72-63-04-400-801-G00	Accessory Gearbox Manual Control Cover Installation (P/B 401)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
79-35-01-400-801-G00	Oil Filter Differential Pressure Sensor Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14847	Kit, Installation - O-Ring, AGB Part #: 956A8610G01 Supplier: 58828
SPL-14893	Guide Pins - Installation, Lubrication Unit Part #: 956A8673G02 Supplier: 58828
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D00601 [CP2101]	Vaseline - Graphite Mineral	
D00672 [CP5070]	Vaseline - Pure Mineral	V V-P-236
G51335 [CP2674]	Lint-Free Cloth	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
8	Gasket seal	79-21-03-01-040	PTF ALL
11	Gasket seal	79-21-03-01-020	PTF ALL
27	Gasket seal	79-21-03-01-080	PTF ALL
31	Gasket seal	79-21-03-01-100	PTF ALL
34	Gasket C seal	72-00-00-12-200	PTF ALL
38	Gasket seal	79-21-03-01-140	PTF ALL
41	Gasket seal	79-21-03-01-120	PTF ALL
44	Gasket seal	79-21-03-01-160	PTF ALL
47	Gasket seal	79-21-03-01-180	PTF ALL

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(Continued)

AMM Item	Description	AIPC Reference	AIPC Effectivity
51	Preformed packing	79-21-03-01-230	PTF ALL
52	Preformed packing	79-21-03-01-210	PTF ALL
53	Preformed packing	79-21-03-01-220	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2

H. Lubrication Unit Installation

SUBTASK 79-21-03-490-001-G00

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

WARNING BE CAREFUL WHEN YOU MOVE THE COMPONENT. THE COMPONENT IS HEAVY. INJURIES TO PERSONS CAN OCCUR.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION THE AGB SHAFT SLEEVE CAN STAY ATTACHED TO THE LUBE PUMP DRIVE-SHAFT WHEN YOU REMOVE THE LUBE PUMP FROM THE AGB. MAKE SURE THAT THE AGB SHAFT SLEEVE STAYS ON THE AGB. IF YOU DO NOT INSTALL THE AGB SHAFT SLEEVE, THE ENGINE WILL NOT HAVE SUFFICIENT LUBRICATION WHEN YOU OPERATE AND DAMAGE TO THE ENGINE CAN OCCUR.

(1) Install the lubrication unit [1].

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Remove the protective covers on the openings.

(c) Make sure that the AGB shaft sleeve [55] is installed on the AGB [35].

(d) Lubricate the splines of the drive shaft [50] with engine oil, D00599 [CP2442].

(e) Lubricate the nine bolts [48] with graphite mineral vaseline, D00601 [CP2101].

(f) Lubricate the new preformed packing [51], new preformed packing [52], and new preformed packing [53] with pure mineral vaseline, D00672 [CP5070].

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(g) Use the O-ring installation kit, SPL-14847, to install the new preformed packing [52] into the groove of the drive shaft [50].

(h) Install the new preformed packing [53] into the groove of the lubrication unit [1].

(i) Install the new preformed packing [51] into the groove of the lubrication unit [1].

(j) Install the guide pins, SPL-14893, through the three bolt holes of the AGB input pad [54].

CAUTION

MAKE SURE THAT YOU INSTALL THE AGB SHAFT SLEEVE IN THE AGB DRIVE GEAR BEFORE YOU INSTALL THE LUBRICATION UNIT ON THE AGB. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(k) Make sure that the AGB shaft sleeve [55] is installed in the AGB [35] drive gear before the lubrication unit [1] installation on the AGB [35].

(l) Align the drive shaft [50] of the lubrication unit [1] with the splines of the AGB input pad [54] as follows:

1) If it is necessary, turn the AGB shaft (TASK 72-00-00-910-801-G00).

(m) Carefully align the lubrication unit [1] with the guide pins and push the drive shaft [50] of the lubrication unit [1] into the AGB input pad [54].

(n) Make sure that the mating flange of the lubrication unit [1] is fully engaged with the AGB input pad [54].

(o) Install the six bolts [48] and six washers [49] that attach the lubrication unit [1] to the AGB input pad [54].

(p) Remove the guide pins.

(q) Install the remaining three bolts [48] and three washers [49] that attach the lubrication unit [1] to the AGB input pad [54].

(r) Tighten the bolts [48] to 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m), in a criss-cross pattern.

(s) If the AGB manual control cover has been removed, do this task: Accessory Gearbox Manual Control Cover Installation, TASK 72-63-04-400-801-G00.

SUBTASK 79-21-03-090-001-G00

(2) Connect the oil tube [45] to the lubrication unit [1].

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Remove the protective covers on the openings.

(c) Lubricate each side of the elastomer part of a new gasket seal [47] with pure mineral vaseline, D00672 [CP5070], or engine oil, D00599 [CP2442].

(d) Lubricate the three bolts [46] and pressure faces with graphite mineral vaseline, D00601 [CP2101].

(e) Install the new gasket seal [47] and oil tube [45] on the lubrication unit [1].

(f) Install the three bolts [46] that attach the oil tube [45] to the lubrication unit [1].

(g) Tighten the three bolts [46] to 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m).

(h) Remove the unwanted pure mineral vaseline, D00672 [CP5070], and graphite mineral vaseline, D00601 [CP2101], with a lint-free cloth, G51335 [CP2674].

SUBTASK 79-21-03-410-001-G00

(3) Connect the oil tube [42] to the lubrication unit [1].

(a) Make sure that the mating faces are clean and clear of unwanted materials.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Remove the protective covers on the openings.

(c) Lubricate each side of the elastomer part of a new gasket seal [44] with pure mineral vaseline, D00672 [CP5070], or engine oil, D00599 [CP2442].

(d) Lubricate the three bolts [43] and pressure faces with graphite mineral vaseline, D00601 [CP2101].

(e) Install the new gasket seal [44] and oil tube [42] on the lubrication unit [1].

(f) Install the three bolts [43] that attach the oil tube [42] to the lubrication unit [1].

(g) Tighten the bolts [43] to 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m).

(h) Remove the unwanted pure mineral vaseline, D00672 [CP5070], and graphite mineral vaseline, D00601 [CP2101], with a lint-free cloth, G51335 [CP2674].

SUBTASK 79-21-03-410-002-G00

(4) Connect the oil tube [39] to the lubrication unit [1].

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Remove the protective covers on the openings.

(c) Lubricate each side of the elastomer part of a new gasket seal [41] with pure mineral vaseline, D00672 [CP5070], or engine oil, D00599 [CP2442].

(d) Lubricate the three bolts [40] and pressure faces with graphite mineral vaseline, D00601 [CP2101].

(e) Install the new gasket seal [41] and oil tube [39] on the lubrication unit [1].

(f) Install the three bolts [40] that attach the oil tube [39] to the lubrication unit [1].

(g) Tighten the bolts [40] to 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m).

(h) Remove the unwanted pure mineral vaseline, D00672 [CP5070], and graphite mineral vaseline, D00601 [CP2101], with a lint-free cloth, G51335 [CP2674].

SUBTASK 79-21-03-410-003-G00

(5) Connect the oil tube [32] to the lubrication unit [1].

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Remove the protective covers on the openings.

(c) Lubricate each side of the elastomer part of a new gasket seal [38] with pure mineral vaseline, D00672 [CP5070], or engine oil, D00599 [CP2442].

(d) Lubricate the three bolts [37] and pressure faces with graphite mineral vaseline, D00601 [CP2101].

(e) Install the new gasket seal [38] and oil tube [32] on the lubrication unit [1].

(f) Install the three bolts [37] that attach the oil tube [32] to the lubrication unit [1].

(g) Tighten the bolts [37] to 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m).

(h) Remove the unwanted pure mineral vaseline, D00672 [CP5070], and graphite mineral vaseline, D00601 [CP2101], with a lint-free cloth, G51335 [CP2674].

SUBTASK 79-21-03-410-004-G00

(6) Connect the oil tube [32] to the AGB [35].

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Remove the protective covers on the openings.

(c) Lubricate the new gasket C seal [34] with engine oil, D00599 [CP2442].

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(d) Install the new gasket C seal [34] and oil tube [32] in a correct position on the lubrication unit [1].

(e) Install the four bolts [33] that attach the oil tube [32] to the lubrication unit [1].

(f) Tighten the bolts [33] in a criss cross pattern to 84.1 in-lb (9.5 N·m) - 92.9 in-lb (10.5 N·m).

SUBTASK 79-21-03-410-005-G00

(7) Connect the oil tube [29] to the lubrication unit [1].

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Remove the protective covers on the openings.

(c) Lubricate each side of the elastomer part of a new gasket seal [31] with pure mineral vaseline, D00672 [CP5070], or engine oil, D00599 [CP2442].

(d) Lubricate the three bolts [30] and pressure faces with graphite mineral vaseline, D00601 [CP2101].

(e) Install the new gasket seal [31] and oil tube [29] on the lubrication unit [1].

(f) Install the three bolts [30] that attach the oil tube [29] to the lubrication unit [1].

(g) Tighten the bolts [30] to 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m).

(h) Remove the unwanted pure mineral vaseline, D00672 [CP5070], and graphite mineral vaseline, D00601 [CP2101], with a lint-free cloth, G51335 [CP2674].

SUBTASK 79-21-03-410-006-G00

(8) Connect the oil tube [22] to the lubrication unit [1].

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Remove the protective covers on the openings.

(c) Lubricate each side of the elastomer part of a new gasket seal [27] with pure mineral vaseline, D00672 [CP5070], or engine oil, D00599 [CP2442].

(d) Lubricate the two bolts [25], pressure faces, and bolt [26] with graphite mineral vaseline, D00601 [CP2101].

(e) Install the new gasket seal [27] and oil tube [22] on the lubrication unit [1].

(f) Install the bolt [26] that attach the oil tube [22] to the lubrication unit [1].

(g) Tighten the bolt [26] to 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m).

(h) Install the bracket [28] on the oil tube [22].

(i) Install the two bolts [25] that attach the bracket [28] to the oil tube [22].

(j) Tighten the two bolts [25] to 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m).

(k) Remove the unwanted pure mineral vaseline, D00672 [CP5070], and graphite mineral vaseline, D00601 [CP2101], with a lint-free cloth, G51335 [CP2674].

SUBTASK 79-21-03-410-007-G00

(9) Install the electrical harness HJ3A [23] on the clip [24].

SUBTASK 79-21-03-410-008-G00

(10) Connect the oil tube [22] and oil tube [29] to the bracket [21].

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Remove the protective covers on the openings.

(c) Install the clamp [20] and oil tube [29] on the bracket [21].

(d) Install the clamp [19] and oil tube [22] on the bracket [21].

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(e) Install the nut [17] and bolt [18] that attach the clamp [19] and clamp [20] to the bracket [21].

(f) Tighten the nut [17] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

SUBTASK 79-21-03-410-009-G00

(11) Connect the oil tube [22] and oil tube [29] to the bracket [16].

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Remove the protective covers on the openings.

(c) Install the clamp [15] and oil tube [29] on the bracket [16].

(d) Install the clamp [14] and oil tube [22] on the bracket [16].

(e) Install the nut [12] and bolt [13] that attach the clamp [14] and the clamp [15] to the bracket [16].

(f) Tighten the nut [12] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

SUBTASK 79-21-03-410-010-G00

(12) Connect the oil tube [9] to the lubrication unit [1].

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Remove the protective covers on the openings.

(c) Lubricate each side of the elastomer part of a new gasket seal [11] with pure mineral vaseline, D00672 [CP5070], or engine oil, D00599 [CP2442].

(d) Lubricate the four bolts [10] and pressure faces with graphite mineral vaseline, D00601 [CP2101].

(e) Install the new gasket seal [11] and oil tube [9] on the lubrication unit [1].

(f) Install the four bolts [10] that attach the oil tube [9] to the lubrication unit [1].

(g) Tighten the bolts [10] in a criss cross pattern to 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m).

(h) Remove the unwanted pure mineral vaseline, D00672 [CP5070], and graphite mineral vaseline, D00601 [CP2101], with a lint-free cloth, G51335 [CP2674].

SUBTASK 79-21-03-410-011-G00

(13) Connect the oil tube [6] to the lubrication unit [1].

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Remove the protective covers on the openings.

(c) Lubricate each side of the elastomer part of a new gasket seal [8] with pure mineral vaseline, D00672 [CP5070], or engine oil, D00599 [CP2442].

(d) Lubricate the four bolts [7] and pressure faces with graphite mineral vaseline, D00601 [CP2101].

(e) Install the new gasket seal [8] and oil tube [6] on the lubrication unit [1].

(f) Install the four bolts [7] that attach the oil tube [6] to the lubrication unit [1].

(g) Tighten the bolts [7] in a criss cross pattern to 110 in-lb (12.4 N·m) - 120 in-lb (13.6 N·m).

(h) Remove the unwanted pure mineral vaseline, D00672 [CP5070], and graphite mineral vaseline, D00601 [CP2101], with a lint-free cloth, G51335 [CP2674].

SUBTASK 79-21-03-100-001-G00

(14) Do this task: Oil Filter Differential Pressure Sensor Installation, TASK 79-35-01-400-801-G00.

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SUBTASK 79-21-03-610-002-G00

(15) Do this task: Accessory Gearbox Assembly Oil Replenishing, TASK 72-63-01-610-802-G00.

SUBTASK 79-21-03-610-001-G00

(16) Do this task: Engine Servicing (Oil Replenishing), TASK 12-13-11-600-801.

SUBTASK 79-21-03-420-001-G00

(17) Clean the work area and clear the tools and other items.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 79-21-03-410-012-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the left thrust reverser:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

(b) Close the applicable left Fan Cowl Panel (TASK 71-11-04-410-801-G00).

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 79-21-03-860-002-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

J. Lubrication Unit Installation Test

SUBTASK 79-21-03-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SCAVENGE SCREEN PLUGS - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) A removal of the scavenge screen plugs

(2) An installation of the scavenge screen plugs

(3) An inspection of the scavenge screen plugs.

TASK 79-21-04-000-801-G00

2. Scavenge Screen Plugs Removal

(Figure 201 or Figure 202)

A. General

(1) This task provides the instructions to remove the scavenge screen plugs.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

F. Prepare for the Removal

SUBTASK 79-21-04-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 79-21-04-860-013-G00

(2) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
423	Left Fan Cowl, Engine 2

G. Scavenge Screen Plugs Removal

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN SB 79-0007

CAUTION DO NOT REMOVE THE HEXAGONAL MAGNETIC CHIP PLUG FROM THE SCAVENGE SCREEN PLUG. IF YOU DO NOT OBEY, DAMAGE TO THE ENGINE CAN OCCUR.

CAUTION DO NOT USE THE HEXAGONAL MAGNETIC CHIP PLUG TO REMOVE THE SCAVENGE SCREEN PLUGS. IF YOU DO NOT OBEY, AN OIL LEAK CAN OCCUR.

PTF ALL

SUBTASK 79-21-04-010-003-G00

(1) Remove the single stage scavenge screen plug [1] as follows:

PTF ALL; PRE LEAP-1B SERVICE BULLETIN SB 79-0007

(a) Do not remove the hexagonal magnetic chip plug [6] and magnetic bar for inspection.

NOTE: If you remove the hexagonal magnetic chip plug and magnetic bar, the scavenge screen sleeve will not catch the particles.

PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Remove the scavenge screen plug bolt [9] that attaches the single stage scavenge screen plug [1] to the lubrication unit [5].

(c) Remove the single stage scavenge screen plug [1] from the lubrication unit [5].

(d) Remove the O-ring [10] from the groove on the single stage scavenge screen plug [1].

(e) Discard the O-ring [10].

(f) Install the protective covers on the single stage scavenge screen plug [1] and lubrication unit [5] port.

SUBTASK 79-21-04-010-004-G00

(2) Remove the single stage scavenge screen plug [2] as follows:

PTF ALL; PRE LEAP-1B SERVICE BULLETIN SB 79-0007

(a) Do not remove the hexagonal magnetic chip plug [7] and magnetic bar for inspection.

NOTE: If you remove the hexagonal magnetic chip plug and magnetic bar, the scavenge screen sleeve will not catch the particles.

PTF ALL

(b) Remove the scavenge screen plug bolt [11] that attaches the single stage scavenge screen plug [2] to the lubrication unit [5].

(c) Remove the single stage scavenge screen plug [2] from the lubrication unit [5].

(d) Remove the O-ring [12] from the groove on the single stage scavenge screen plug [2].

(e) Discard the O-ring [12].

(f) Install the protective covers on the single stage scavenge screen plug [2] and lubrication unit [5] port.

SUBTASK 79-21-04-010-005-G00

(3) Remove the double stage scavenge screen plug [3] as follows:

PTF ALL; PRE LEAP-1B SERVICE BULLETIN SB 79-0007

(a) Do not remove the hexagonal magnetic chip plug [19] and magnetic bar for inspection.

NOTE: If you remove the hexagonal magnetic chip plug and magnetic bar, the scavenge screen sleeve will not catch the particles.

PTF ALL

(b) Remove the scavenge screen plug bolt [13] that attaches the double stage scavenge screen plug [3] to the lubrication unit [5].

(c) Remove the double stage scavenge screen plug [3] from the lubrication unit [5].

(d) Remove the O-ring [14] from the groove on the double stage scavenge screen plug [3].

(e) Remove the O-ring [15] from the groove on the double stage scavenge screen plug [3].

(f) Discard the O-ring [14] and O-ring [15].

(g) Install the protective covers on the double stage scavenge screen plug [3] and lubrication unit [5] port.

SUBTASK 79-21-04-010-001-G00

(4) Remove the double stage scavenge screen plug [4] as follows:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; PRE LEAP-1B SERVICE BULLETIN SB 79-0007

(a) Do not remove the hexagonal magnetic chip plug [20] and magnetic bar for inspection.

NOTE: If you remove the hexagonal magnetic chip plug and magnetic bar, the scavenge screen sleeve will not catch the particles.

PTF ALL

(b) Remove the scavenge screen plug bolt [16] that attaches the double stage scavenge screen plug [4] to the lubrication unit [5].

(c) Remove the double stage scavenge screen plug [4] from the lubrication unit [5].

(d) Remove the O-ring [17] from the groove on the double stage scavenge screen plug [4].

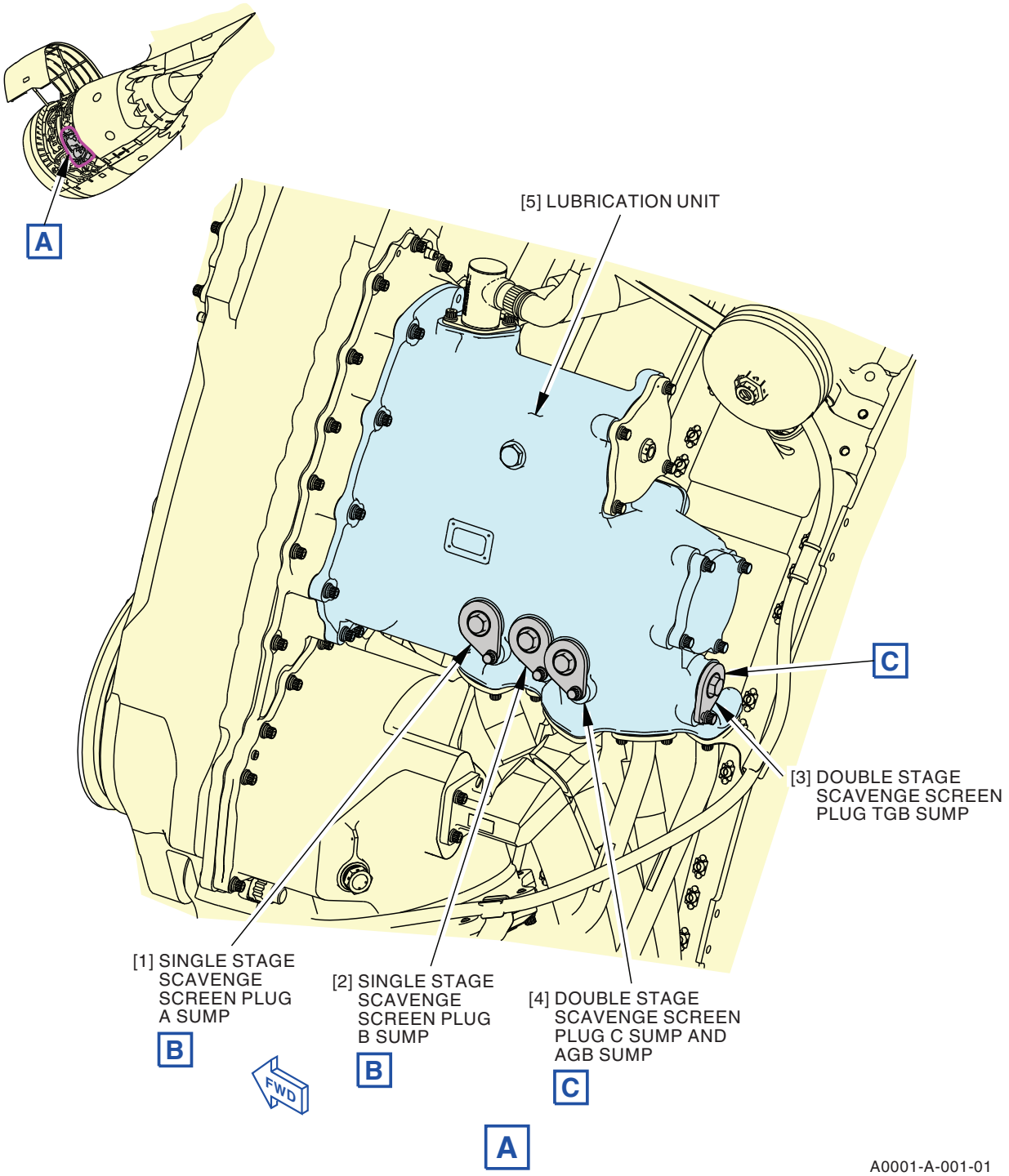
(e) Remove the O-ring [18] from the groove on the double stage scavenge screen plug [4].

(f) Discard the O-ring [17] and O-ring [18].

(g) Install the protective covers on the double stage scavenge screen plug [4] and lubrication unit [5] port.

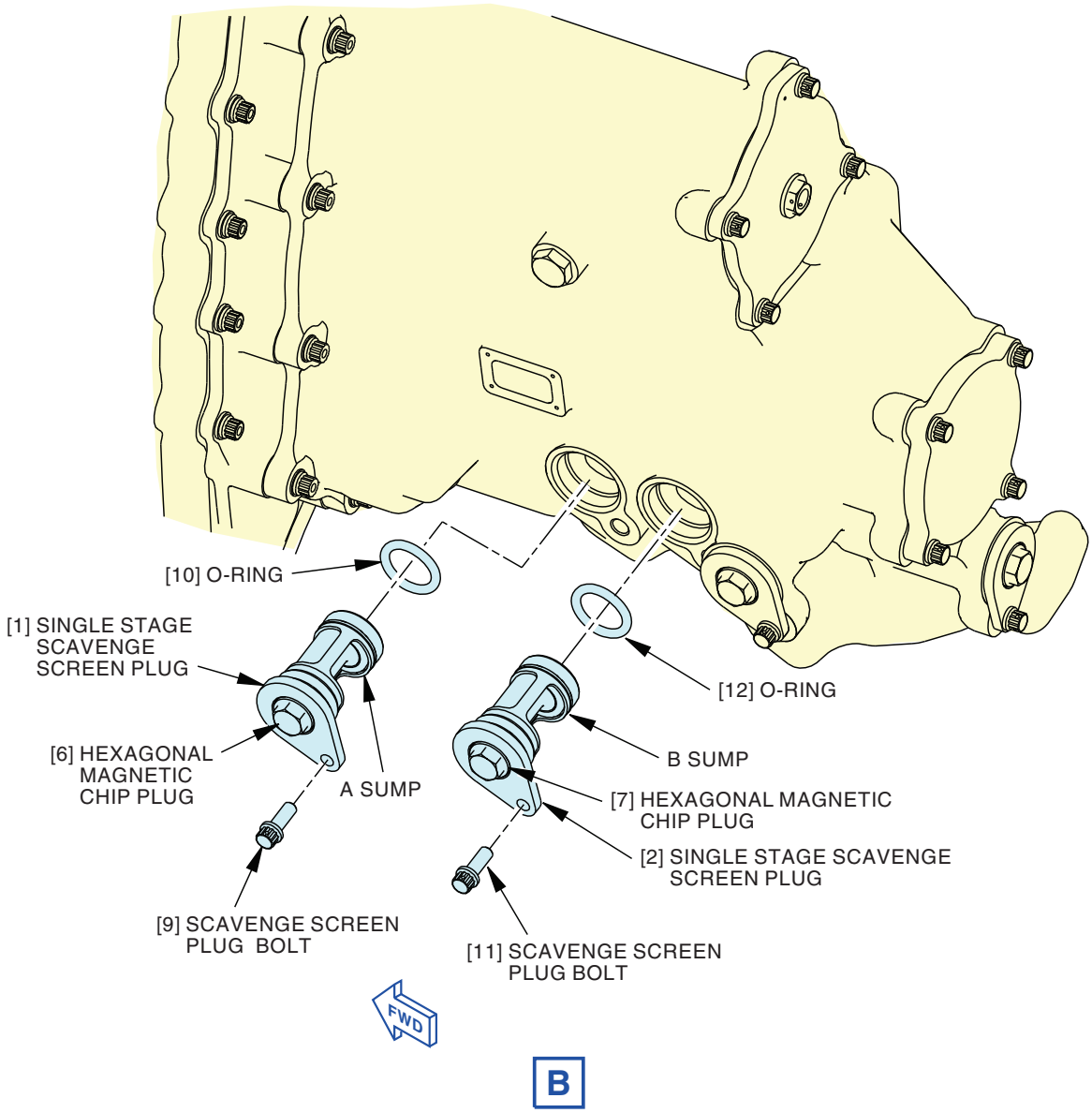
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Scavenge Screen Plugs Installation
Figure 201/79-21-04-990-804-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

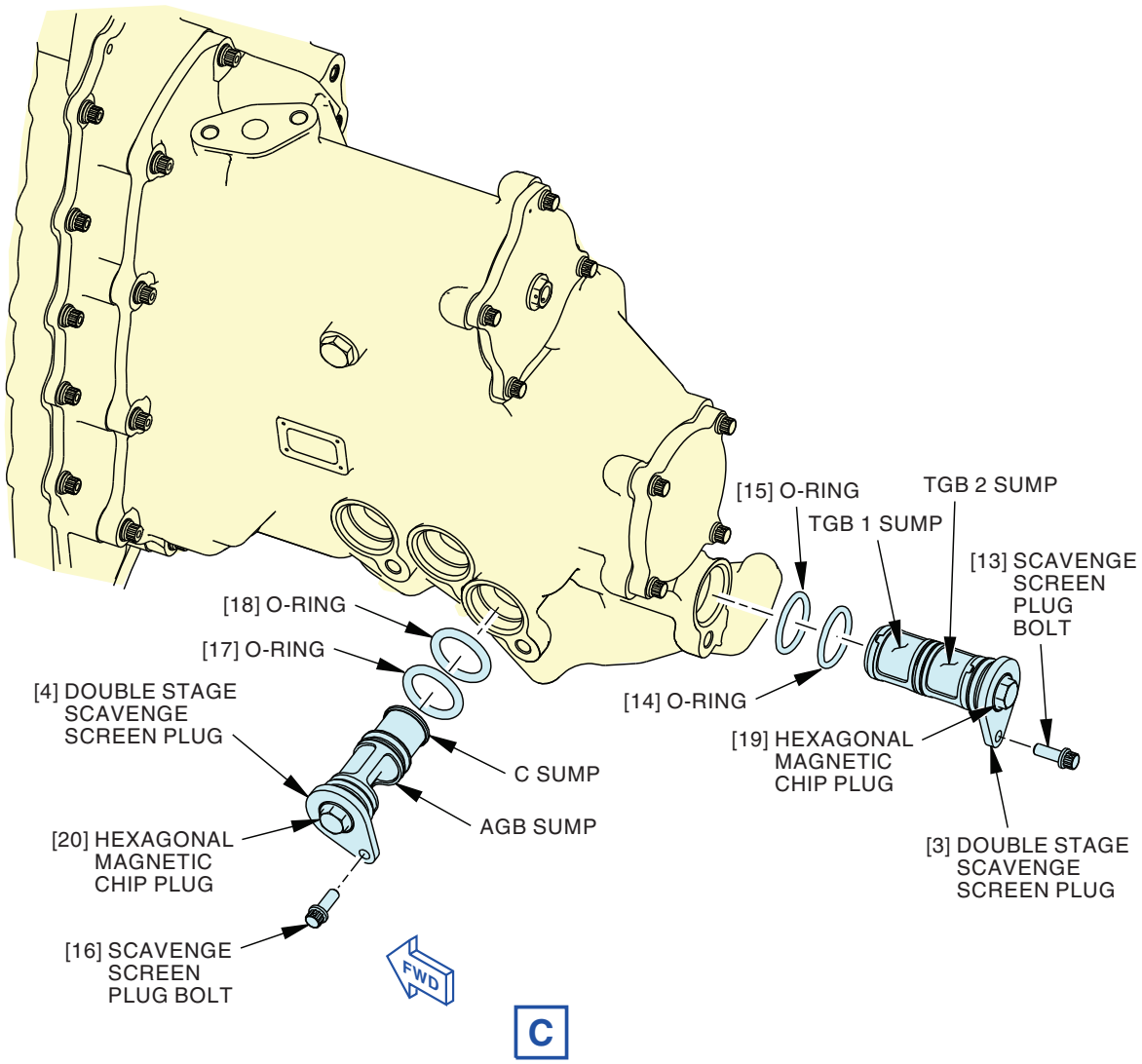


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Scavenge Screen Plugs Installation
Figure 201/79-21-04-990-804-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

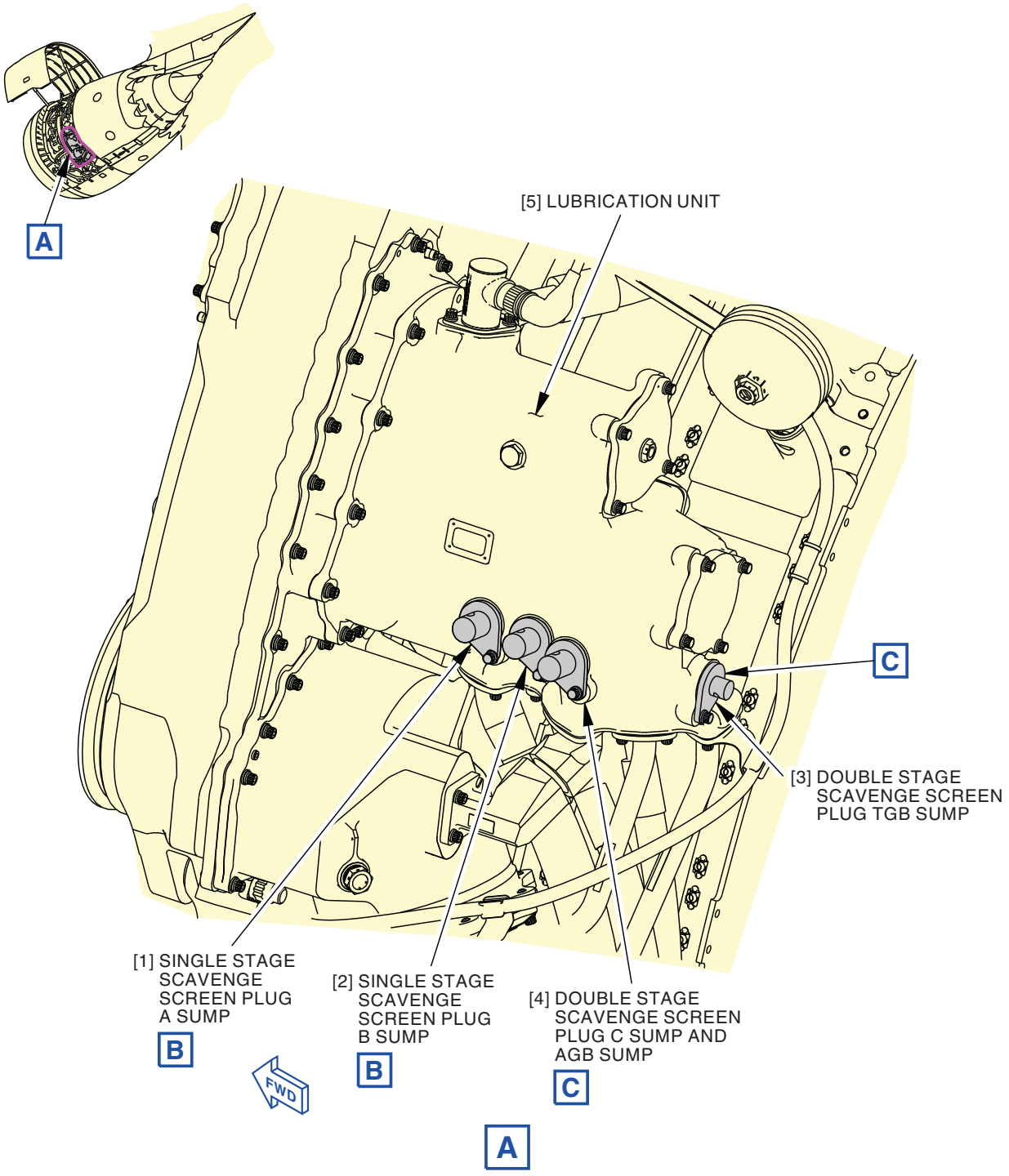


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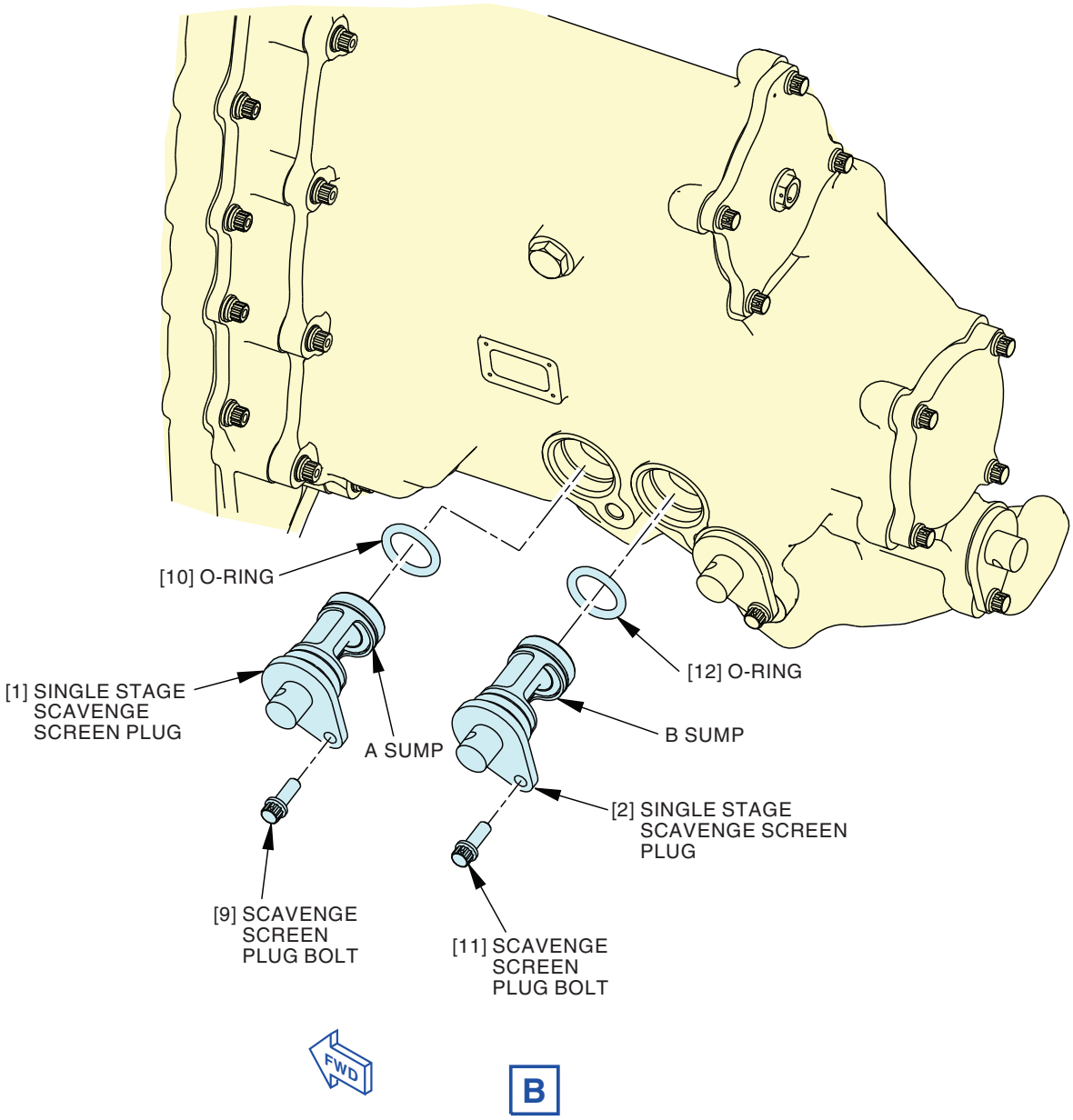
Scavenge Screen Plugs Installation
Figure 201/79-21-04-990-804-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Scavenge Screen Plugs Installation
Figure 202/79-21-04-990-806-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

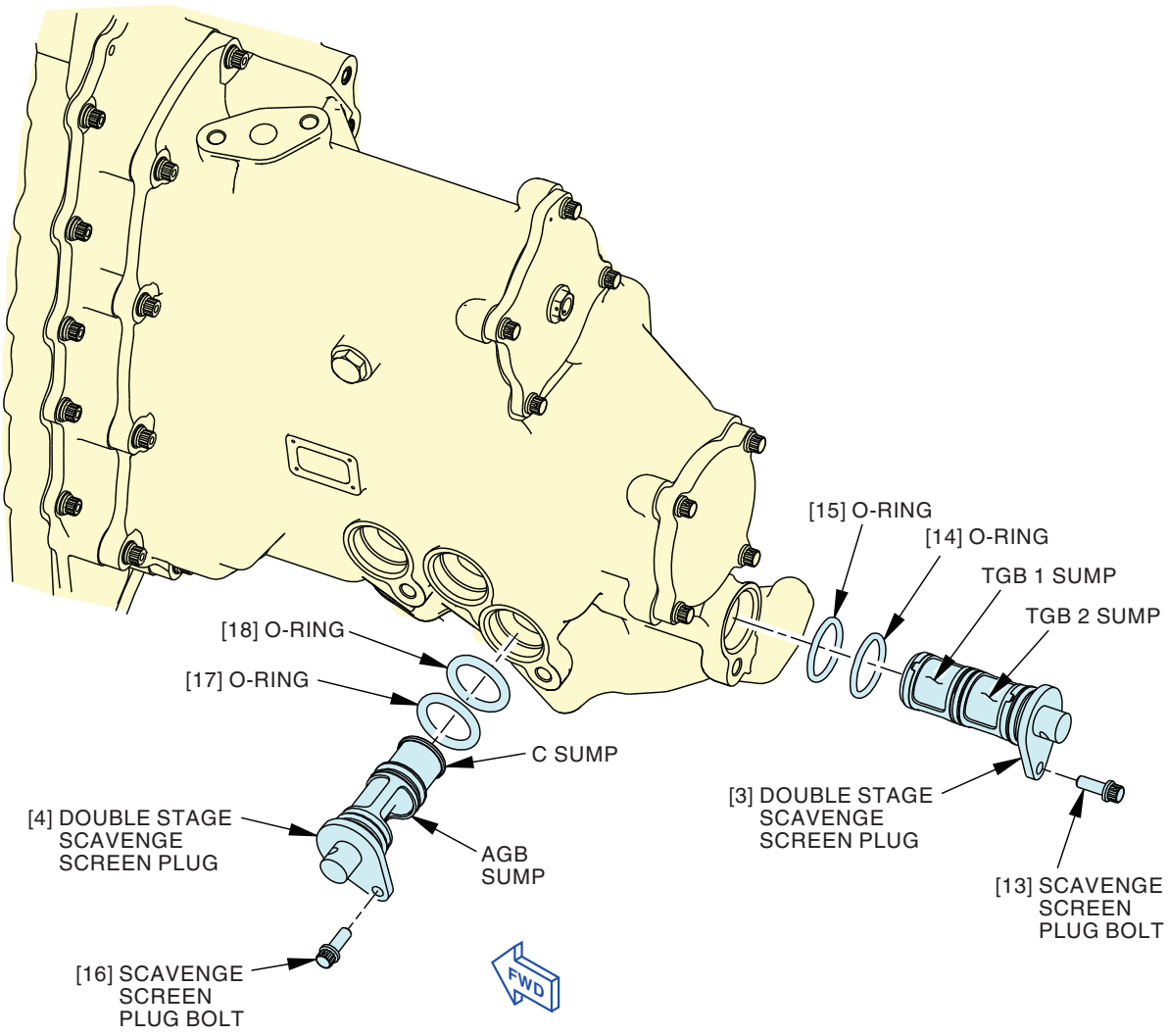


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Scavenge Screen Plugs Installation
Figure 202/79-21-04-990-806-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Scavenge Screen Plugs Installation
Figure 202/79-21-04-990-806-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 79-21-04-400-801-G00

3. Scavenge Screen Plugs Installation

(Figure 201 or Figure 202)

A. General

(1) This task provides the instructions to install the scavenge screen plugs.

B. References

Reference	Title
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00672 [CP5070]	Vaseline - Pure Mineral	V V-P-236

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
10	O-ring	79-21-04-01-035	PTF ALL
12	O-ring	79-21-04-01-035	PTF ALL
14	O-ring	79-21-04-01-225	PTF ALL
15	O-ring	79-21-04-01-222	PTF ALL
17	O-ring	79-21-04-01-125	PTF ALL
18	O-ring	79-21-04-01-122	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

H. Scavenge Screen Plugs Installation

SUBTASK 79-21-04-010-002-G00

WARNING

DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION

DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Install the scavenge screen plugs as follows:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the mating faces are clean and clear of unwanted materials.

CAUTION DO NOT REMOVE THE HEXAGONAL MAGNETIC CHIP PLUG FROM THE SCAVENGE SCREEN PLUG. IF YOU DO NOT OBEY, DAMAGE TO THE ENGINE CAN OCCUR.

CAUTION DO NOT USE THE HEXAGONAL MAGNETIC CHIP PLUG TO REMOVE THE SCAVENGE SCREEN PLUGS. IF YOU DO NOT OBEY, AN OIL LEAK CAN OCCUR.

(b) Install the double stage scavenge screen plug [4] as follows:

1) Remove the protective covers from the double stage scavenge screen plug [4] and lubrication unit [5] port.
2) Lubricate the new O-ring [17] and new O-ring [18] with pure mineral vaseline, D00672 [CP5070].
3) Install the new O-ring [17] on the groove on the double stage scavenge screen plug [4].

CAUTION MAKE SURE THAT YOU USE THE CORRECT PART NUMBERS WHEN YOU REPLACE THE O-RING IN EACH SCAVENGE-SCREEN GROOVE LOCATION. THE PART NUMBERS OF THE DIFFERENT O-RINGS ARE ALMOST THE SAME NUMBER. IF YOU DO NOT USE THE CORRECT PART NUMBERS, DAMAGE CAN OCCUR.

4) Install the new O-ring [18] on the groove on the double stage scavenge screen plug [4].
5) Install the double stage scavenge screen plug [4] on the lubrication unit [5].
6) Install the scavenge screen plug bolt [16] that attaches the double stage scavenge screen plug [4] to the lubrication unit [5].
7) Tighten the scavenge screen plug bolt [16] to 67.45 in-lb (7.62 N·m) - 74.55 in-lb (8.42 N·m).

CAUTION DO NOT REMOVE THE HEXAGONAL MAGNETIC CHIP PLUG FROM THE SCAVENGE SCREEN PLUG. IF YOU DO NOT OBEY, DAMAGE TO THE ENGINE CAN OCCUR.

CAUTION DO NOT USE THE HEXAGONAL MAGNETIC CHIP PLUG TO REMOVE THE SCAVENGE SCREEN PLUGS. IF YOU DO NOT OBEY, AN OIL LEAK CAN OCCUR.

(c) Install the double stage scavenge screen plug [3] as follows:

1) Remove the protective covers from the double stage scavenge screen plug [3] and lubrication unit [5] port.
2) Lubricate the new O-ring [14] and new O-ring [15] with pure mineral vaseline, D00672 [CP5070].
3) Install the new O-ring [14] on the groove on the double stage scavenge screen plug [3].

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CAUTION MAKE SURE THAT YOU USE THE CORRECT PART NUMBERS WHEN YOU REPLACE THE O-RING IN EACH SCAVENGE-SCREEN GROOVE LOCATION. THE PART NUMBERS OF THE DIFFERENT O-RINGS ARE ALMOST THE SAME NUMBER. IF YOU DO NOT USE THE CORRECT PART NUMBERS, DAMAGE CAN OCCUR.

4) Install the new O-ring [15] on the groove on the double stage scavenge screen plug [3].
5) Install the double stage scavenge screen plug [3] on the lubrication unit [5].
6) Install the scavenge screen plug bolt [13] that attaches the double stage scavenge screen plug [3] to the lubrication unit [5].
7) Tighten the scavenge screen plug bolt [13] to 67.45 in-lb (7.62 N·m) - 74.55 in-lb (8.42 N·m).

CAUTION DO NOT REMOVE THE HEXAGONAL MAGNETIC CHIP PLUG FROM THE SCAVENGE SCREEN PLUG. IF YOU DO NOT OBEY, DAMAGE TO THE ENGINE CAN OCCUR.

CAUTION DO NOT USE THE HEXAGONAL MAGNETIC CHIP PLUG TO REMOVE THE SCAVENGE SCREEN PLUGS. IF YOU DO NOT OBEY, AN OIL LEAK CAN OCCUR.

(d) Install the single stage scavenge screen plug [2] as follows:

1) Remove the protective covers from the single stage scavenge screen plug [2] and lubrication unit [5] port.
2) Lubricate the new O-ring [12] with pure mineral vaseline, D00672 [CP5070].
3) Install the new O-ring [12] on the groove on the single stage scavenge screen plug [2].
4) Install the single stage scavenge screen plug [2] on the lubrication unit [5].
5) Install the scavenge screen plug bolt [11] that attaches the single stage scavenge screen plug [2] to the lubrication unit [5].
6) Tighten the scavenge screen plug bolt [11] to 67.45 in-lb (7.62 N·m) - 74.55 in-lb (8.42 N·m).

CAUTION DO NOT REMOVE THE HEXAGONAL MAGNETIC CHIP PLUG FROM THE SCAVENGE SCREEN PLUG. IF YOU DO NOT OBEY, DAMAGE TO THE ENGINE CAN OCCUR.

CAUTION DO NOT USE THE HEXAGONAL MAGNETIC CHIP PLUG TO REMOVE THE SCAVENGE SCREEN PLUGS. IF YOU DO NOT OBEY, AN OIL LEAK CAN OCCUR.

(e) Install the single stage scavenge screen plug [1] as follows:

1) Remove the protective covers from the single stage scavenge screen plug [1] and lubrication unit [5] port.
2) Lubricate the new O-ring [10] with pure mineral vaseline, D00672 [CP5070].

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3) Installs the new O-ring [10] on the groove on the single stage scavenge screen plug [1].

4) Install the single stage scavenge screen plug [1] on the lubrication unit [5].

5) Install the scavenge screen plug bolt [9] that attaches the single stage scavenge screen plug [1] to the lubrication unit [5].

6) Tighten the scavenge screen plug bolt [9] to 67.45 in-lb (7.62 N·m) - 74.55 in-lb (8.42 N·m).

(f) Clean the work area and clear the tools and other items.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 79-21-04-860-014-G00

(1) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 79-21-04-860-007-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

J. Scavenge Screen Plugs Installation Test

SUBTASK 79-21-04-730-001-G00

(1) Do the applicable tests given in the Power Plant Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

TASK 79-21-04-211-801-G00

4. Scavenge Screen Plug Inspection

(Figure 203 or Figure 204 or Figure 205)

A. General

(1) This task includes steps to perform the inspection of the scavenge screen plug.

B. References

Reference	Title
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
79-00-00-220-801-G00	Engine Oil System (Oil Debris Monitoring System Sensor (ODMS), Scavenge Screen Plugs, Starter Magnetic Plug, and Oil Filter) Detailed Dimensional Check (P/B 601)
80-11-09-200-801-G00	Engine Starter Magnetic Plug Inspection (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE
STD-1280	Source - Air, Regulated, Dry Filtered, 0-30 PSIG

D. Consumable Materials

Reference	Description	Specification
B00682 [CP2011]	Solvent - Stoddard	MIL-PRF-680 Type I, II or III
B50295 [CP2010]	Solvent - cleaning, White Spirit	
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

G. Prepare for the Inspection

CAUTION

SOME OF THE SCAVENGE SCREENS PLUGS ARE DOUBLE STAGES. TAKE CARE TO DO NOT MIX PARTICLES DURING DOUBLE STAGES SCAVENGE SCREENS PLUGS MAINTENANCE. A MIX OF THE PARTICLES COULD LEAD TO AN INACCURATE INTERPRETATION OF SERVICEABLE CRITERIA, DAMAGE CAN OCCUR.

CAUTION

TAG THE REMOVED PARTICLES WITH THE RIGHT SUMP NAME TO PREVENT AN ERROR IN IDENTIFICATION OF THE SOURCE OF THE MATERIAL DURING VISUAL CHARACTERIZATION AND LABORATORY ANALYSIS. AN ERROR IN IDENTIFICATION OF THE SOURCE OF THE MATERIAL COULD LEAD TO AN INACCURATE INTERPRETATION OF SERVICEABLE CRITERIA AND DAMAGE CAN OCCUR.

CAUTION

CFM DOES NOT RECOMMEND THE USE OF ADHESIVE TAPE. IF YOU USE A PIECE OF ADHESIVE TAPE TO COLLECT THE PARTICLES, MAKE SURE THE ADHESIVE IS NOT TOO STRONG. IF THE ADHESIVE ON THE TAPE IS TOO STRONG, IT CAN CAUSE DAMAGE.

CAUTION

DO NOT USE A TOO STRONG POWER MAGNET TO REMOVE THE PARTICLES FROM THE SCAVENGE SCREENS PLUGS. IF YOU USE A TOO STRONG POWER MAGNET, IT CAN DECREASE THE MAGNETIC PERFORMANCE OF THE MAGNETIC CHIP DETECTOR. IF YOU DO NOT OBEY, DAMAGE CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

NOTE: Remove and examine the scavenge screen plug one by one to be sure of the origin of the particles.

NOTE: Remove, tag and examine one scavenge screen plug at a time. This is to prevent errors in identification of the source of the particles on the scavenge screen plug.

NOTE: Characteristics of magnetic bar retriever without impact on scavenge screen plug magnetic field:

- BHmax = 5.5 MGOe (max. energy product of magnetic field strength and magnetic flux density)
- Br = 12.8 kG (remanence, residual magnetism left behind after an external magnetic field is removed)
- Hc = 640 Oe (coercivity, measure of ability to withstand an external magnetic field without becoming demagnetized).

SUBTASK 79-21-04-860-009-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 79-21-04-860-010-G00

(2) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

H. Scavenge Screen Plug Inspection

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION

DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

TAG THE REMOVED PARTICLES WITH THE RIGHT SUMP NAME TO PREVENT AN ERROR IN IDENTIFICATION OF THE SOURCE OF THE MATERIAL DURING VISUAL CHARACTERIZATION AND LABORATORY ANALYSIS. AN ERROR IN IDENTIFICATION OF THE SOURCE OF THE MATERIAL COULD LEAD TO AN INACCURATE INTERPRETATION OF SERVICEABLE CRITERIA AND DAMAGE CAN OCCUR.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN SB 79-0007

CAUTION

DO NOT REMOVE THE HEXAGONAL MAGNETIC CHIP PLUG FROM THE SCAVENGE SCREEN PLUG. IF YOU DO NOT OBEY, DAMAGE TO THE ENGINE CAN OCCUR.

CAUTION

DO NOT USE THE HEXAGONAL MAGNETIC CHIP PLUG TO REMOVE THE SCAVENGE SCREEN PLUGS. IF YOU DO NOT OBEY, AN OIL LEAK CAN OCCUR.

PTF ALL

SUBTASK 79-21-04-200-002-G00

(1) Examine the single stage scavenge screen plug [2], single stage scavenge screen plug [4], double stage scavenge screen plug [6], and double stage scavenge screen plug [9] for debris.

NOTE: Remove and inspect the scavenge screen plugs one by one to be sure of the origin of the particles.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN SB 79-0007

(a) Do not remove the hexagonal magnetic chip plug [12], hexagonal magnetic chip plug [13], hexagonal magnetic chip plug [16], hexagonal magnetic chip plug [17] and magnetic bars for inspection.

NOTE: If you remove the hexagonal magnetic chip plugs and magnetic bars, the scavenge screen sleeve will not catch the particles.

PTF ALL

(b) Remove the single stage scavenge screen plug [2], single stage scavenge screen plug [4], double stage scavenge screen plug [6], and double stage scavenge screen plug [9], from the lubrication unit lube and scavenge pump [1], do this task: Scavenge Screen Plugs Removal, TASK 79-21-04-000-801-G00.

(c) Visually inspect the O-ring [15] of each scavenge screen plug.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) If the O-ring [15] is partially cut, inspect the lubrication unit lube and scavenge pump [1] scavenge cavity to check that the cavity is free of any piece(s) of O-ring [15] (Figure 203 or Figure 204 or Figure 205).
   a) If there is a piece of O-ring [15] in the lubrication unit lube and scavenge pump [1] scavenge cavity, remove and keep it.
   b) Make sure that the piece of O-ring [15] aligns with the cut O-ring [18].
      <1> If the piece of O-ring [15] aligns with the O-ring [18], discard O-ring [15].
      <2> If the piece of O-ring [15] does not align with the O-ring [18], do these steps:
         <a> Take pictures of the piece of the O-ring [15].
         <b> Record dimension of the O-ring [18] section and estimated radius.
         <c> Do this task: Engine Oil System (Oil Debris Monitoring System Sensor (ODMS), Scavenge Screen Plugs, Starter Magnetic Plug, and Oil Filter) Detailed Dimensional Check, TASK 79-00-00-220-801-G00.

NOTE: If the piece of O-ring [15] is not removed during the inspection of the scavenge screen plug, there is a possibility to have a piece of O-ring [15] left in the lubrication unit lube and scavenge pump scavenge cavity. If this piece of O-ring [15] is not removed before scavenge screen plug installation, it will be trapped in the screen during the next engine operation. At the next scavenge screen plug inspection, this piece of O-ring [15] will be found and if the O-ring [15] is not cut during the visual inspection, this will cause confusion and potentially lead to unnecessary actions to troubleshoot the problem.

(d) Visually inspect the single stage scavenge screen plug [2], single stage scavenge screen plug [4], double stage scavenge screen plug [6], and double stage scavenge screen plug [9] for debris.

(e) If you find some magnetic material on the TGB1 sump [7], TGB2 sump [8], or AGB sump [11] scavenge screen plugs, examine the starter magnetic plug for contamination, do this task: Engine Starter Magnetic Plug Inspection, TASK 80-11-09-200-801-G00.

(f) If the single stage scavenge screen plug [2], single stage scavenge screen plug [4], double stage scavenge screen plug [6], and double stage scavenge screen plug [9] are free from debris, do the following steps:

WARNING

DO NOT GET THE SOLVENT IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE IN THE FUMES FROM THE SOLVENT. PUT ON PROTECTIVE SPLASH GOGGLES AND GLOVES WHEN YOU USE THE SOLVENT. KEEP THE SOLVENT AWAY FROM SPARKS, FLAME, AND HEAT. THE SOLVENT IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

1) Clean the single stage scavenge screen plug [2], single stage scavenge screen plug [4], double stage scavenge screen plug [6], and double stage scavenge screen plug [9] with solvent, B00682 [CP2011].
2) Dry the single stage scavenge screen plug [2], single stage scavenge screen plug [4], double stage scavenge screen plug [6], and double stage scavenge screen plug [9] with a 0-30 psig dry filtered regulated air source, STD-1280.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) Make sure that the single stage scavenge screen plug [2], single stage scavenge screen plug [4], double stage scavenge screen plug [6], and double stage scavenge screen plug [9] are free of all particles.

4) Install the single stage scavenge screen plug [2], single stage scavenge screen plug [4], double stage scavenge screen plug [6], and double stage scavenge screen plug [9], do this task: Scavenge Screen Plugs Installation, TASK 79-21-04-400-801-G00.

(g) If there are debris particles on single stage scavenge screen plug [2], single stage scavenge screen plug [4], double stage scavenge screen plug [6], or double stage scavenge screen plug [9], carefully remove them and keep the particles.

1) If it is possible, take pictures of the particles on the concerned scavenge screen sump (either A sump [3], B sump [5], TGB1 sump [7], TGB2 sump [8], C sump [10], or AGB sump [11]).

a) Make sure that you identify each of the concerned scavenge screen sump before remove them.

2) Carefully remove and keep the particles as follows:

CAUTION DO NOT USE A TOO STRONG POWER MAGNET TO REMOVE THE PARTICLES FROM THE SCAVENGE SCREENS PLUGS. IF YOU USE A TOO STRONG POWER MAGNET, IT CAN DECREASE THE MAGNETIC PERFORMANCE OF THE MAGNETIC CHIP DETECTOR. IF YOU DO NOT OBEY, DAMAGE CAN OCCUR.

a) Use cotton wiper, G00034, on a thin sheet of paper or/and a magnetic bar retriever.

NOTE: Characteristics of magnetic bar retriever without impact on scavenge screen plug magnetic field:

- BHmax = 5.5 MGOe (max. energy product of magnetic field strength and magnetic flux density)
- Br = 12.8 kG (remanence, residual magnetism left behind after an external magnetic field is removed)
- Hc = 640 Oe (coercivity, measure of ability to withstand an external magnetic field without becoming demagnetized).

b) Use a magnet to separate the magnetic particles.

WARNING DO NOT GET THE SOLVENT IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE IN THE FUMES FROM THE SOLVENT. PUT ON PROTECTIVE SPLASH GOGGLES AND GLOVES WHEN YOU USE THE SOLVENT. KEEP THE SOLVENT AWAY FROM SPARKS, FLAME, AND HEAT. THE SOLVENT IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

<1> Remove the oil from the collected particles with solvent, B00682 [CP2011], or White Spirit, B50295 [CP2010].

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<2> Keep the material collected from each scavenge screen separately with the following information attached: ESN, TSN/CSN, source of particles (sump), and removal date.

3) Clean the single stage scavenge screen plug [2], single stage scavenge screen plug [4], double stage scavenge screen plug [6], and double stage scavenge screen plug [9] with solvent, B00682 [CP2011].

4) Make sure that single stage scavenge screen plug [2], single stage scavenge screen plug [4], double stage scavenge screen plug [6], and double stage scavenge screen plug [9] are free of all particles.

5) Dry the single stage scavenge screen plug [2], single stage scavenge screen plug [4], double stage scavenge screen plug [6], and double stage scavenge screen plug [9] with a 0-30 psig dry filtered regulated air source, STD-1280.

6) Install the single stage scavenge screen plug [2], single stage scavenge screen plug [4], double stage scavenge screen plug [6], and double stage scavenge screen plug [9], do this task: Scavenge Screen Plugs Installation, TASK 79-21-04-400-801-G00.

WARNING

DO NOT GET THE SOLVENT IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE IN THE FUMES FROM THE SOLVENT. PUT ON PROTECTIVE SPLASH GOGGLES AND GLOVES WHEN YOU USE THE SOLVENT. KEEP THE SOLVENT AWAY FROM SPARKS, FLAME, AND HEAT. THE SOLVENT IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

7) Do the analysis of the particles, do this task: Engine Oil System (Oil Debris Monitoring System Sensor (ODMS), Scavenge Screen Plugs, Starter Magnetic Plug, and Oil Filter) Detailed Dimensional Check, TASK 79-00-00-220-801-G00.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 79-21-04-860-011-G00

(1) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 79-21-04-860-012-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

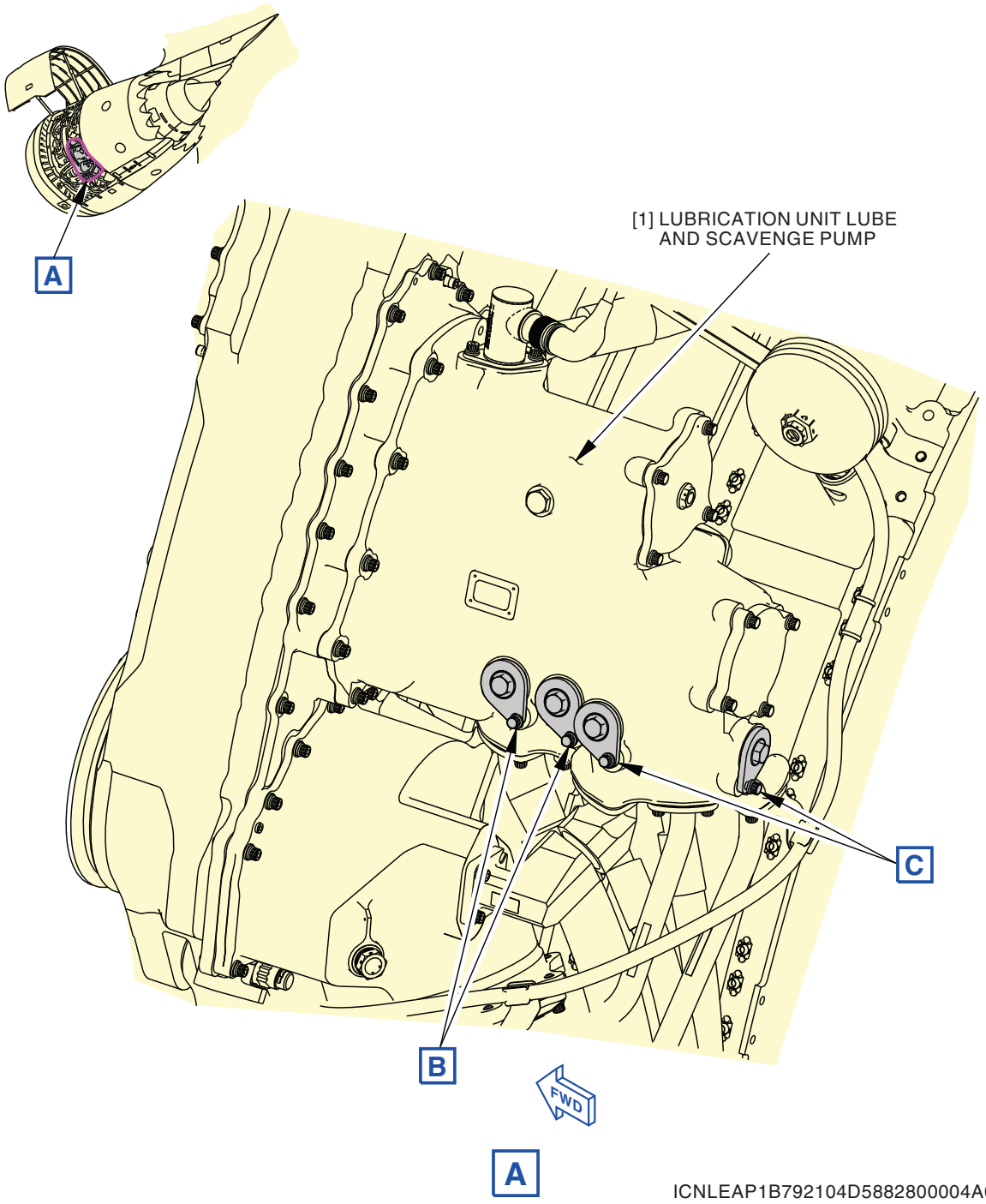
J. Scavenge Screen Plug Test

SUBTASK 79-21-04-710-002-G00

(1) Do the applicable tests given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

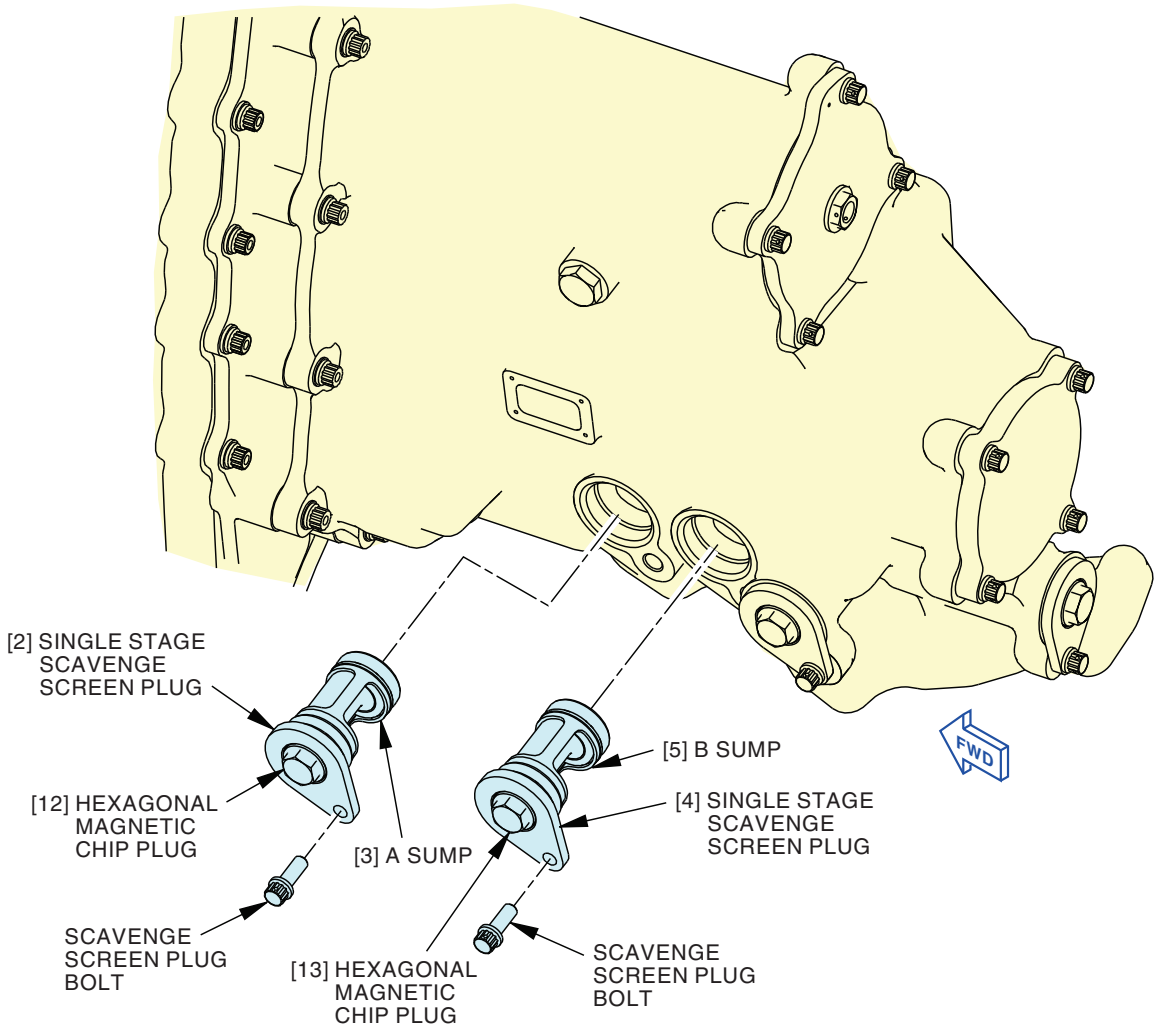
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Scavenge Screen Plug Inspection
Figure 203/79-21-04-990-802-G00 (Sheet 1 of 3)

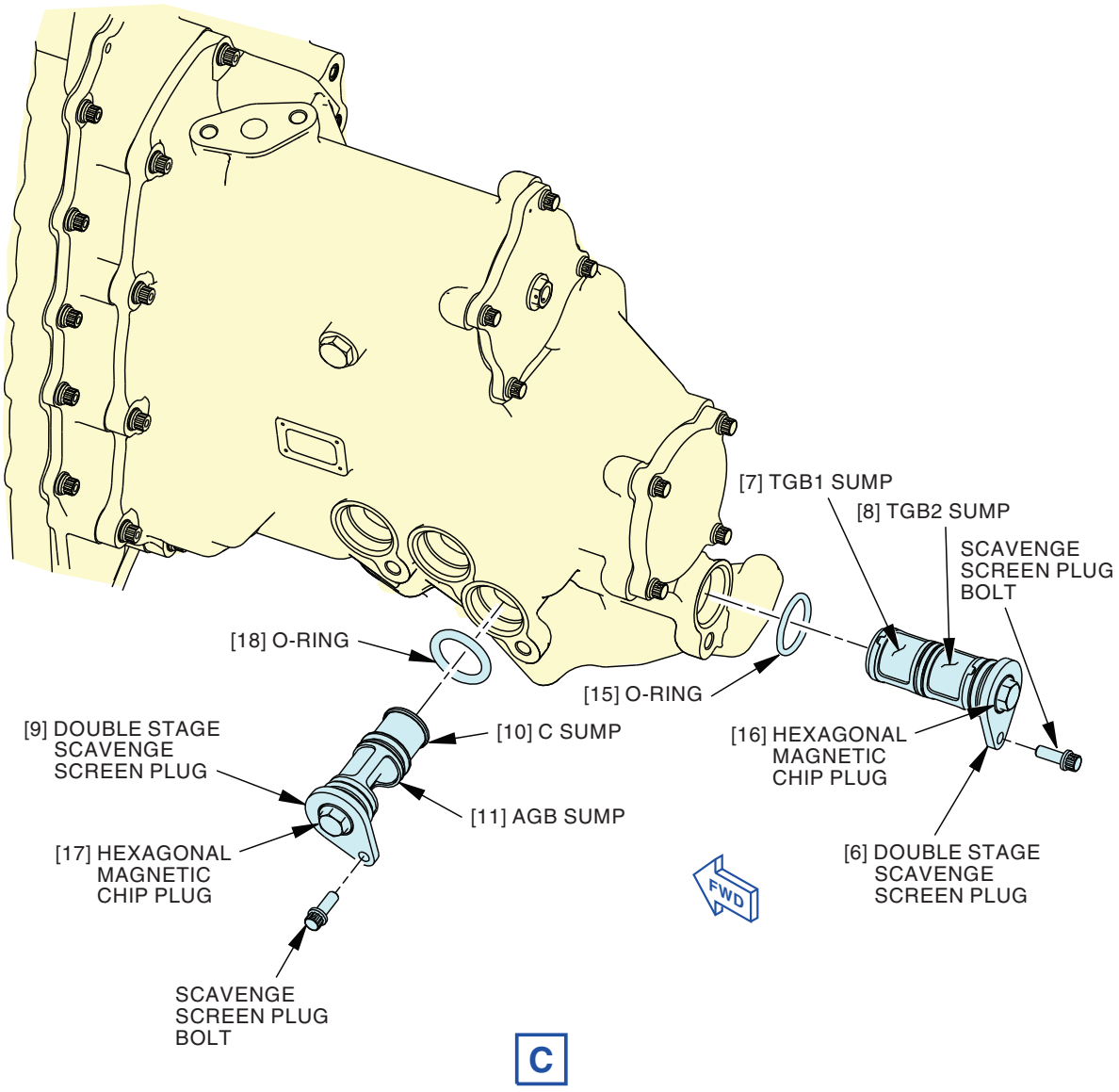
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICNLEAP1B792104D5882800005A00201
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Scavenge Screen Plug Inspection
Figure 203/79-21-04-990-802-G00 (Sheet 2 of 3)

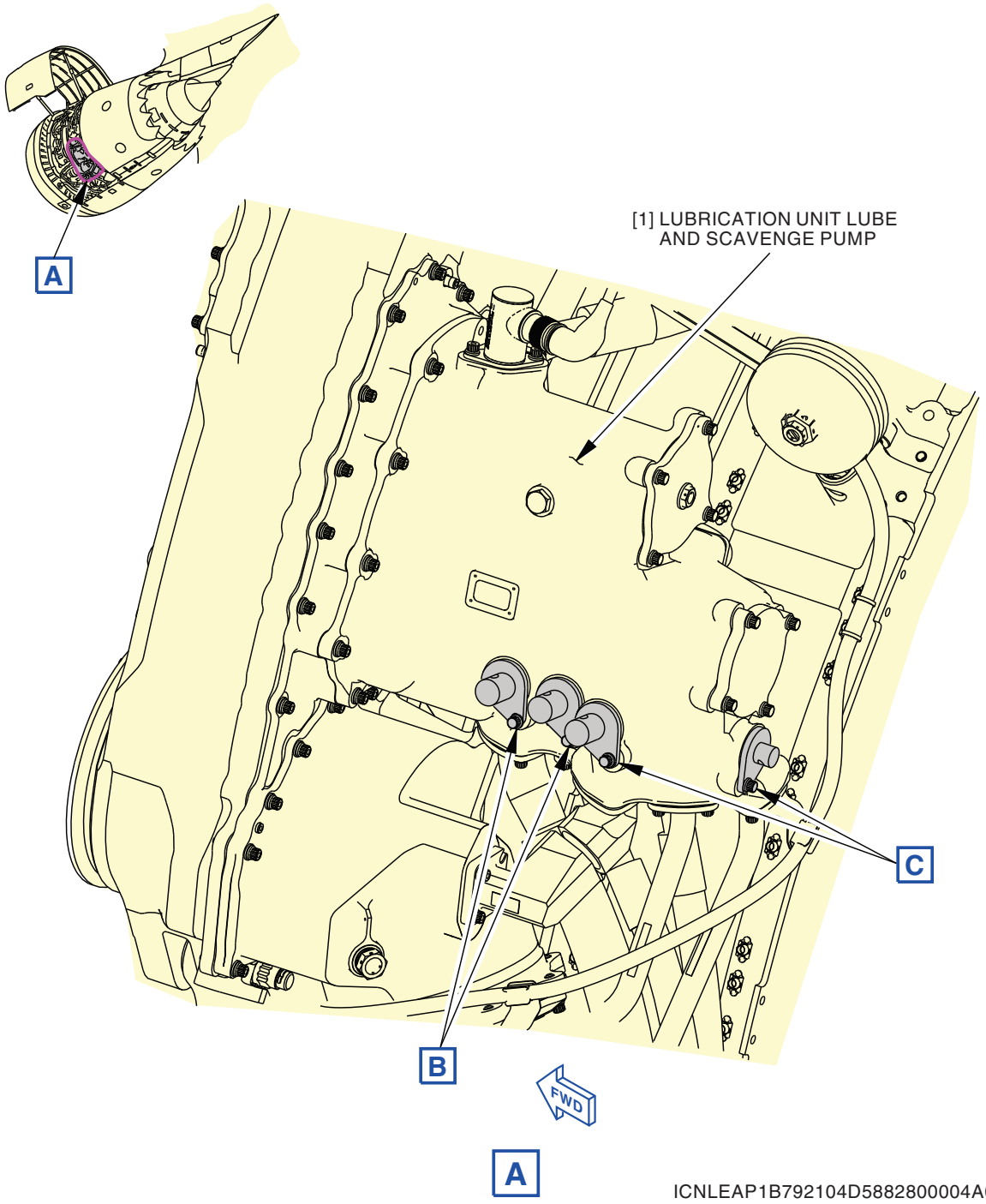
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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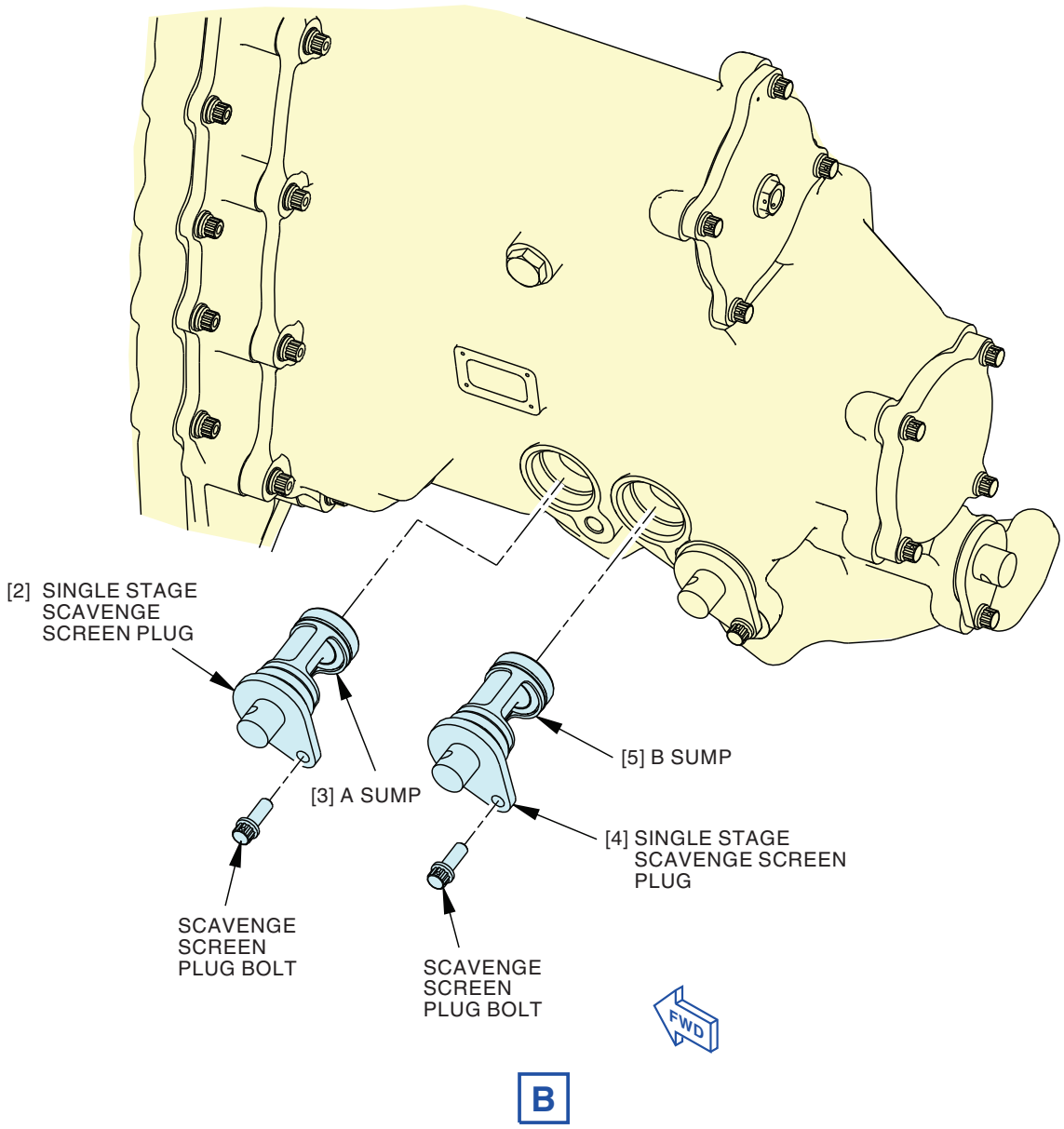
Scavenge Screen Plug Inspection
Figure 203/79-21-04-990-802-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Scavenge Screen Plug Inspection
Figure 204/79-21-04-990-807-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

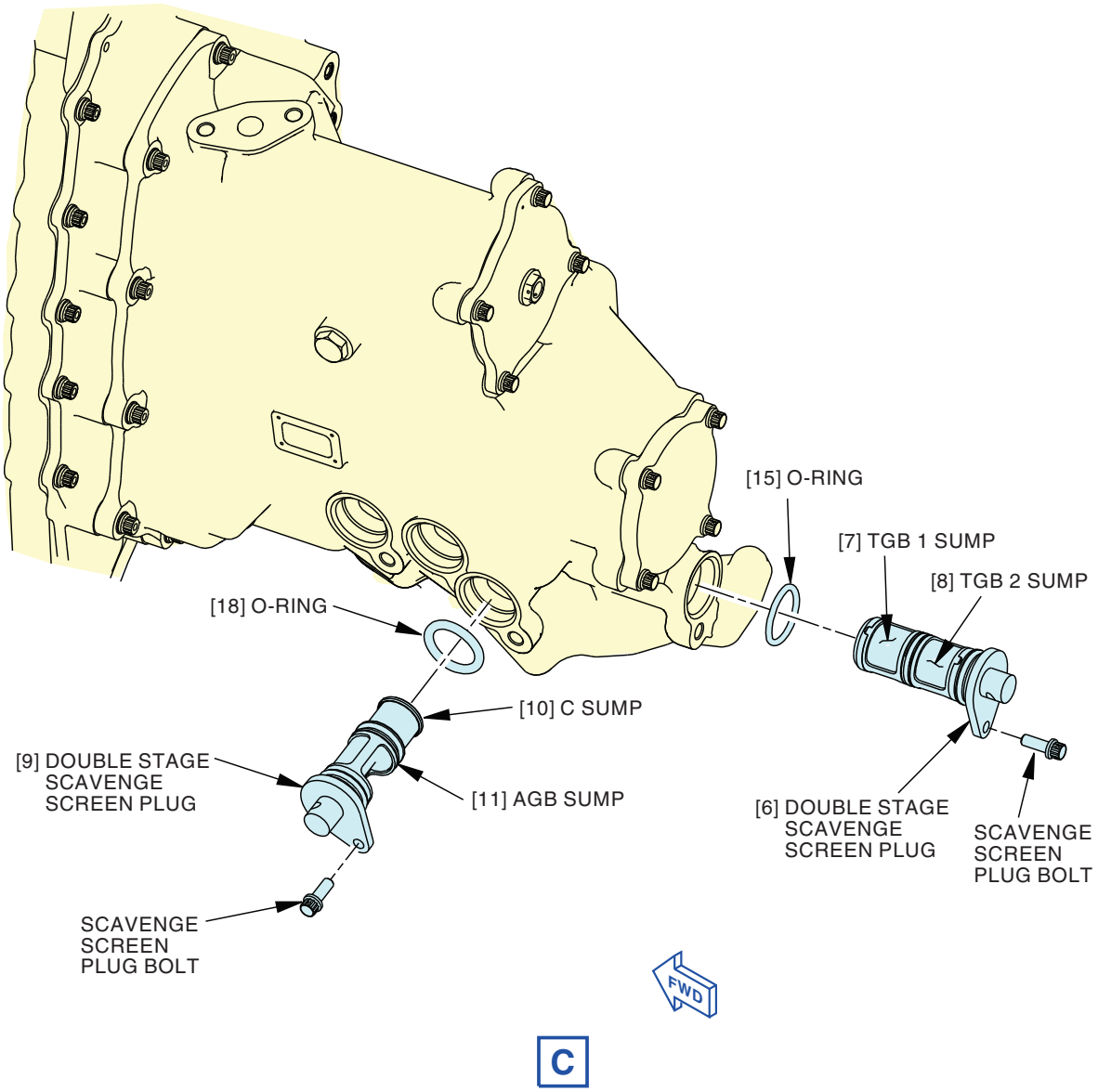


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2979251 S0000757355_V1

Scavenge Screen Plug Inspection
Figure 204/79-21-04-990-807-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



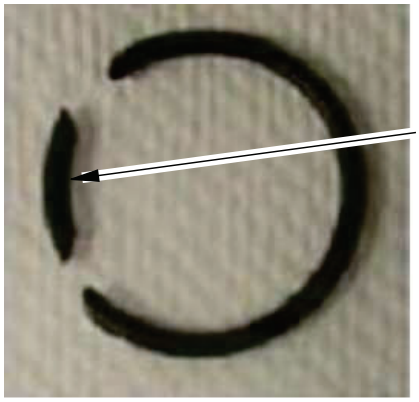
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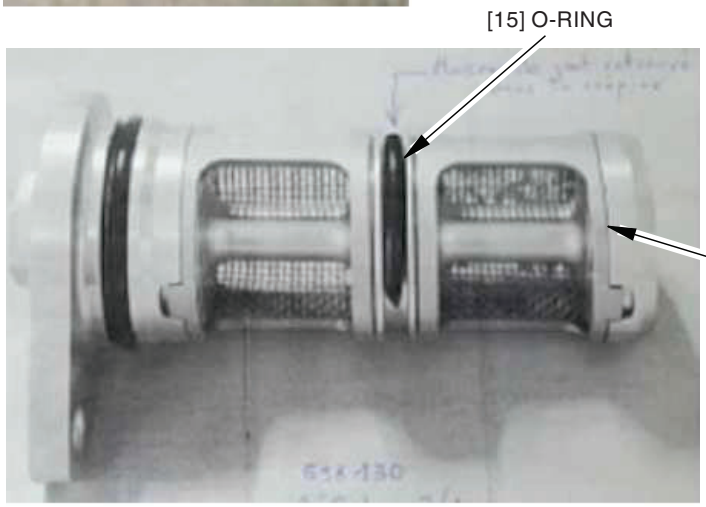
2979255 S0000757359_V1

Scavenge Screen Plug Inspection
Figure 204/79-21-04-990-807-G00 (Sheet 3 of 3)

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DOUBLE STAGE SCAVENGE PLUGS WITH CUT O-RING (EXAMPLE GRAPHIC)

2898960 S0000691671_V2

Scavenge Screen Plug Inspection
Figure 205/79-21-04-990-808-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OIL FILTER CARTRIDGE ELEMENT - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the oil filter cartridge element

(2) An installation of the oil filter cartridge element.

TASK 79-21-07-000-801-G00

2. Oil Filter Cartridge Element Removal

(Figure 401)

A. General

(1) This task gives the instructions on how to remove the oil filter cartridge element.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
79-21-03-000-801-G00	Lubrication Unit Removal (P/B 401)
79-21-03-400-801-G00	Lubrication Unit Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-199	Container - 2 U.S. Gallon (7.6 l) (oil resistant)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	Preformed packing	79-21-07-01-020	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

H. Prepare for the Removal

SUBTASK 79-21-07-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 79-21-07-860-002-G00

(2) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

I. Oil Filter Cartridge Element Removal

SUBTASK 79-21-07-020-001-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Remove the oil filter cartridge element [4] as follows:

(a) Put a container, STD-199, below the lubrication unit lube and scavenge pump [1].

(b) Drain the oil from the lubrication unit lube and scavenge pump [1] as follows:

1) Cut the safety cable [11] from the drain plug [2].

2) Remove the drain plug [2] from the lubrication unit lube and scavenge pump [1] as follows:

a) Remove the drain plug [2] from the lubrication unit lube and scavenge pump [1].

b) Remove and discard the preformed packing [3] from the groove of the drain plug [2].

c) Let the oil drain into the container, STD-199.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Install the drain plug [2] on the lubrication unit lube and scavenge pump [1] as follows:

1) Make sure that the mating faces are clean and clear of unwanted materials.

2) Lubricate the threads of the drain plug [2] and the new preformed packing [3] with engine oil, D00599 [CP2442].

3) Install the new preformed packing [3] onto the groove of the drain plug [2].

4) Install the drain plug [2] in a correct position on the lubrication unit lube and scavenge pump [1].

5) Tighten the drain plug [2] to 67.2 in-lb (7.6 N·m) - 74.3 in-lb (8.4 N·m).

NOTE: Safety cable will be installed after the installation of the oil filter cartridge cover.

(d) Cut the safety cable [6] from the filter drain plug [5].

(e) Loosen the filter drain plug [5] from the oil filter cartridge cover [8].

(f) Remove the two bolts [9], two bolts [13], and two nuts [14] that attach the oil filter cartridge cover [8] to the lubrication unit lube and scavenge pump [1].

(g) Remove the oil filter cartridge cover [8] and oil filter cartridge element [4] from the lubrication unit lube and scavenge pump [1].

(h) Remove the oil filter cartridge element [4] from the oil filter cartridge cover [8] as follows:

1) Remove the filter drain plug [5] from the oil filter cartridge cover [8].

2) Remove and discard the preformed packing [7] from the groove of the filter drain plug [5].

3) Remove and discard the oil filter cartridge element [4] from the oil filter cartridge cover [8].

(i) Remove and discard the preformed packing [10] from the groove of the oil filter cartridge cover [8].

(j) Inspect the two inserts on the housing at the oil filter cartridge cover [8] location for indications such as insert pulled out, insert movement, impacted thread insert.

1) If any, replace the lubrication unit lube and scavenge pump [1]:

- Lubrication Unit Removal, TASK 79-21-03-000-801-G00
- Lubrication Unit Installation, TASK 79-21-03-400-801-G00.

(k) Inspect the oil filter cartridge cover [8] flange in contact with the lubrication unit lube and scavenge pump [1].

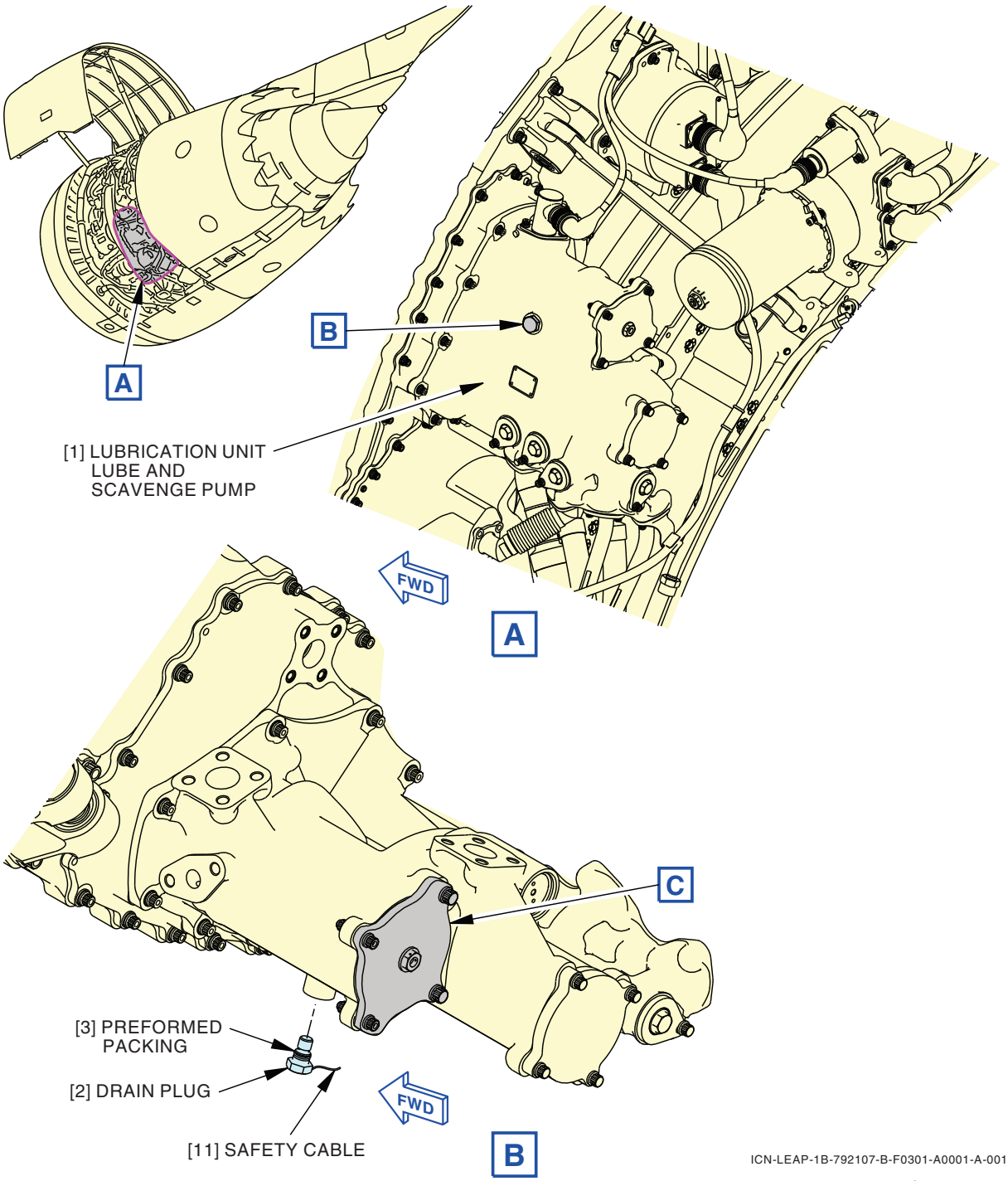
1) If there is any marks on the oil filter cartridge cover [8] at the location of the two inserts on the housing, replace the lubrication unit lube and scavenge pump [1] (Figure 401):

- Lubrication Unit Removal, TASK 79-21-03-000-801-G00
- Lubrication Unit Installation, TASK 79-21-03-400-801-G00.

(l) Install the protective cover on the lubrication unit lube and scavenge pump [1].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

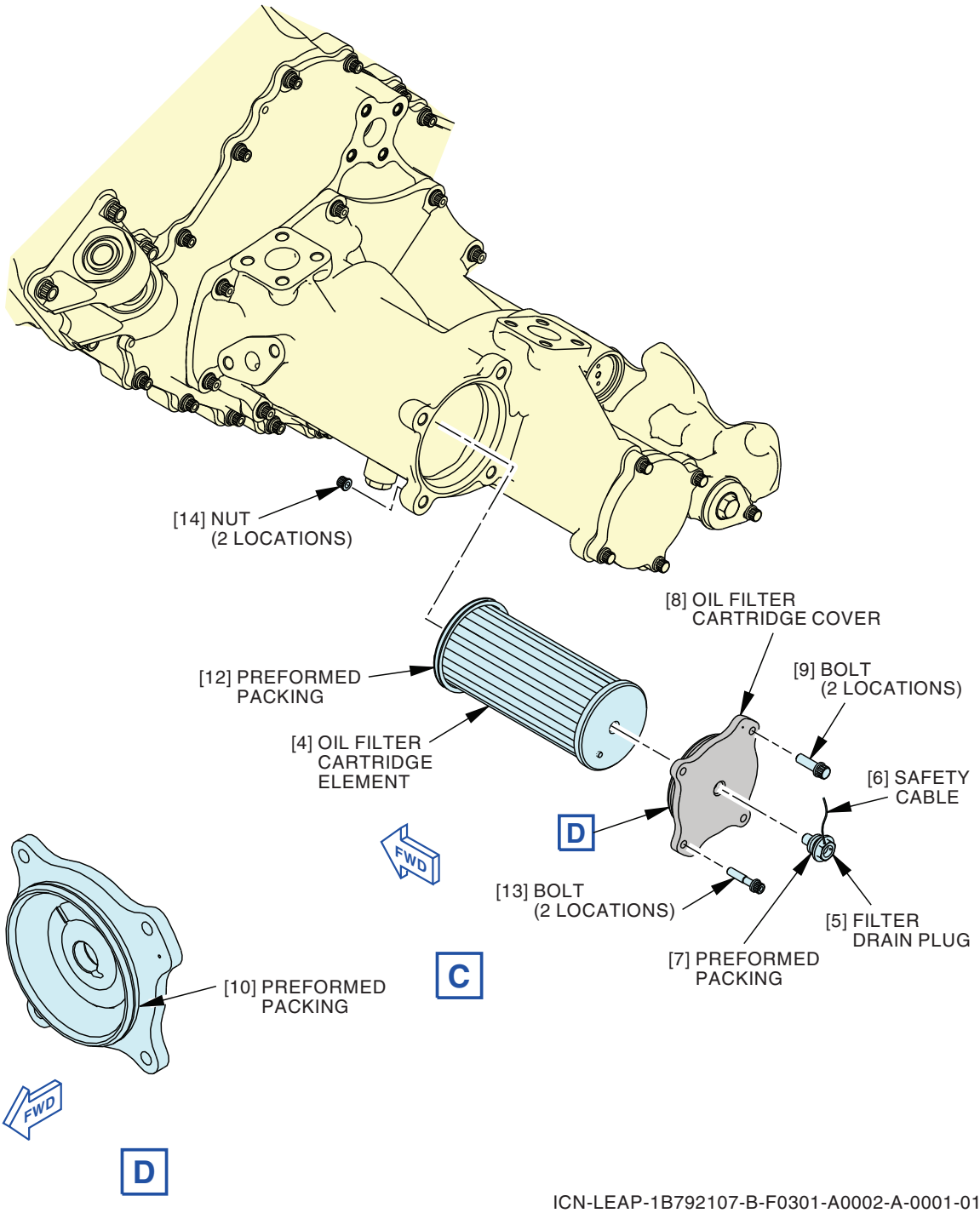


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Oil Filter Cartridge Element Installation
Figure 401/79-21-07-990-801-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Oil Filter Cartridge Element Installation
Figure 401/79-21-07-990-801-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 79-21-07-400-801-G00

3. Oil Filter Cartridge Element Installation

(Figure 401)

A. General

(1) This task provides the instructions on how to install the oil filter cartridge element.

B. References

Reference	Title
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00601 [CP2101]	Vaseline - Graphite Mineral	
D00672 [CP5070]	Vaseline - Pure Mineral	V V-P-236
G50065 [CP8006]	Cable, Safety, Stainless Steel, 0.032 inch (0.8 mm) Diameter	M50 TF 9 CL-A
G50375	Kit - Safety Cable, 321 CRES - 0.032 Inch (0.81 mm) Diameter, (Contains both Cable and Ferrule)	BACC13AT3K, AMS 5689

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
4	Oil filter cartridge element	79-21-07-01-100	PTF ALL
7	Preformed packing	79-21-07-01-040	PTF ALL
10	Preformed packing	79-21-07-01-090	PTF ALL
12	Preformed packing	79-21-07-01-110	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

H. Oil Filter Cartridge Element Installation

SUBTASK 79-21-07-420-001-G00

WARNING

DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Install the oil filter cartridge element [4] as follows:

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Remove the protective cover from the lubrication unit lube and scavenge pump [1].

(c) Lubricate the threads of the filter drain plug [5], bolts [9], bolts [13], and nuts [14] with graphite mineral vaseline, D00601 [CP2101].

(d) Lubricate the new preformed packing [7] and new preformed packing [10] with pure mineral vaseline, D00672 [CP5070].

(e) Install the preformed packing [10] into the groove of the oil filter cartridge cover [8].

(f) Install the new preformed packing [12] into the groove of the new oil filter cartridge element [4].

(g) Install the oil filter cartridge element [4] on the oil filter cartridge cover [8] as follows:

1) Install the preformed packing [7] into the groove of the filter drain plug [5].

2) Put the oil filter cartridge element [4] in position on the oil filter cartridge cover [8].

3) Install the filter drain plug [5] on the oil filter cartridge cover [8].

4) Tighten the filter drain plug [5] to 67.2 in-lb (7.6 N·m) - 74.3 in-lb (8.4 N·m) .

(h) Install the oil filter cartridge cover [8] and oil filter cartridge element [4] on the lubrication unit lube and scavenge pump [1].

(i) Install the bolts [9], bolts [13], and nuts [14] that attach the oil filter cartridge cover [8] to the lubrication unit lube and scavenge pump [1].

WARNING YOU MUST APPLY THE TORQUE SPECIFIED FOR BOLTS AND NUTS. IF YOU DO NOT OBEY, AN OIL LEAK CAN OCCUR. THE LEAK CAN CAUSE AN IN-FLIGHT SHUT DOWN (IFSD).

(j) Tighten the bolts [9] and nuts [14] to 67.2 in-lb (7.6 N·m) - 74.3 in-lb (8.4 N·m).

(k) Install a 0.032 in. (0.813 mm) safety cable [11] from safety cable kit, G50375, or safety cable, G50065 [CP8006], on the drain plug [2].

NOTE: The safety cable is anchored on the oil filter cartridge cover.

(l) Install a 0.032 in. (0.813 mm) safety cable [6] from safety cable kit, G50375, or safety cable, G50065 [CP8006], on the filter drain plug [5].

(m) Clean the work area and clear the tools you used to do the task.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 79-21-07-860-003-G00

(1) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 79-21-07-860-004-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

J. Oil Filter Cartridge Element Installation Test

SUBTASK 79-21-07-710-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OIL FILTER CARTRIDGE ELEMENT- INSPECTION/CHECK

1. General

A. This procedure has this task:

(1) Oil Filter Cartridge Element Inspection.

TASK 79-21-07-200-801-G00

2. Oil Filter Cartridge Element Inspection

A. General

(1) This task gives the instructions to do the inspection of the oil filter cartridge element.

B. References

Reference	Title
79-21-04-211-801-G00	Scavenge Screen Plug Inspection (P/B 201)
79-21-07-000-801-G00	Oil Filter Cartridge Element Removal (P/B 401)
79-21-07-400-801-G00	Oil Filter Cartridge Element Installation (P/B 401)

C. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

D. Oil Filter Cartridge Element Inspection

SUBTASK 79-21-07-210-001-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Do the inspection of oil filter cartridge element:

(a) Do this task: Oil Filter Cartridge Element Removal, TASK 79-21-07-000-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Examine the lubrication unit inner wall of the oil filter housing for contaminants or debris.

1) If you find any contaminants or debris, do a visual inspection of the scavenge screen plug for debris (TASK 79-21-04-211-801-G00).

(c) Examine the removed oil filter element as follows:

1) Take the filter element in both hands and put two pleats apart to show the contaminants at the bottom of these pleats. Do this at several locations around the cartridge.

2) If you find any contaminants or debris, do a visual inspection of the scavenge screen plug for debris (TASK 79-21-04-211-801-G00).

(d) If an oil filter impending or Actual By-Pass Faults is triggered on the engine, place oil filter on hold.

1) Do not discard the oil filter.

NOTE: It could be requested for investigation by CFM.

(e) Install a new oil filter cartridge element, do this task: Oil Filter Cartridge Element Installation, TASK 79-21-07-400-801-G00.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OIL DEBRIS MONITORING SYSTEM SENSOR - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Oil debris monitoring system sensor removal

(2) Oil debris monitoring system sensor installation.

TASK 79-21-12-000-801-G00

2. Oil Debris Monitoring System Sensor Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the Oil Debris Monitoring System (ODMS) sensor.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

F. Prepare for the Removal

SUBTASK 79-21-12-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 79-21-12-010-001-G00

(2) On the applicable engine, open the right fan cowl panel (TASK 71-11-04-010-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Open these access panels:

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

G. Oil Debris Monitoring System Sensor Removal

SUBTASK 79-21-12-010-002-G00

WARNING DO NOT TOUCH THE COMPONENTS OF THE OIL SYSTEM IF THE ENGINE IS HOT. THESE COMPONENTS STAY HOTTER THAN OTHER COMPONENTS. HOT COMPONENTS CAN BURN YOU.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Remove the ODMS sensor [1] as follows:

(a) Remove the two nuts [9] from the two bolts [6] that attach the ODMS sensor [1] to the air/oil separator [8].

NOTE: The bolts [6] are captive to the ODMS sensor.

(b) Remove the ODMS sensor [1] from the air/oil separator [8].

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(c) Use a teflon-jawed pliers, STD-664 to disconnect the electrical connector [4] from oil debris monitoring unit [3] (TASK 70-00-01-910-803-G00).

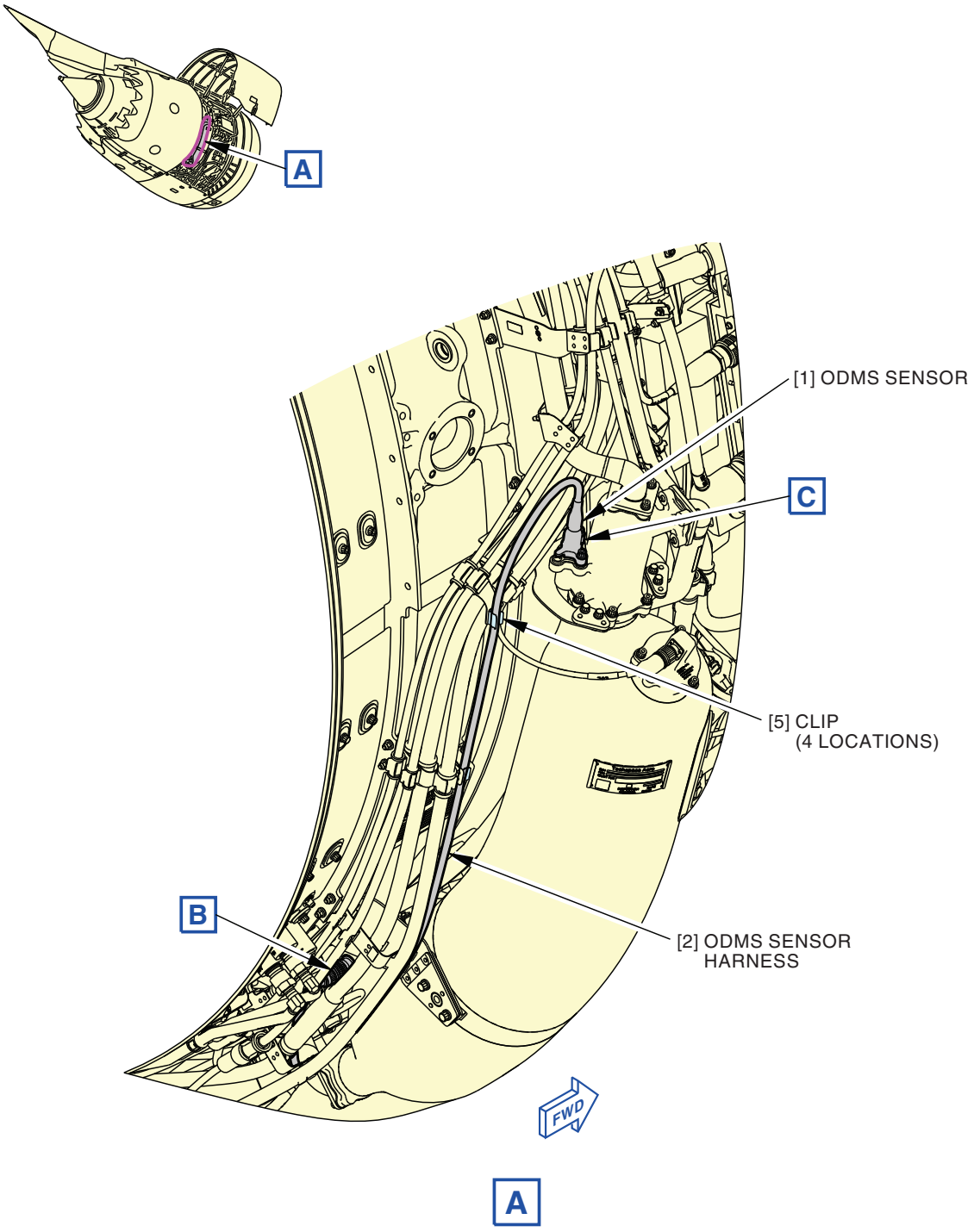
(d) Remove and discard the preformed packing [7] from the ODMS sensor [1].

(e) Remove the ODMS sensor [1] and the ODMS sensor harness [2] from the four clips [5].

(f) Install protective covers on the ODMS sensor [1] and the air/oil separator [8].

END OF TASK

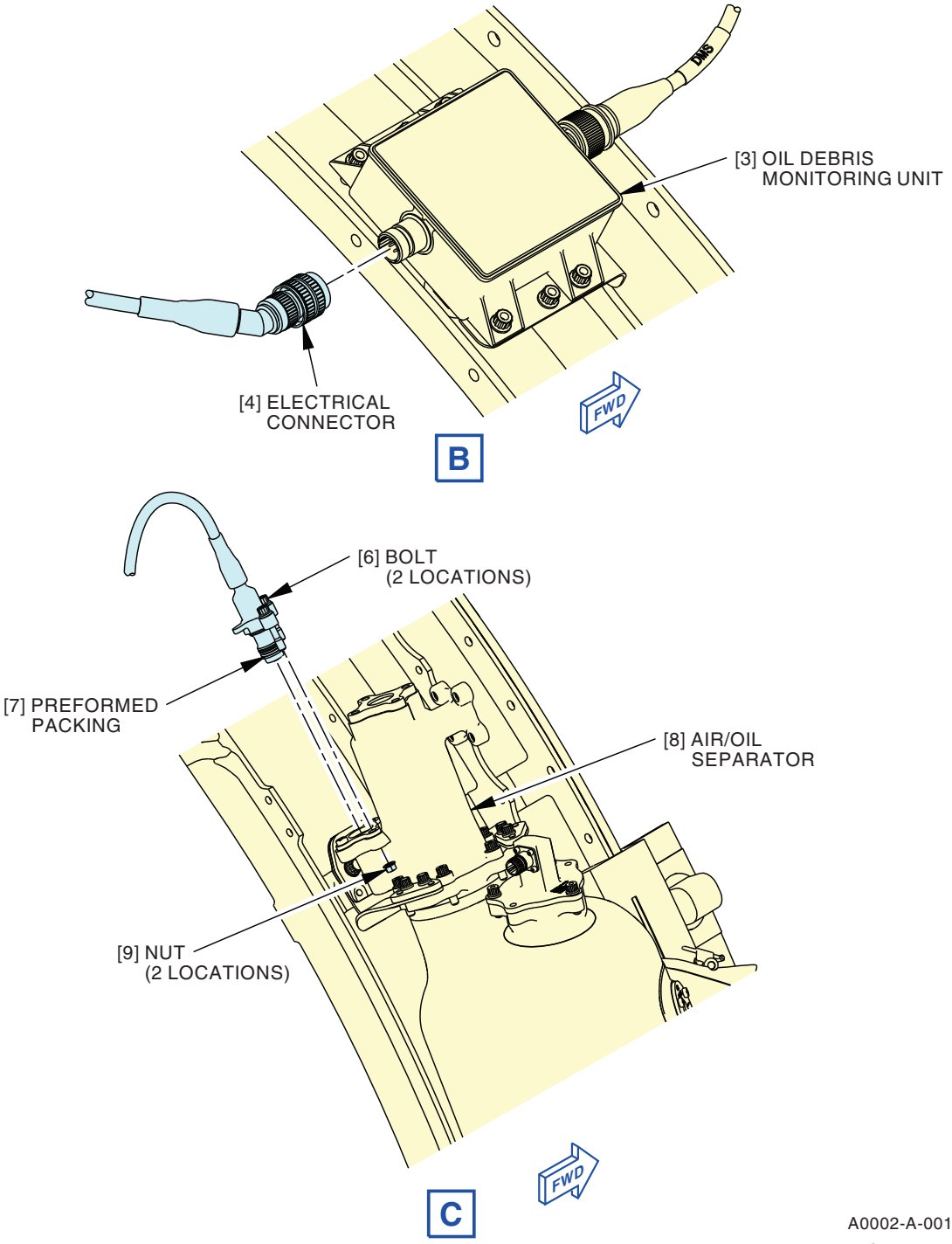
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Oil Debris Monitoring System (ODMS) Sensor Installation
Figure 401/79-21-12-990-801-G00 (Sheet 1 of 2)

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Oil Debris Monitoring System (ODMS) Sensor Installation
Figure 401/79-21-12-990-801-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 79-21-12-400-801-G00

3. Oil Debris Monitoring System Sensor Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the Oil Debris Monitoring System (ODMS) sensor.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D00672 [CP5070]	Vaseline - Pure Mineral	V V-P-236

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	ODMS sensor	79-21-12-01-030	PTF ALL
7	Preformed packing	79-21-12-01-020	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

H. Oil Debris Monitoring System Sensor Installation

SUBTASK 79-21-12-410-001-G00



WARNING

DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.



CAUTION

DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Install the ODMS sensor [1] as follows:

(a) Remove the protective covers from the ODMS sensor [1] and the air/oil separator [8].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Make sure that the mating surfaces are clean and clear of unwanted materials.

(c) Lubricate the threads of the two nuts [9] with engine oil, D00599 [CP2442].

(d) Lubricate the new preformed packing [7] with mineral vaseline, D00672 [CP5070].

(e) Install the new preformed packing [7] on the ODMS sensor [1].

(f) Install the ODMS sensor [1] on the air/oil separator [8] with the two bolts [6] and nuts [9].

(g) Tighten the nuts [9] to 100.7 in-lb (11.4 N·m) - 111.3 in-lb (12.6 N·m).

(h) Install the ODMS sensor harness [2] on the four clips [5].

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTOR AND RECEPTACLE ARE CLEAN WHEN YOU CONNECT THEM. DIRTY CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

DO NOT USE PLIERS THAT HAVE METAL JAWS TO TIGHTEN THE ELECTRICAL CONNECTOR. DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(i) Use a teflon-jawed pliers, STD-664 to connect the electrical connector [4] to the oil debris monitoring unit [3] (TASK 70-00-01-910-803-G00).

SUBTASK 79-21-12-160-001-G00

(2) Clean the work area and clear the tools used to do the task.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 79-21-12-410-002-G00

(1) On the applicable engine, close the right fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

SUBTASK 79-21-12-860-002-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

J. Oil Debris Monitoring System Sensor Installation Test

SUBTASK 79-21-12-710-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OIL DEBRIS MONITORING SYSTEM UNIT - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Oil debris monitoring system unit removal

(2) Oil debris monitoring system unit installation.

TASK 79-21-14-000-801-G00

2. Oil Debris Monitoring System Unit Removal

(Figure 401)

A. General

(1) This task provides the instructions on how to remove the oil debris monitoring system unit.

(2) The oil debris monitoring system unit is referred to as the ODMS unit.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

F. Prepare for the Removal

SUBTASK 79-21-14-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 79-21-14-860-002-G00

(2) On the applicable engine, open the right fan cowl panel (TASK 71-11-04-010-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Open these access panels:

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

G. Oil Debris Monitoring System Unit Removal

SUBTASK 79-21-14-020-001-G00

(1) Remove the ODMS unit [1] as follows:

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(a) Use the teflon-jawed pliers, STD-664 to disconnect the electrical connector [2] and the electrical connector HJ6B [3] from the ODMS unit [1] as follows (TASK 70-00-01-910-803-G00):

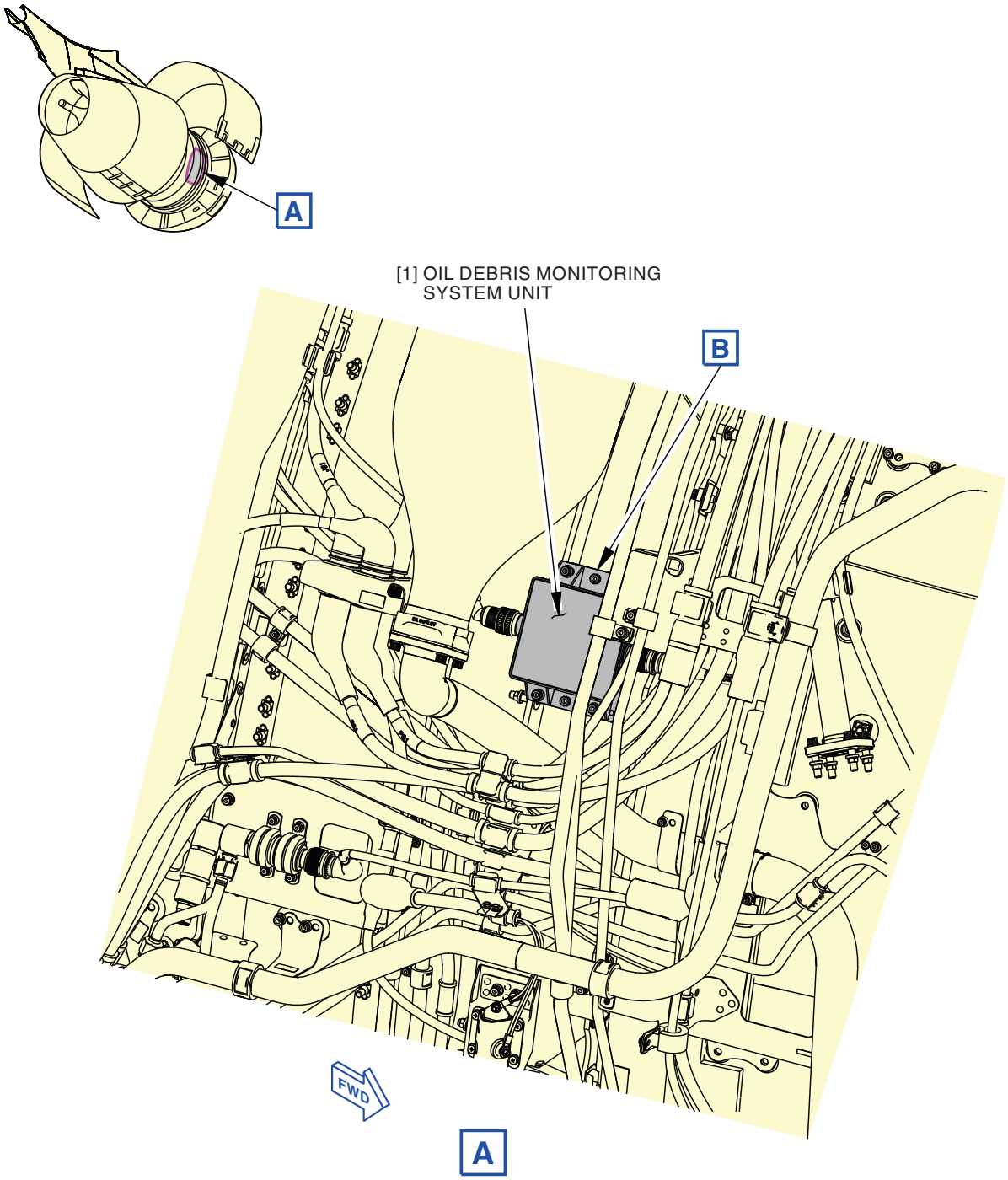
1) Disconnect the oil debris monitoring system sensor electrical connector [2] from the ODMS unit [1].
2) Disconnect the electrical connector HJ6B [3] from the ODMS unit [1].
3) Install the protective covers on the electrical connector [2], the electrical connector HJ6B [3], and the electrical receptacles on the ODMS unit [1].

(b) Remove the four bolts [4] from the ODMS unit [1].

(c) Remove the ODMS unit [1] from the mounting bracket [5]

END OF TASK

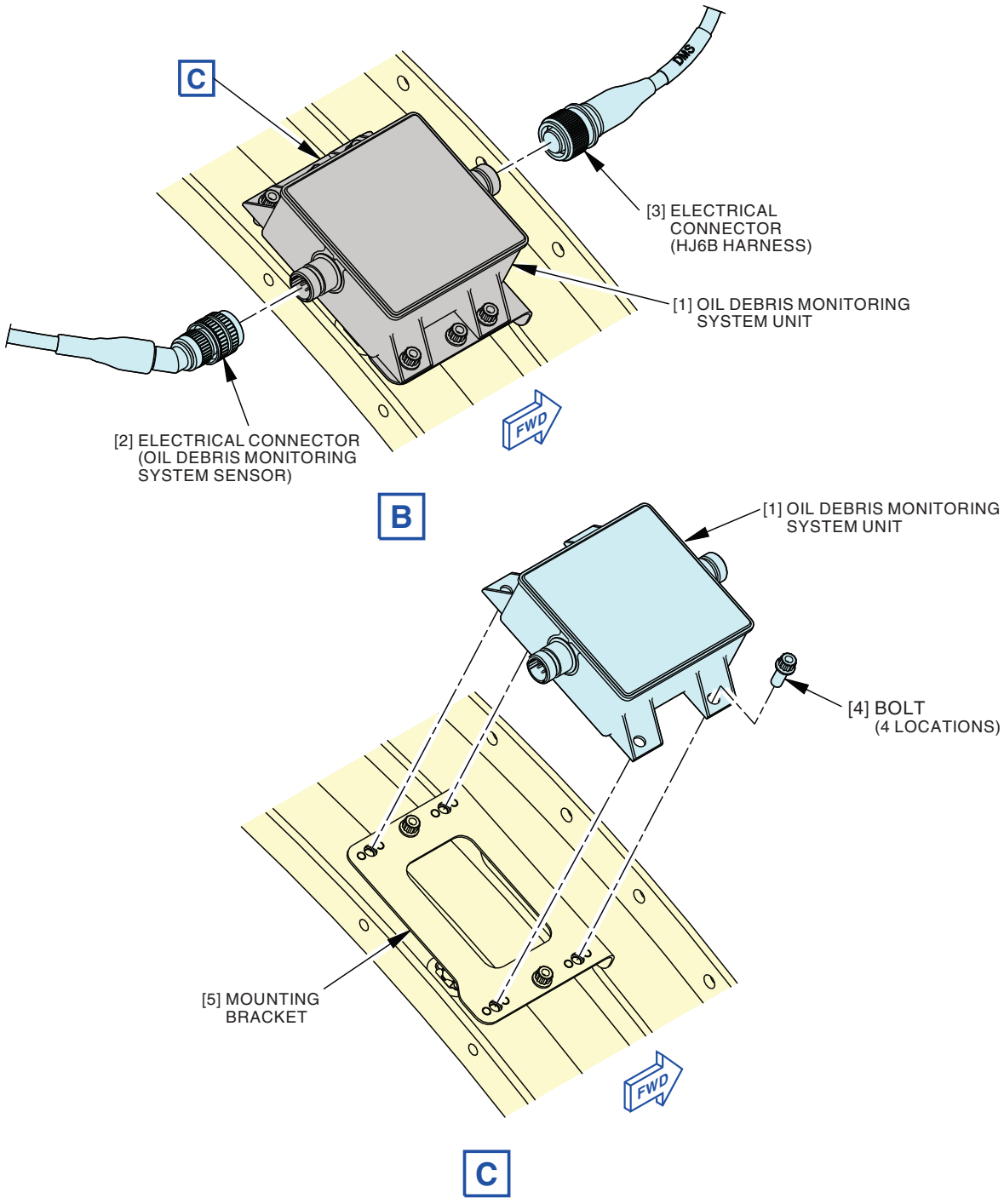
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Oil Debris Monitoring System Unit Installation
Figure 401/79-21-14-990-801-G00 (Sheet 1 of 2)

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Oil Debris Monitoring System Unit Installation
Figure 401/79-21-14-990-801-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 79-21-14-400-801-G00

3. Oil Debris Monitoring System Unit Installation

(Figure 401)

A. General

(1) This task provides the instructions on how to install the oil debris monitoring system unit.

(2) The oil debris monitoring system unit is referred to as the ODMS unit.

(3) You must do the tests that are listed in the power plant test reference table after you install the ODMS unit.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	ODMS unit	79-21-14-01-020	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

H. Oil Debris Monitoring System Unit Installation

SUBTASK 79-21-14-420-001-G00

(1) Install the ODMS unit [1] as follows:

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Lubricate the threads of the four bolts [4] with the engine oil, D00599 [CP2442].

(c) Install the ODMS unit [1] in a correct position on the mounting bracket [5].

(d) Install the four bolts [4] that attach the ODMS unit [1] on the mounting bracket [5].

(e) Tighten the bolts [4] to 67.45 in-lb (7.6 N·m) - 74.55 in-lb (8.4 N·m).

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CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(f) Use the teflon-jawed pliers, STD-664 to connect the electrical connector [2] and the electrical connector HJ6B [3] to the ODMS unit [1] as follows (TASK 70-00-01-910-803-G00):

1) Remove the protective covers from the electrical connector [2], the electrical connector HJ6B [3], and the electrical receptacles on the ODMS unit [1].

2) Connect the oil debris monitoring system sensor electrical connector [2] to the ODMS unit [1].

3) Connect the electrical connector HJ6B [3] to the ODMS unit [1].

4) Tighten the electrical connector [2] and the electrical connector HJ6B [3].

(g) Clean the work area and clear the tools and other items.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 79-21-14-860-005-G00

(1) On the applicable engine, close the right fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

SUBTASK 79-21-14-860-004-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

J. Oil Debris Monitoring System Unit Installation Test

SUBTASK 79-21-14-710-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OIL EDUCTOR VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Oil eductor valve removal

(2) Oil eductor valve installation.

TASK 79-21-17-000-801-G00

2. Oil Eductor Valve Removal

(Figure 401 and Figure 402)

A. General

(1) This task gives the instructions on how to remove the oil eductor valve.

(2) The oil eductor valve is installed at the 7:00 o'clock position on the turbine center frame.

(3) You must open the left thrust reverser to get access to the oil educator valve.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14947	Protector - SACOC Part #: 956A8681G02 Supplier: 58828

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 79-21-17-010-001-G00

(1) Do these tasks in sequence to safely open the left thrust reverser on the applicable engine:

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WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left fan cowl panel (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left thrust reverser (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

SUBTASK 79-21-17-480-001-G00

(2) Install the protector, SPL-14947 on the engine oil coolers.

G. Oil Eductor Valve Removal

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

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(WARNING PRECEDES)

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION MAKE SURE THAT YOU INSTALL THE SACOC PROTECTOR TO PREVENT DAMAGE TO THE OIL COOLER.

SUBTASK 79-21-17-020-001-G00

(1) Disconnect the eductor air tube [3] B-nut from the oil eductor valve [5].

SUBTASK 79-21-17-020-002-G00

(2) Disconnect the eductor air tube [1] B-nut from the oil eductor valve [5].

SUBTASK 79-21-17-020-003-G00

(3) Remove the three bolts [4] that attach the oil eductor valve [5] to the eductor valve support bracket [2].

SUBTASK 79-21-17-020-004-G00

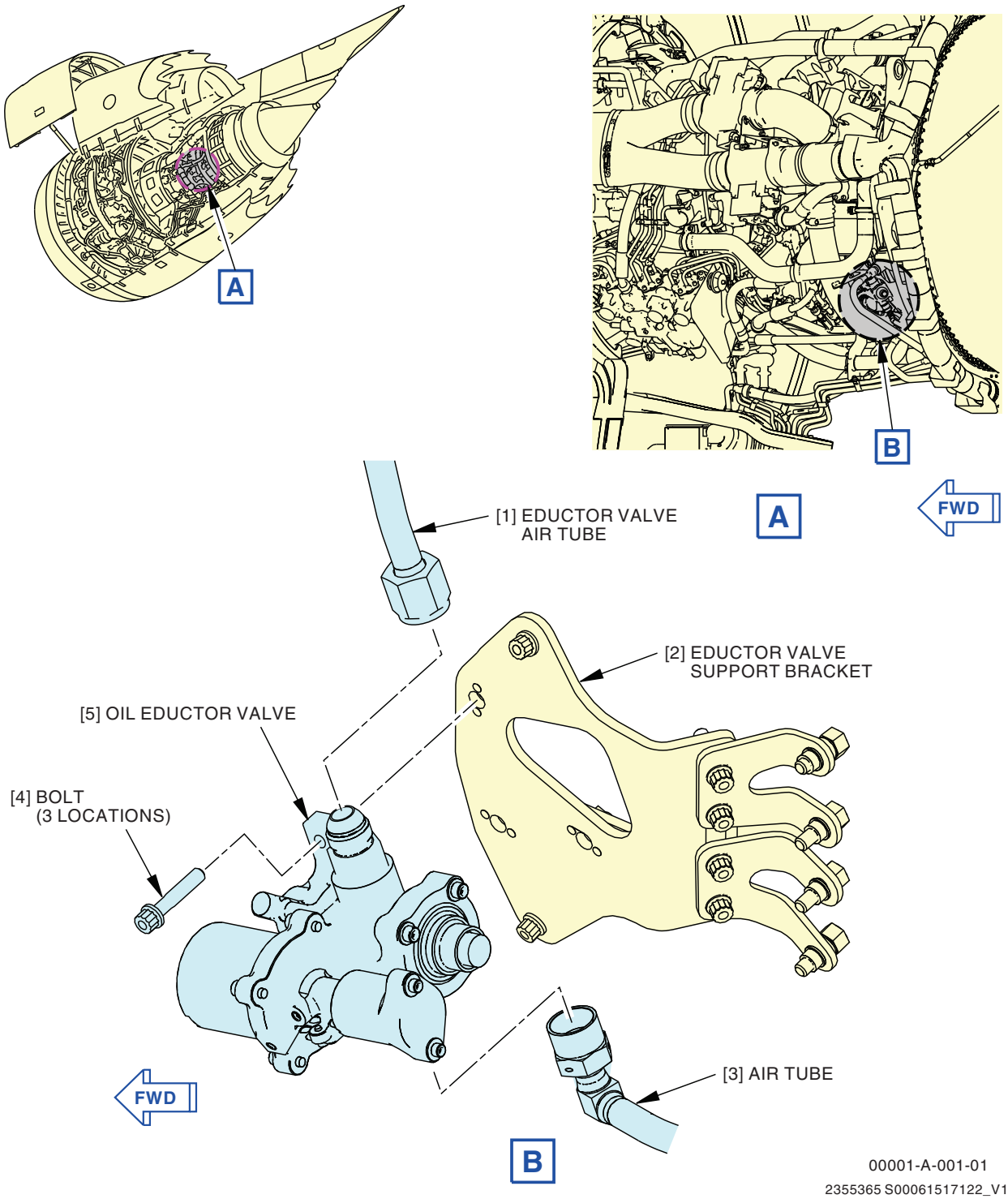
(4) Remove the oil eductor valve [5] from the engine.

SUBTASK 79-21-17-020-005-G00

(5) Install the protective covers on the eductor air tube [3], the eductor air tube [1], and the oil eductor valve [5].

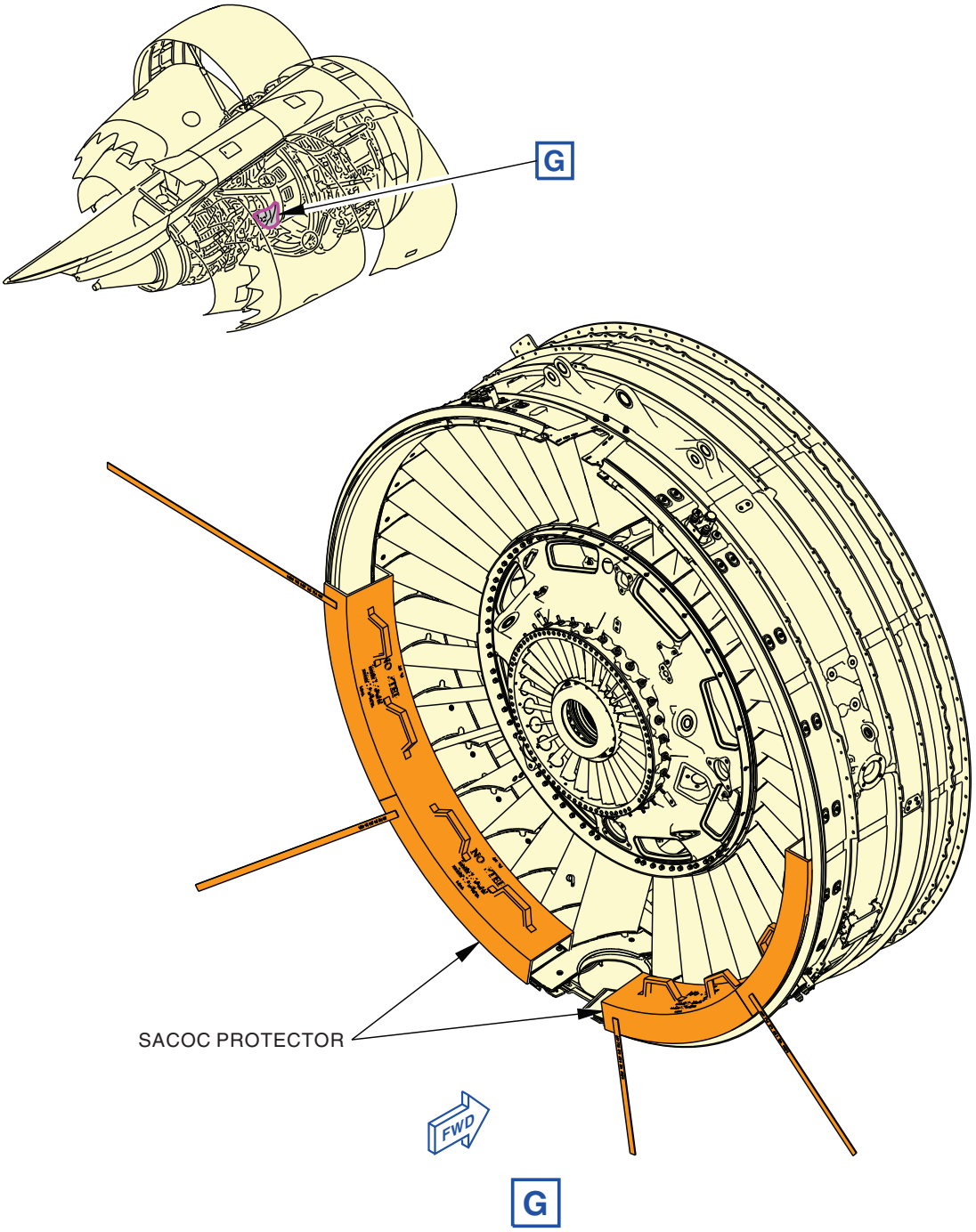
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Oil Eductor Valve Installation
Figure 401/79-21-17-990-801-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



SACOC Protector Installation
Figure 402/79-21-17-990-802-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 79-21-17-400-801-G00

3. Oil Eductor Valve Installation

(Figure 401 and Figure 402)

A. General

(1) This task provides the instructions on how to install the oil eductor valve.

(2) You must do the tests that are listed in the power plant test reference table after you install the oil eductor valve.

B. References

Reference	Title
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14947	Protector - SACOC Part #: 956A8681G02 Supplier: 58828

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
5	Oil eductor valve	79-21-17-01-020	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2

H. Oil Eductor Valve Installation

SUBTASK 79-21-17-420-001-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Remove the protective covers from the eductor air tube [3], the eductor air tube [1], and the oil eductor valve [5].

SUBTASK 79-21-17-420-002-G00

(2) Put the oil eductor valve [5] in position on the eductor valve support bracket [2].

SUBTASK 79-21-17-410-001-G00

(3) Put a light coat of the oil, D00599 [CP2442] on the threads of the three bolts [4].

SUBTASK 79-21-17-420-003-G00

(4) Install the three bolts [4] that attach the oil eductor valve [5] to the eductor valve support bracket [2].

(a) Tighten the three bolts [4] to 106 in-lb (12 N·m) - 124 in-lb (14 N·m).

SUBTASK 79-21-17-420-004-G00

(5) Connect the eductor air tube [1] B-nut to the oil eductor valve [5].

(a) Tighten the eductor air tube [1] B-nut to 460 in-lb (52 N·m) - 540 in-lb (61 N·m).

SUBTASK 79-21-17-420-005-G00

(6) Connect the eductor air tube [3] B-nut to the oil eductor valve [5].

(a) Tighten the eductor air tube [3] B-nut to 450 in-lb (51 N·m) - 550 in-lb (62 N·m).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 79-21-17-080-001-G00

(1) Remove the protector, SPL-14947 from the engine.

SUBTASK 79-21-17-410-002-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do these tasks in sequence to safely close the left thrust reverser:

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(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

(b) Close the applicable left Fan Cowl Panel (TASK 71-11-04-410-801-G00).

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

J. Oil Eductor Valve Installation Test

SUBTASK 79-21-17-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

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AIR OIL SEPARATOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the air oil separator

(2) An installation of the air oil separator.

TASK 79-21-21-000-801-G00

2. Air Oil Separator Removal

(Figure 401)

A. General

(1) This task provides the instructions on how to remove the air oil separator.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
79-21-12-000-801-G00	Oil Debris Monitoring System Sensor Removal (P/B 401)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE
STD-1055	Container - Oil Resistant, 5 Gallon (19 Liter)

D. Access Panels

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

E. Prepare for the Removal

SUBTASK 79-21-21-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 79-21-21-010-001-G00

(2) On the applicable engine, open the right fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Air Oil Separator Removal

SUBTASK 79-21-21-010-002-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

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WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Remove the air oil separator [1] as follows:

(a) Put an oil resistant container (5 gal)(19 Liter), STD-1055, below all the tube openings and air oil separator [1].

(b) Remove the oil debris monitoring system sensor [3] (TASK 79-21-12-000-801-G00).

(c) Disconnect the oil tube [4] from the air oil separator [1] as follows:

1) Remove the bolt [5] from the clamp [6] that attaches the oil tube [4] to the bracket [7].

2) Remove the bolt [8] from the clamp [9] that attaches the oil tube [4] to the bracket [10].

3) Remove the four bolts [11] from the oil tube [4].

a) Do not move the flange [16] of the check valve air oil separator [1].

4) Disconnect the oil tube [4] from the air oil separator [1].

5) Remove and discard the gasket seal [12] from the oil tube [4].

6) Install the protective caps on the openings.

(d) Disconnect the oil tube [13] from the air oil separator [1]:

1) Remove the four bolts [14] from the oil tube [13].

2) Disconnect the oil tube [13] from the air oil separator [1].

3) Remove and discard the gasket seal [15] from the oil tube [13].

4) Install the protective caps on the openings.

(e) Remove the six bolts [17] from the air oil separator [1].

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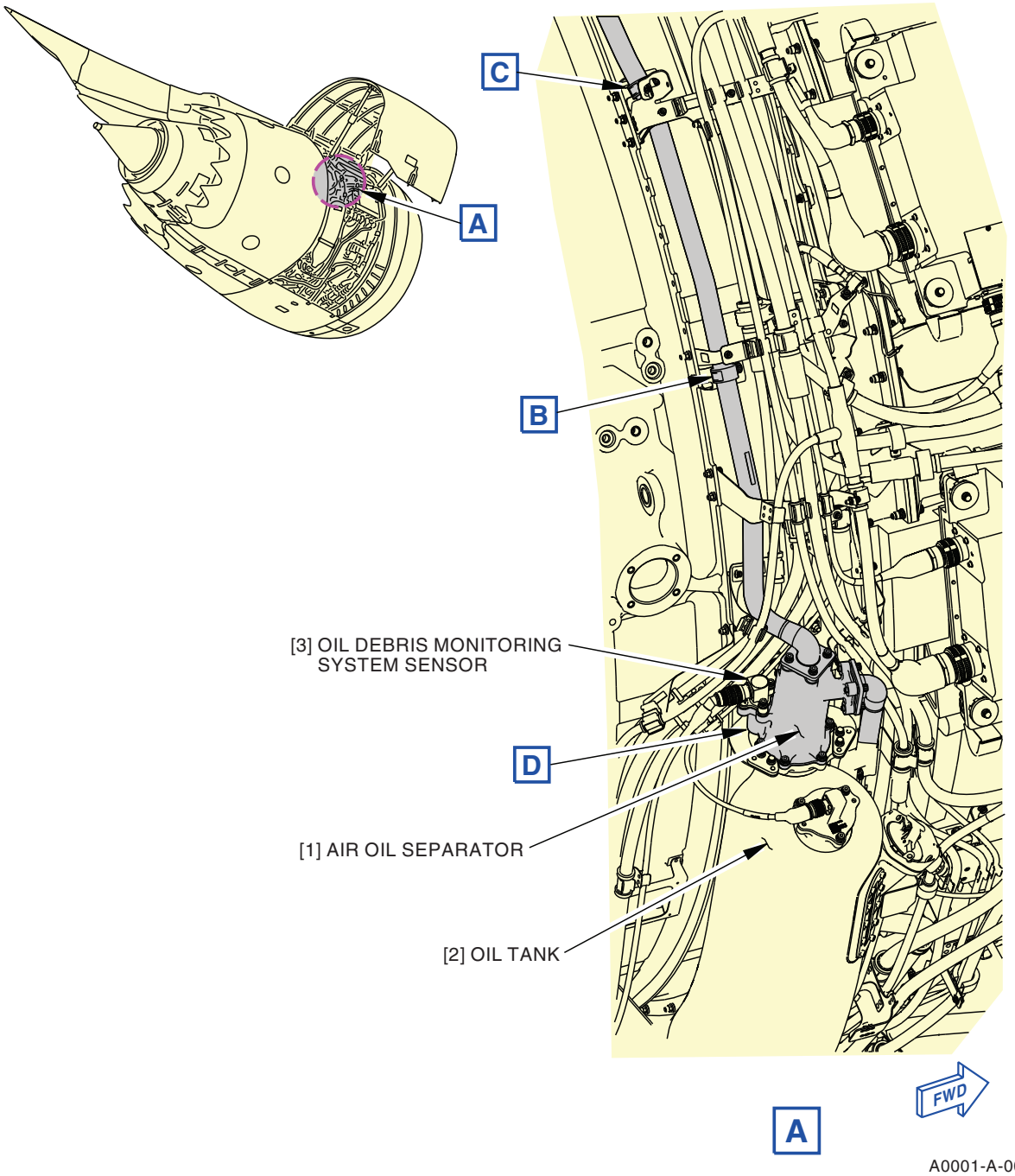
(f) Remove the air oil separator [1] from the oil tank [2].

(g) Remove and discard the preformed packing [18] from the air oil separator [1].

(h) Install the protective caps on the openings.

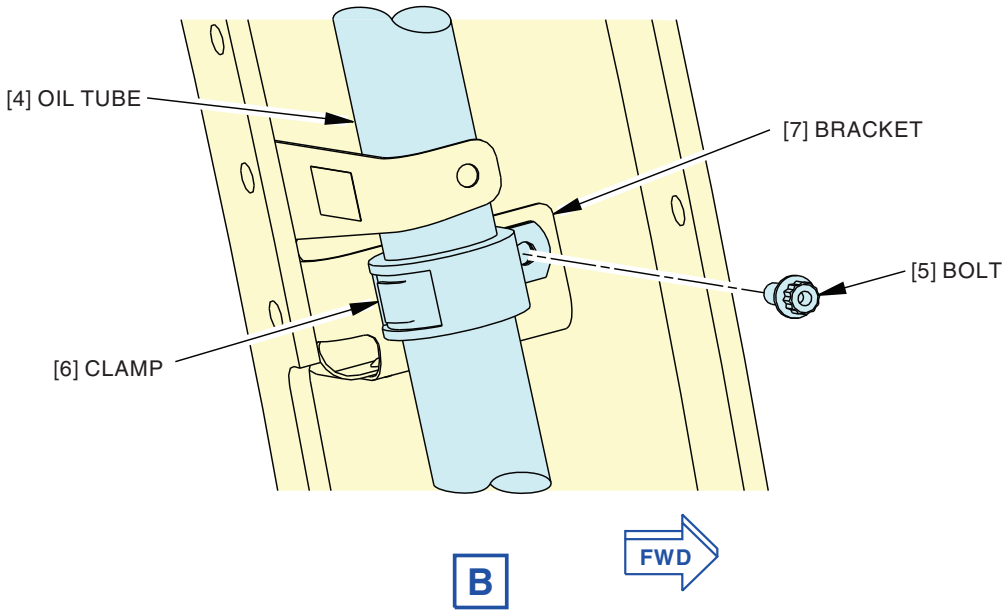
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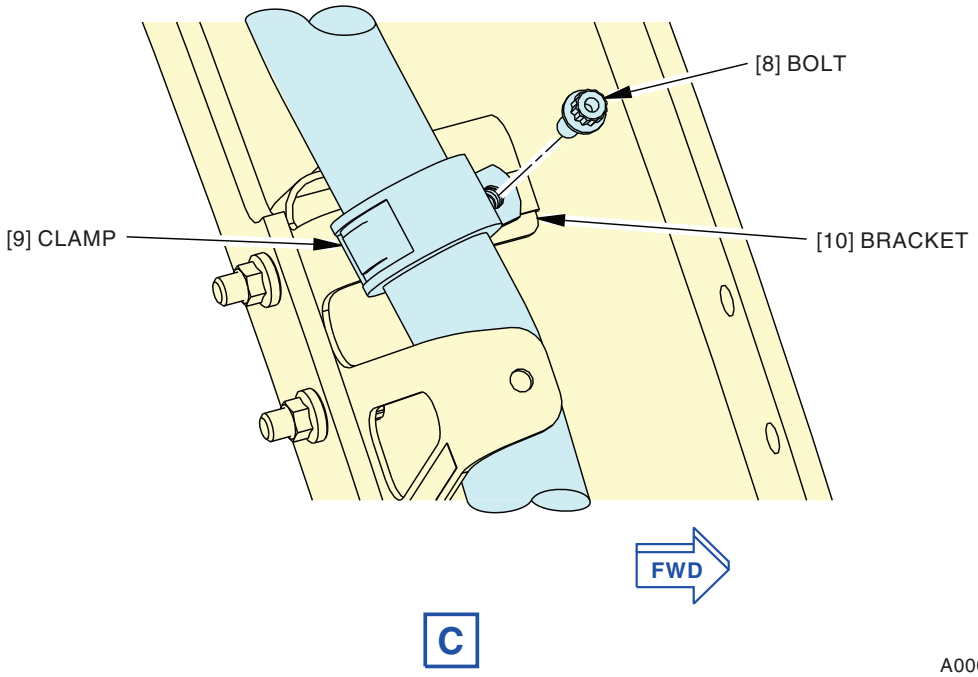
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Air Oil Separator Installation
Figure 401/79-21-21-990-801-G00 (Sheet 1 of 5)

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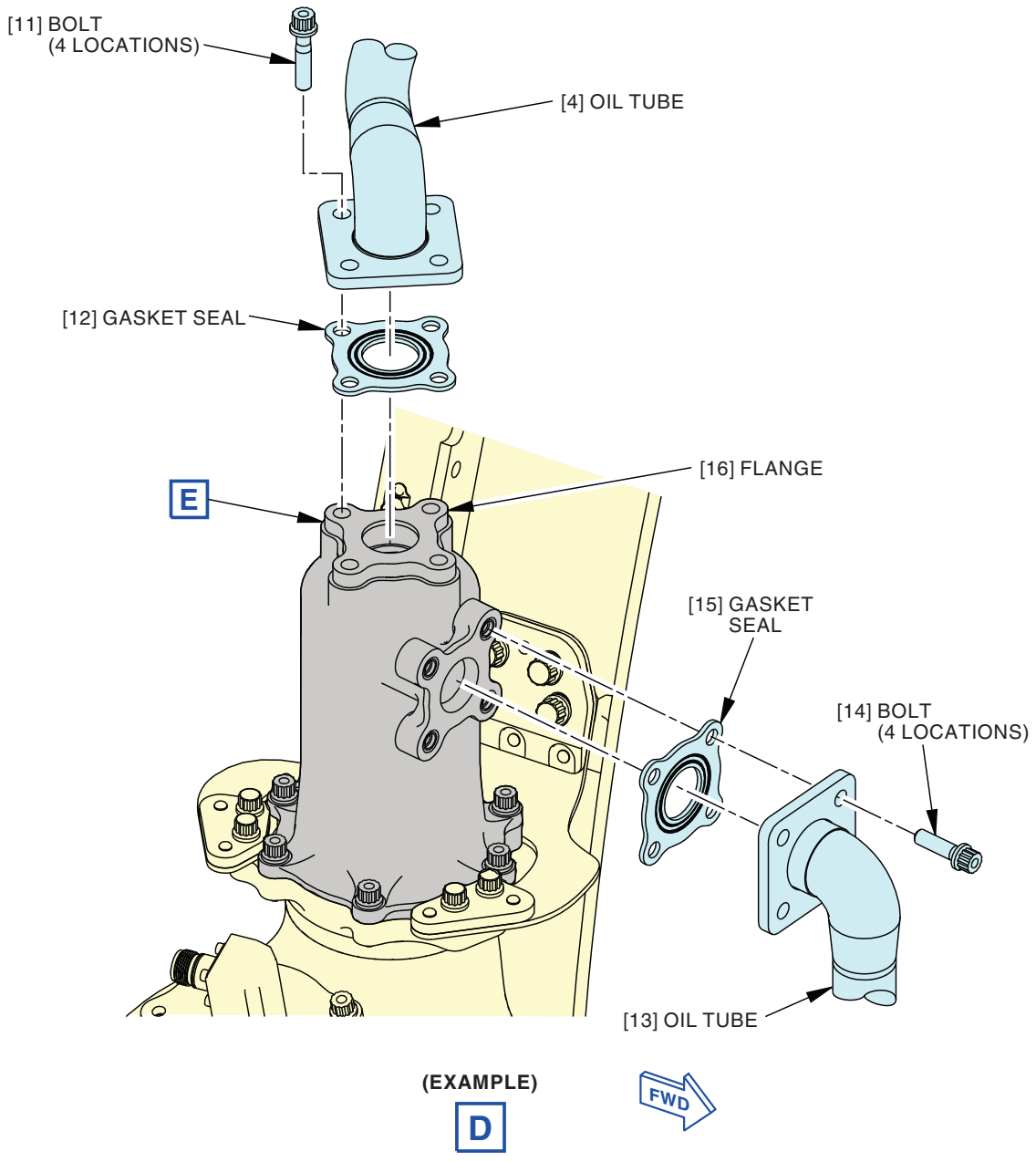




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Air Oil Separator Installation
Figure 401/79-21-21-990-801-G00 (Sheet 2 of 5)

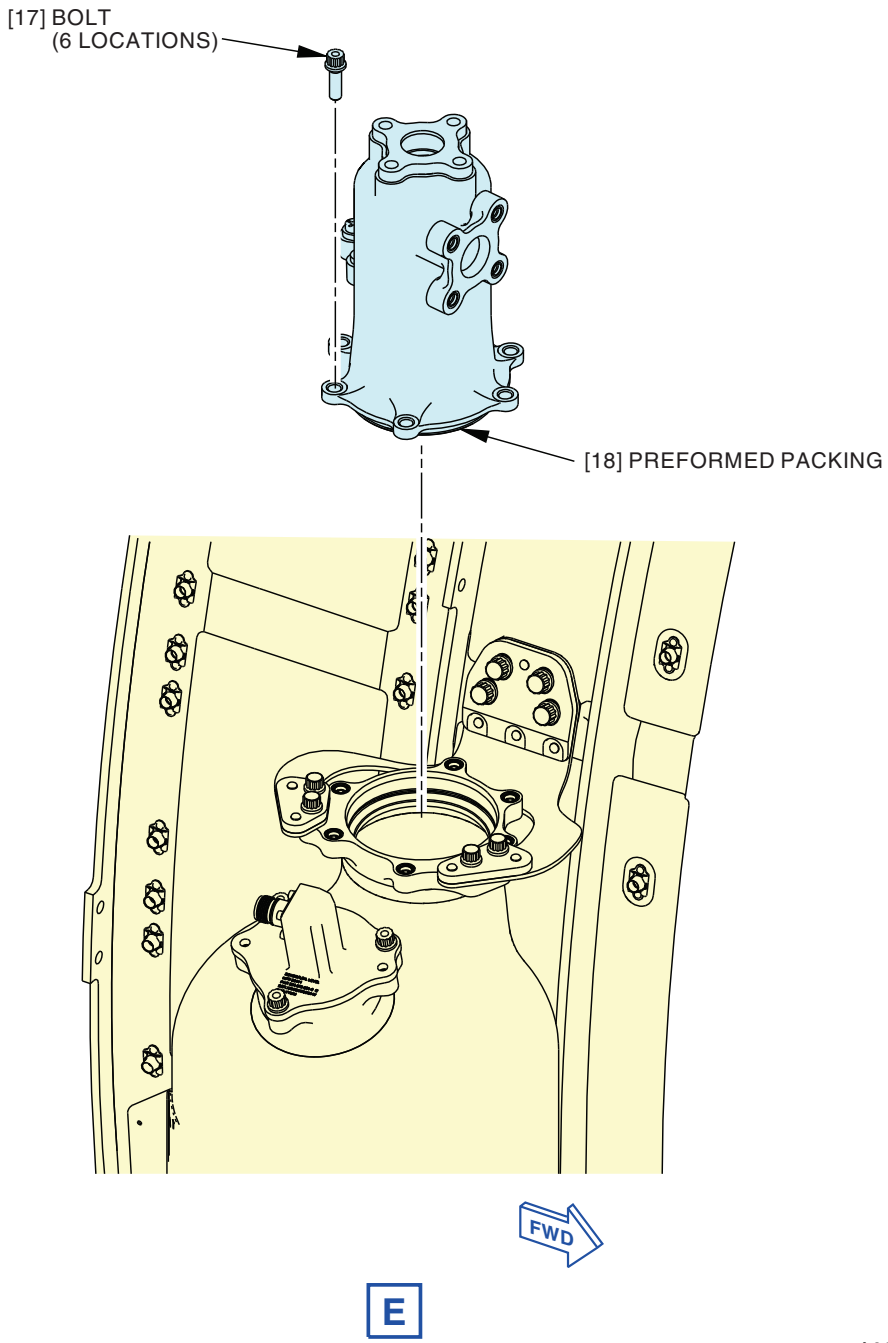
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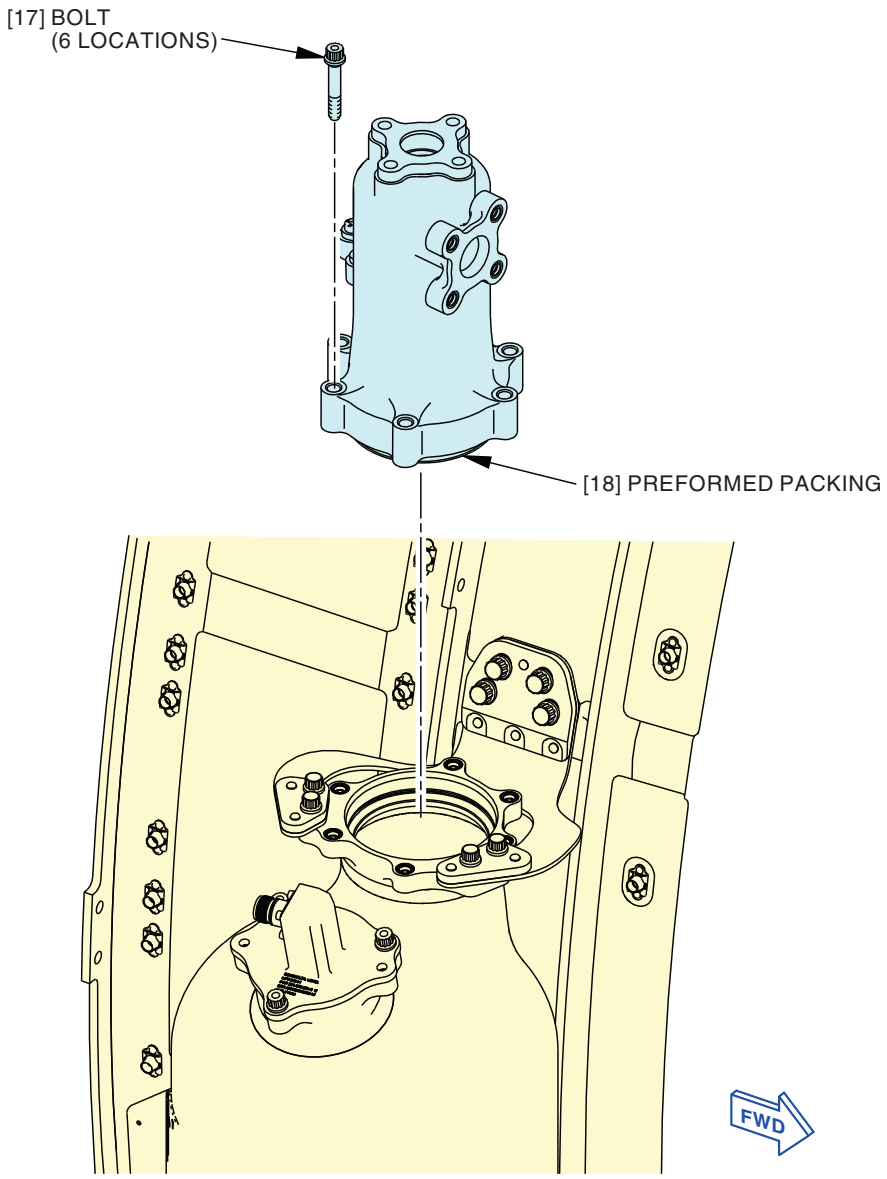
Air Oil Separator Installation
Figure 401/79-21-21-990-801-G00 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Air Oil Separator Installation
Figure 401/79-21-21-990-801-G00 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Air Oil Separator Installation
Figure 401/79-21-21-990-801-G00 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 79-21-21-400-801-G00

3. Air Oil Separator Installation

(Figure 401)

A. General

(1) This task gives the instructions on how to install the air oil separator.

B. References

Reference	Title
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
79-21-12-400-801-G00	Oil Debris Monitoring System Sensor Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D00601 [CP2101]	Vaseline - Graphite Mineral	
D00672 [CP5070]	Vaseline - Pure Mineral	V V-P-236
G51335 [CP2674]	Lint-Free Cloth	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Air oil separator	79-21-21-01-120	PTF ALL
		79-21-21-01-122	PTF ALL
12	Gasket seal	79-21-21-01-080	PTF ALL
15	Gasket seal	79-21-21-01-100	PTF ALL
18	Preformed packing	79-21-21-01-130	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

H. Air Oil Separator Installation

SUBTASK 79-21-21-410-001-G00

WARNING

DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION MAKE SURE THAT YOU INSTALL THE CORRECT BOLT LENGTH WHEN YOU INSTALL THE POST SB 737-CFM-LEAP-1B-79-0012 AIR OIL SEPARATOR. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Install the air oil separator [1]:

(a) Remove the protective covers from the openings.

(b) Make sure that the mating faces are clean and clear of unwanted materials.

(c) Lubricate the threads and pressure faces of the six bolts [17] with graphite mineral vaseline, D00601 [CP2101].

(d) Lubricate the new preformed packing [18] with the mineral vaseline, D00672 [CP5070].

(e) Install the preformed packing [18] on the groove of the air oil separator [1].

(f) Install the air oil separator [1] on the oil tank [2].

(g) Install the six bolts [17] on the air oil separator [1].

(h) Connect the oil tube [13] to the air oil separator [1]:

1) Remove the protective covers from the openings.

2) Make sure that the mating faces are clean and clear of unwanted materials.

3) Lubricate the threads and pressure faces of the four bolts [14] with graphite mineral vaseline, D00601 [CP2101].

4) Lubricate each side of the elastomer part of a new gasket seal [15] with vaseline, D00672 [CP5070], or engine oil, D00599 [CP2442].

5) Install the gasket seal [15] and oil tube [13] on the air oil separator [1].

6) Install the four bolts [14] that attach the oil tube [13] to the air oil separator [1].

(i) Connect the oil tube [4] to the air oil separator [1]:

1) Do not move the flange [16] of the check valve air oil separator [1].

2) Remove the protective covers from the openings.

3) Make sure that the mating faces are clean and clear of unwanted materials.

4) Lubricate the threads and pressure faces of the four bolts [11] with graphite mineral vaseline, D00601 [CP2101].

5) Lubricate each side of the elastomer part of a new gasket seal [12] with vaseline, D00672 [CP5070], or engine oil, D00599 [CP2442].

6) Install the gasket seal [12] and oil tube [4] in a correct position on the air oil separator [1].

7) Install the four bolts [11] that attach the oil tube [4] to the air oil separator [1].

8) Install the bolt [8] on the clamp [9] that attaches the oil tube [4] on the bracket [10].

9) Install the bolt [5] on the clamp [6] that attaches the oil tube [4] on the bracket [7].

(j) Tighten the bolts [17] in a criss cross pattern to 109.25 in-lb (12.34 N·m) - 120.75 in-lb (13.64 N·m).

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(k) Tighten the bolts [14] in a criss cross pattern to 109.25 in-lb (12.34 N·m) - 120.75 in-lb (13.64 N·m).

(l) Tighten the bolts [11] in a criss cross pattern to 109.25 in-lb (12.34 N·m) - 120.75 in-lb (13.64 N·m).

(m) Tighten the bolt [8] to 41.80 in-lb (4.72 N·m) - 46.20 in-lb (5.22 N·m).

(n) Tighten the bolt [5] to 41.80 in-lb (4.72 N·m) - 46.20 in-lb (5.22 N·m).

(o) Remove the unwanted vaseline, D00672 [CP5070], and graphite mineral vaseline, D00601 [CP2101], with a lint-free cloth, G51335 [CP2674].

(p) Install the oil debris monitoring system sensor [3] (TASK 79-21-12-400-801-G00).

(q) Clean the work area and clear the tools used to do the task.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 79-21-21-860-002-G00

(1) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 79-21-21-410-002-G00

(2) On the applicable engine, close the right fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

J. Air Oil Separator Installation Test

SUBTASK 79-21-21-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ANTI LEAK VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the anti leak valve

(2) An installation of the anti leak valve.

TASK 79-21-22-000-801-G00

2. Anti Leak Valve Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the anti leak valve.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
79-21-03-000-801-G00	Lubrication Unit Removal (P/B 401)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 79-21-22-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 79-21-22-010-001-G00

(2) Do these tasks in sequence to safely open the left thrust reverser on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left fan cowl panel (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left thrust reverser (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

G. Anti Leak Valve Removal

SUBTASK 79-21-22-020-001-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE IS STOPPED. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

CAUTION

DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO PAINT AND RUBBER.

(1) Do this task: Lubrication Unit Removal, TASK 79-21-03-000-801-G00.

SUBTASK 79-21-22-020-002-G00

(2) Remove the anti leak valve [1] as follows:

(a) Insert a screwdriver or a dental pick into the removal notch of the retaining ring [2].

(b) Use the screwdriver or the dental pick to pull out the first end of the retaining ring [2].

(c) Manually twist the retaining ring [2] around until it is released from the groove of the lubrication unit [3].

(d) Discard the retaining ring [2].

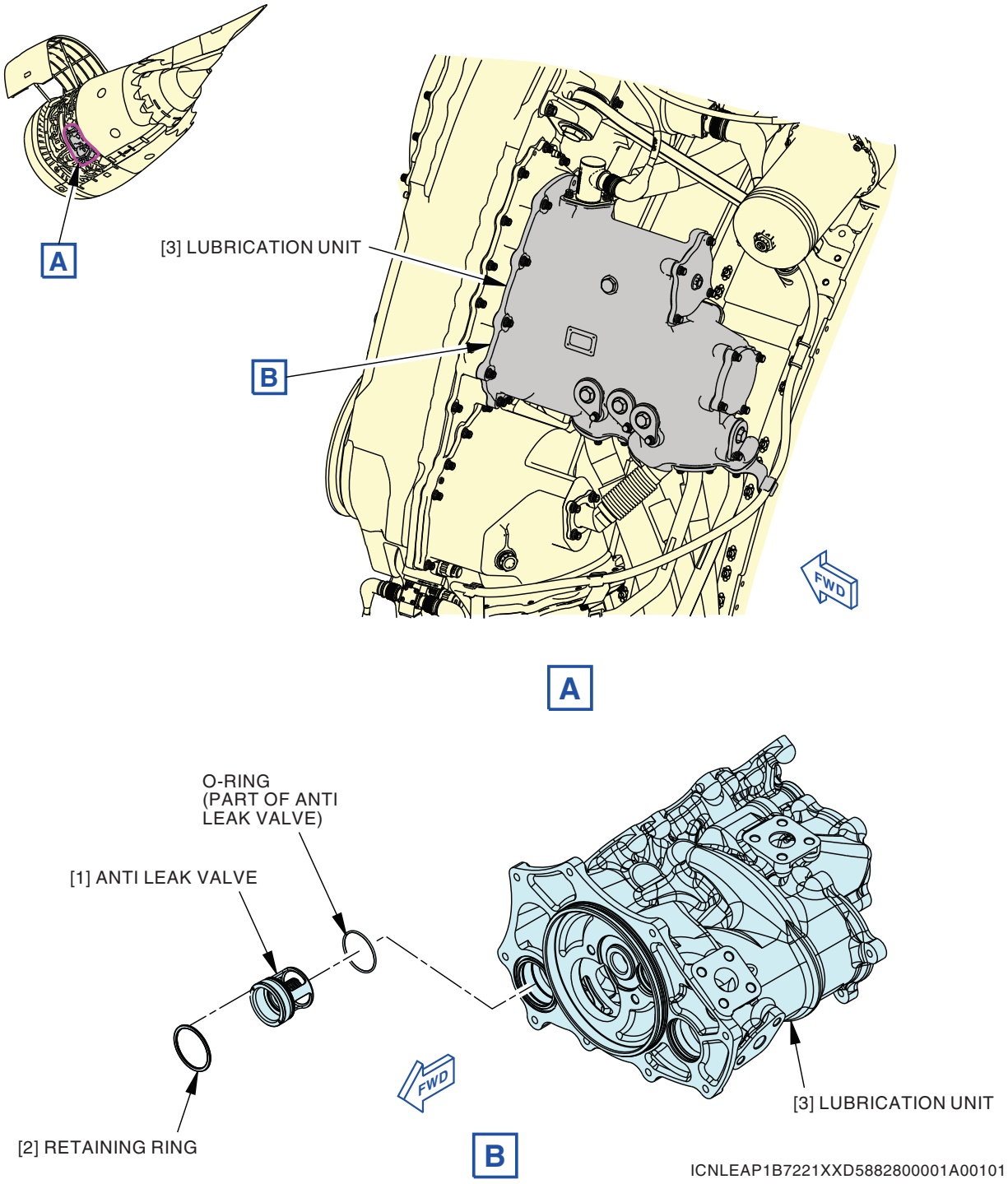
(e) Remove the anti leak valve [1] equipped with its O-ring from the lubrication unit [3].

(f) Discard the anti leak valve [1] equipped with its O-ring.

(g) Install the protective covers on the openings.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Anti Leak Valve Installation
Figure 401/79-21-22-990-801-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 79-21-22-400-801-G00

3. Anti Leak Valve Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the anti leak valve.

B. References

Reference	Title
12-13-11-600-801	Engine Servicing (Oil Replenishing) (P/B 301)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-63-01-610-802-G00	Accessory Gearbox Assembly Oil Replenishing (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
79-21-03-400-801-G00	Lubrication Unit Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00672 [CP5070]	Vaseline - Pure Mineral	V V-P-236

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Anti leak valve	79-21-03-01-370	PTF ALL
2	Retaining ring	79-21-03-01-365	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2

H. Anti Leak Valve Installation

SUBTASK 79-21-22-410-001-G00

WARNING

DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO PAINT AND RUBBER.

(1) Install the anti leak valve [1] as follows:

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Remove the protective covers on the openings.

CAUTION MAKE SURE THAT THE O-RING IS INSTALLED ON THE NEW ANTI LEAK VALVE. IF YOU DO NOT OBEY, DAMAGE TO THE ANTI LEAK VALVE CAN OCCUR.

(c) Lubricate the O-ring of a new anti leak valve [1] with vaseline, D00672 [CP5070].

(d) Install the anti leak valve [1] equipped with its O-ring to the lubrication unit [3].

(e) Install the new retaining ring [2] on the groove of the lubrication unit [3].

SUBTASK 79-21-22-400-001-G00

(2) Do this task: Lubrication Unit Installation, TASK 79-21-03-400-801-G00.

SUBTASK 79-21-22-610-001-G00

(3) Do this task: Accessory Gearbox Assembly Oil Replenishing, TASK 72-63-01-610-802-G00.

SUBTASK 79-21-22-600-001-G00

(4) Do this task: Engine Servicing (Oil Replenishing), TASK 12-13-11-600-801.

SUBTASK 79-21-22-420-001-G00

(5) Clean the work area and clear the tools and other items.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 79-21-22-410-002-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the left thrust reverser:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

(b) Close the applicable left Fan Cowl Panel (TASK 71-11-04-410-801-G00).

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

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SUBTASK 79-21-22-860-002-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

J. Lubrication Unit Installation Test

SUBTASK 79-21-22-730-002-G00

(1) Do the applicable tests given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OIL INDICATING SYSTEM - INSPECTION/CHECK

1. General

A. This procedure contains this task:

(1) Oil indicating system inspection.

TASK 79-30-00-210-801-G00

2. Oil Indicating System Inspection

(Figure 601)

A. General

(1) This task gives the instructions on how to examine the oil indicating system.

(2) The oil indicating system inspection examines these components:

(a) Oil level sensor

(b) Oil pressure and temperature sensor

(c) Oil Filter Differential Pressure (OFDP) sensor.

B. References

Reference	Title
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)
79-31-01-000-801-G00	Oil Level Sensor Removal (P/B 401)
79-31-01-400-801-G00	Oil Level Sensor Installation (P/B 401)
79-32-01-000-801-G00	Oil Pressure and Temperature Sensor Removal (P/B 401)
79-32-01-400-801-G00	Oil Pressure and Temperature Sensor Installation (P/B 401)
79-35-01-000-801-G00	Oil Filter Differential Pressure Sensor Removal (P/B 401)
79-35-01-400-801-G00	Oil Filter Differential Pressure Sensor Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-562	Lens - Magnifying, 10x
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Inspection

SUBTASK 79-30-00-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 79-30-00-010-001-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1

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(Continued)

Number	Name/Location
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

SUBTASK 79-30-00-480-001-G00

(3) Use good lighting and a 10x magnifying lens, STD-562 to help you see damage to the oil indicating system.

NOTE: Reject all parts that have wear or damage more than the applicable limits.

NOTE: Reject the parts that show wear or damage that is more than normal. Repair the minor damage to the applicable parts.

G. Oil Indicating System Inspection

SUBTASK 79-30-00-212-001-G00

(1) Do the general checks of the oil level sensor [1], oil pressure and temperature sensor [5] and oil filter differential pressure sensor [8] as follows:

(a) Do a visual check to the parts for signs of wear, galled material, loose metal, cracks, nicks burrs, dents and other damage.

(b) Do a visual check to the treated parts for damage that shows bare metal.

(c) Do a visual check to the internal housing bores and machined surfaces for scores, wear, or damage.

(d) Do a check of the internal openings to be sure that they are clean and do not have blockages.

(e) Do a check of the threads for rough edges and defects.

NOTE: Make sure that the threads are satisfactory.

SUBTASK 79-30-00-010-002-G00

(2) Do this task: Oil Level Sensor Removal, TASK 79-31-01-000-801-G00.

SUBTASK 79-30-00-212-002-G00

(3) Do the inspection of the oil level sensor [1] zones A, B1, B2, C1, C2, D1, D2 and E for these conditions:

(a) Scratches, nicks and dents

1) Not permitted.

2) Replace the oil level sensor [1].

SUBTASK 79-30-00-410-001-G00

(4) Do this task: Oil Level Sensor Installation, TASK 79-31-01-400-801-G00.

SUBTASK 79-30-00-212-003-G00

(5) Do the inspection of the oil pressure and temperature sensor [5] for these conditions:

(a) Cracks, nicks, dents, scratches and deformations.

1) Not permitted.

2) Replace the oil pressure and temperature sensor [5] (TASK 79-32-01-000-801-G00 and TASK 79-32-01-400-801-G00).

(b) Visible voids and cracks at where the connector [6] and connector [7] are welded to the oil pressure and temperature sensor [5].

1) Not permitted.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Replace the oil pressure and temperature sensor [5] (TASK 79-32-01-000-801-G00 and TASK 79-32-01-400-801-G00).

(c) Dents, cracks and loose or bent pins on the connector [6] and the connector [7].

1) Not permitted.

2) Replace the oil pressure and temperature sensor [5] (TASK 79-32-01-000-801-G00 and TASK 79-32-01-400-801-G00).

SUBTASK 79-30-00-212-004-G00

(6) Do the inspection of the oil filter differential pressure sensor [8] for these conditions:

(a) Cracks, nicks, dents, scratches and deformations.

1) Not permitted if the oil filter differential pressure sensor [8] shows evidence of defects.

2) Replace the oil filter differential pressure sensor [8] (TASK 79-35-01-000-801-G00 and TASK 79-35-01-400-801-G00).

(b) Voids and cracks where the connector [9] is welded to the oil filter differential pressure sensor [8].

1) Not permitted.

2) Replace the oil filter differential pressure sensor [8] (TASK 79-35-01-000-801-G00 and TASK 79-35-01-400-801-G00).

(c) Dents, cracks and loose or bent pins on connector [9].

1) Not permitted.

2) Replace the oil filter differential pressure sensor [8] (TASK 79-35-01-000-801-G00 and TASK 79-35-01-400-801-G00).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 79-30-00-410-002-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 79-30-00-860-002-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

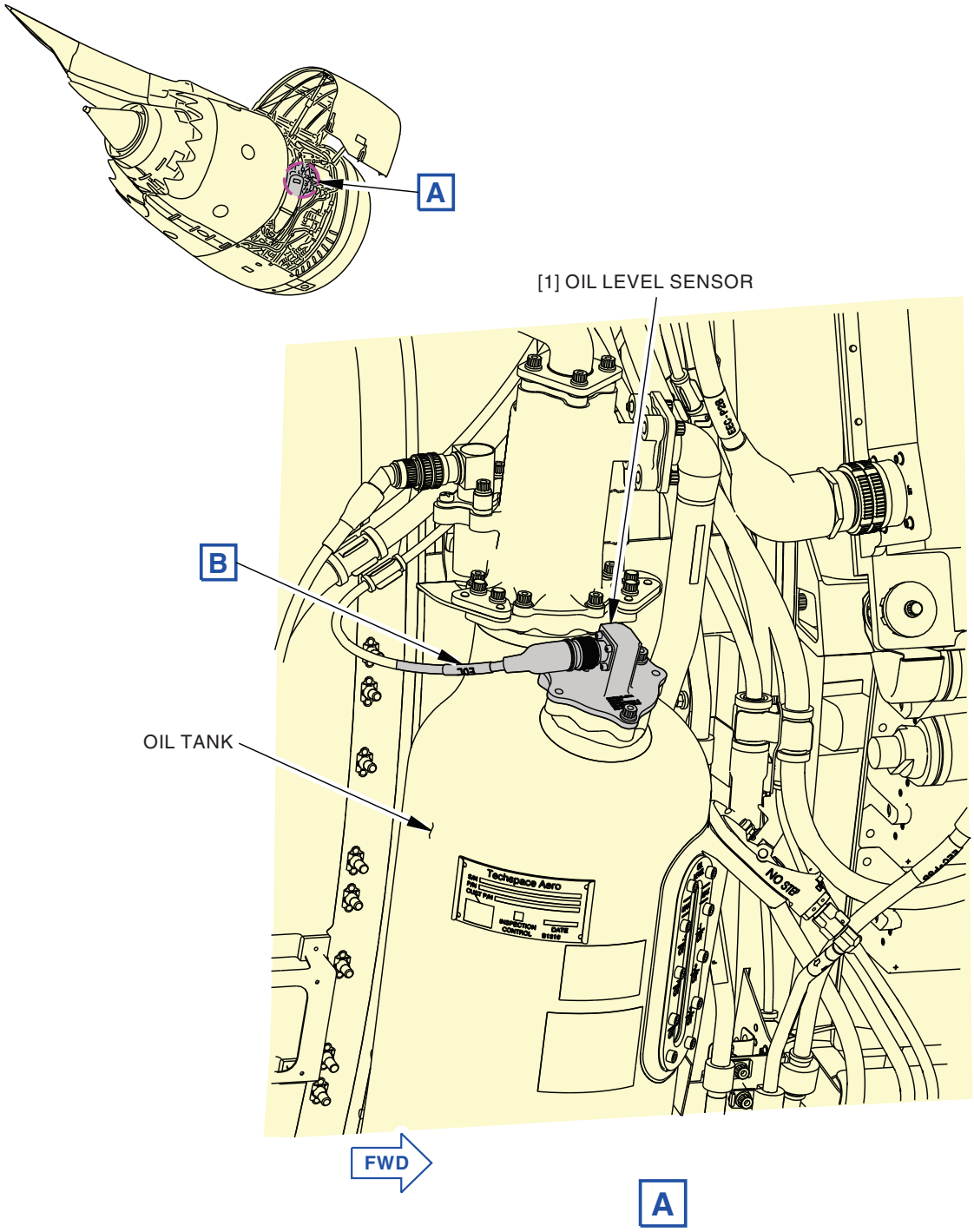
I. Oil Indicating System Tests

SUBTASK 79-30-00-700-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

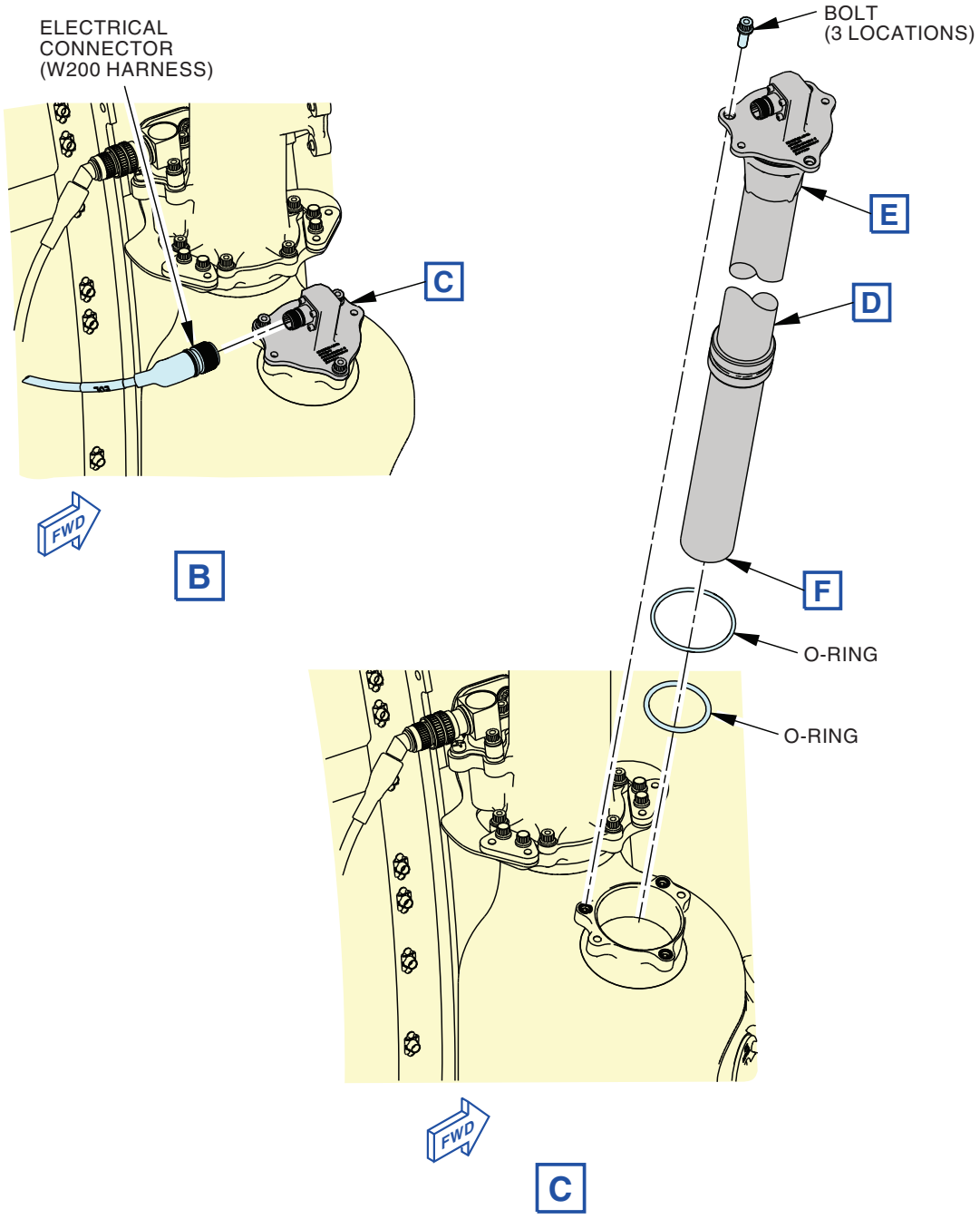
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



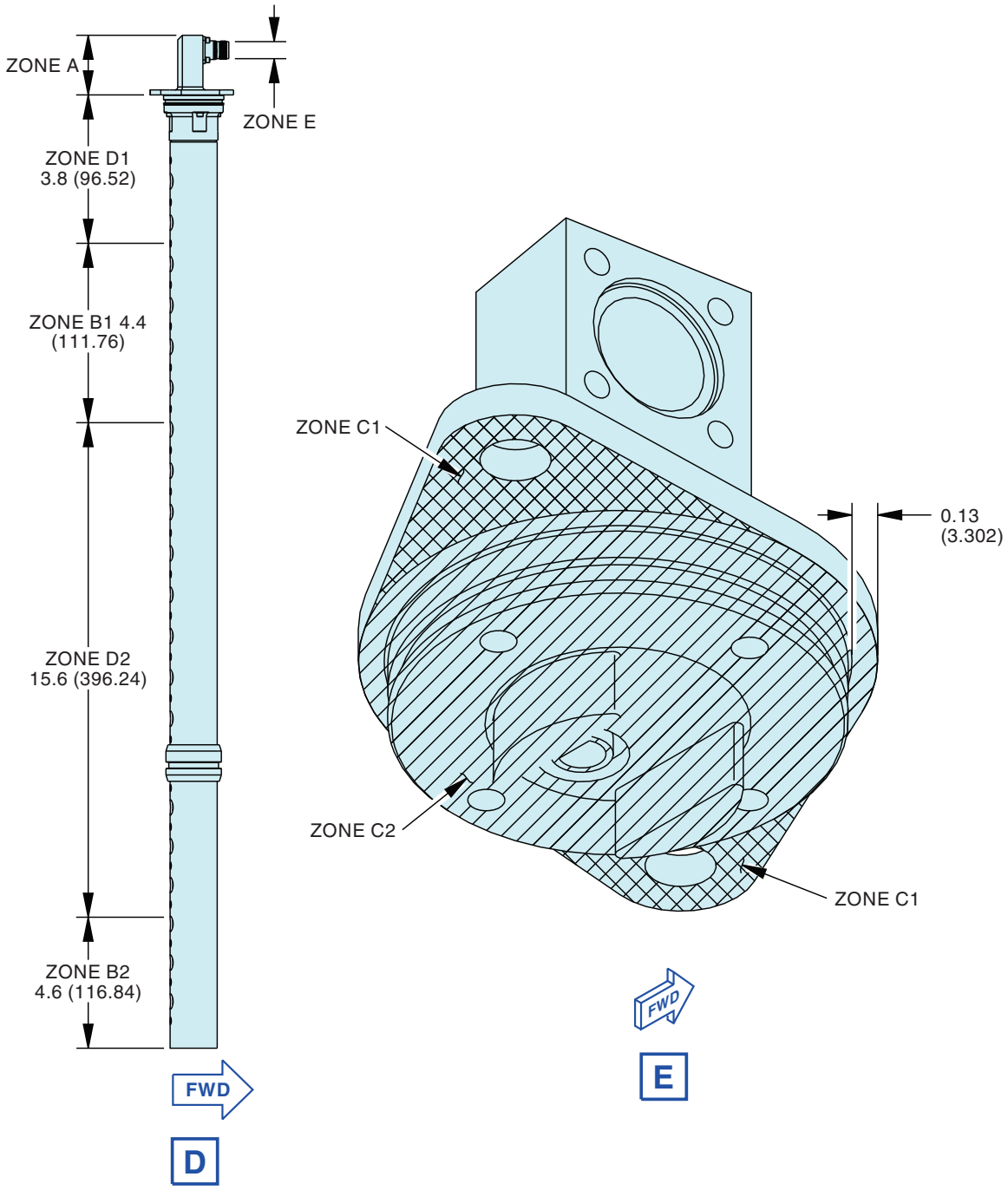
Oil Indicating System Inspection
Figure 601/79-30-00-990-801-G00 (Sheet 1 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Oil Indicating System Inspection
Figure 601/79-30-00-990-801-G00 (Sheet 2 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



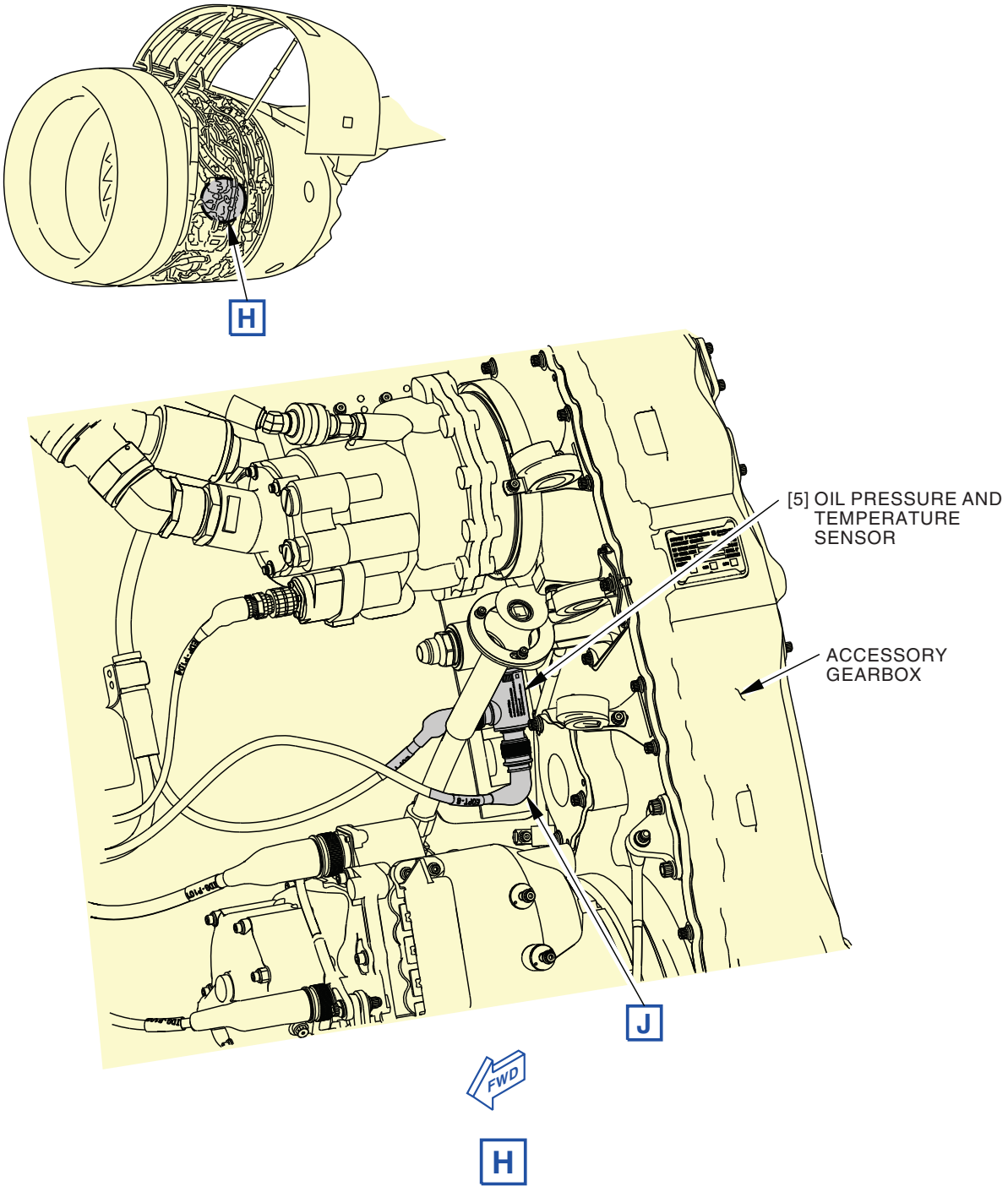
NOTE:

ALL DIMENSIONS ARE IN INCHES (mm).

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Oil Indicating System Inspection
Figure 601/79-30-00-990-801-G00 (Sheet 3 of 7)

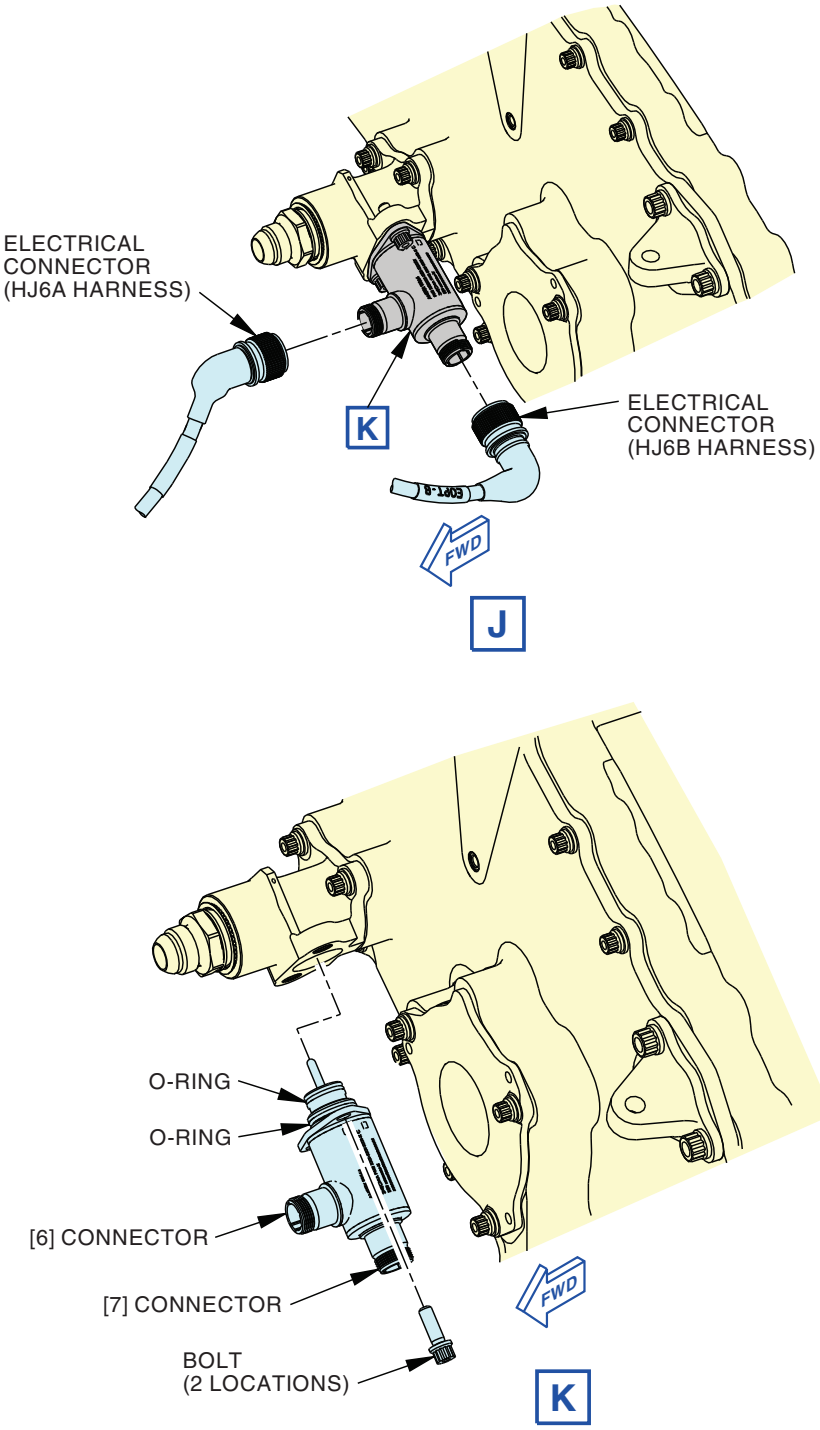
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Oil Indicating System Inspection
Figure 601/79-30-00-990-801-G00 (Sheet 4 of 7)

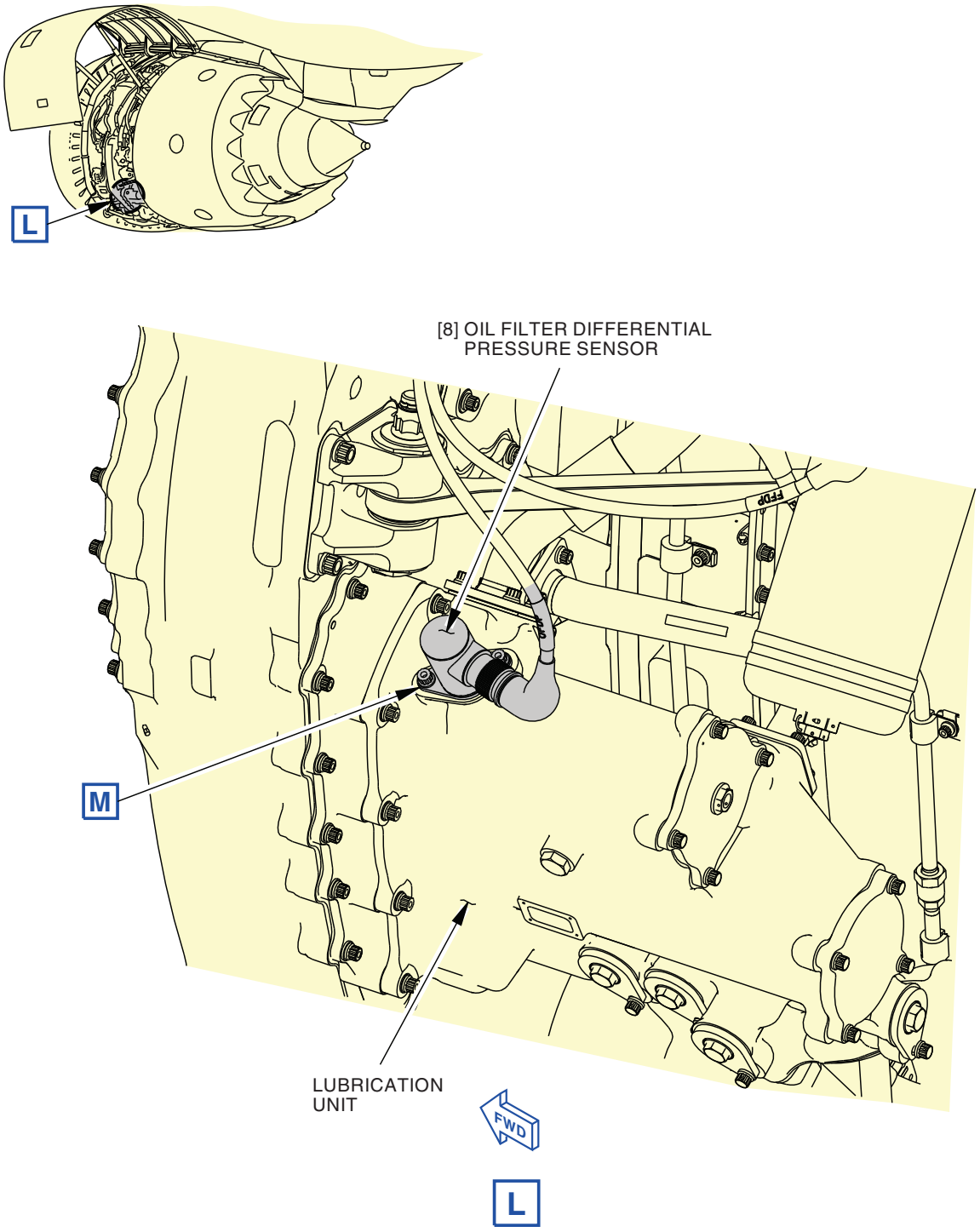
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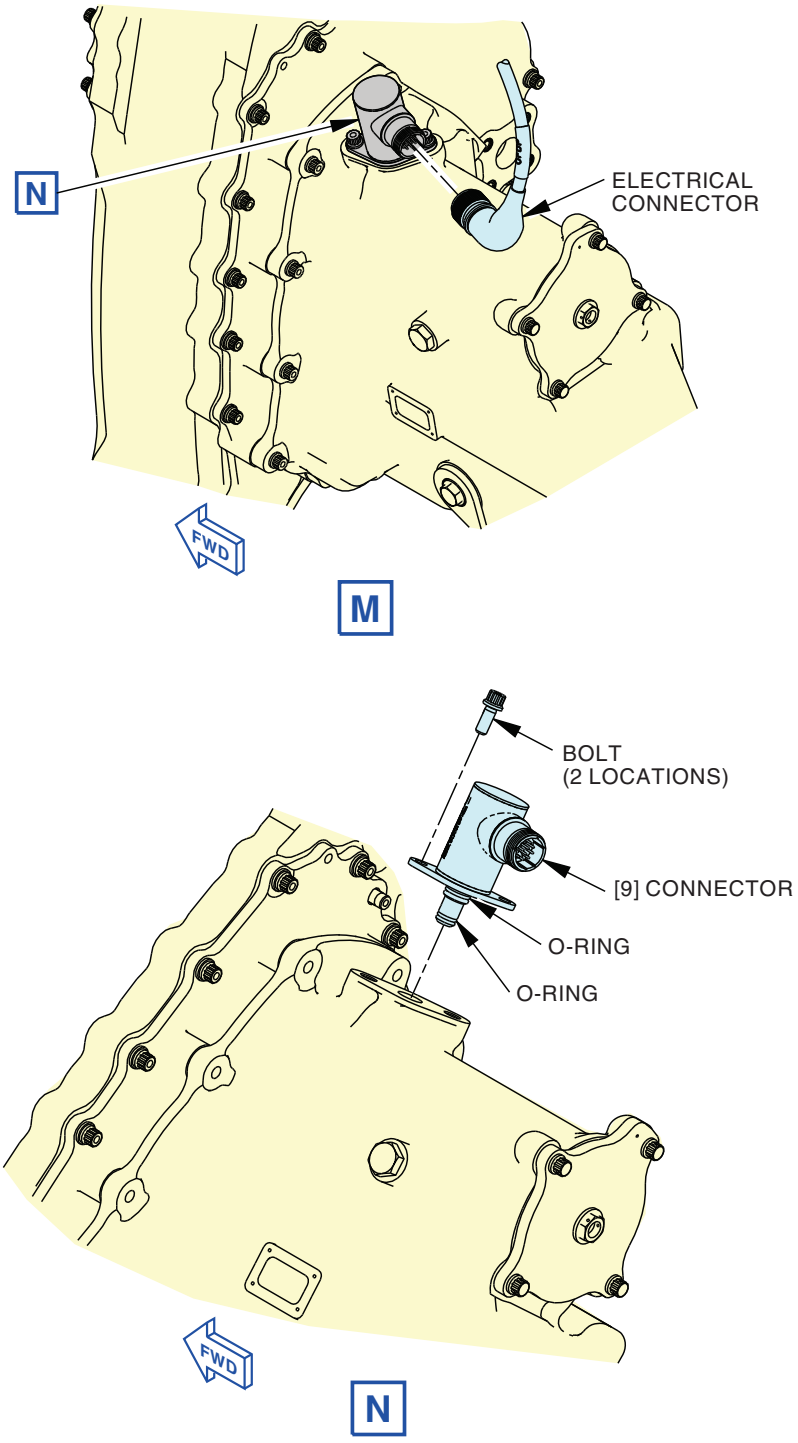
Oil Indicating System Inspection
Figure 601/79-30-00-990-801-G00 (Sheet 5 of 7)

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Oil Indicating System Inspection
Figure 601/79-30-00-990-801-G00 (Sheet 6 of 7)

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2485667 S0000584598_V1

Oil Indicating System Inspection
Figure 601/79-30-00-990-801-G00 (Sheet 7 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OIL LEVEL SENSOR - REMOVAL/INSTALLATION

1. General

A. This procedure contains two tasks:

(1) Oil level sensor removal

(2) Oil level sensor installation.

TASK 79-31-01-000-801-G00

2. Oil Level Sensor Removal

(Figure 401)

A. General

(1) This task gives the instructions on how to remove the oil level sensor.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14904	Tool Set - Jackscrew Part #: 956A1609G01 Supplier: 58828
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE
STD-1055	Container - Oil Resistant, 5 Gallon (19 Liter)

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Oil level sensor	79-31-01-01-020	PTF ALL

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

G. Prepare for the Removal

SUBTASK 79-31-01-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

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1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 79-31-01-010-001-G00

(2) On the applicable engine, open the right fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

H. Oil Level Sensor Removal

SUBTASK 79-31-01-030-001-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Remove the oil level sensor [1] as follows:

(a) Put the oil resistant container (5 gal)(19 Liter), STD-1055 below the oil tank [2].

CAUTION MAKE SURE ELECTRICAL CONNECTORS ARE CLEAN WHEN YOU DISCONNECT THEM. CONTAMINATION OF ELECTRICAL CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

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(CAUTION PRECEDES)

CAUTION

USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(b) Use the teflon-jawed pliers, STD-664 to disconnect the electrical connector [3] from the oil level sensor [1]:

1) Disconnect the electrical connector [3] from the oil level sensor [1] (TASK 70-00-01-910-803-G00).

2) Put the protective covers on the electrical connector [3] and the electrical receptacles on the oil level sensor [1].

(c) Remove the three bolts [4] that attach the oil level sensor [1] on the oil tank [2].

(d) Install the Tool Set, SPL-14904 jackscrews through the extraction bolt holes of the oil level sensor [1].

NOTE: You can use the bolts [4] as an alternative to the Tool Set, SPL-14904 jackscrews.

CAUTION

BE CAREFUL WHEN YOU REMOVE THE OIL LEVEL SENSOR BECAUSE OF ITS LENGTH. DAMAGE TO THE OIL LEVEL SENSOR CAN OCCUR.

(e) Remove the oil level sensor [1] from the oil tank [2].

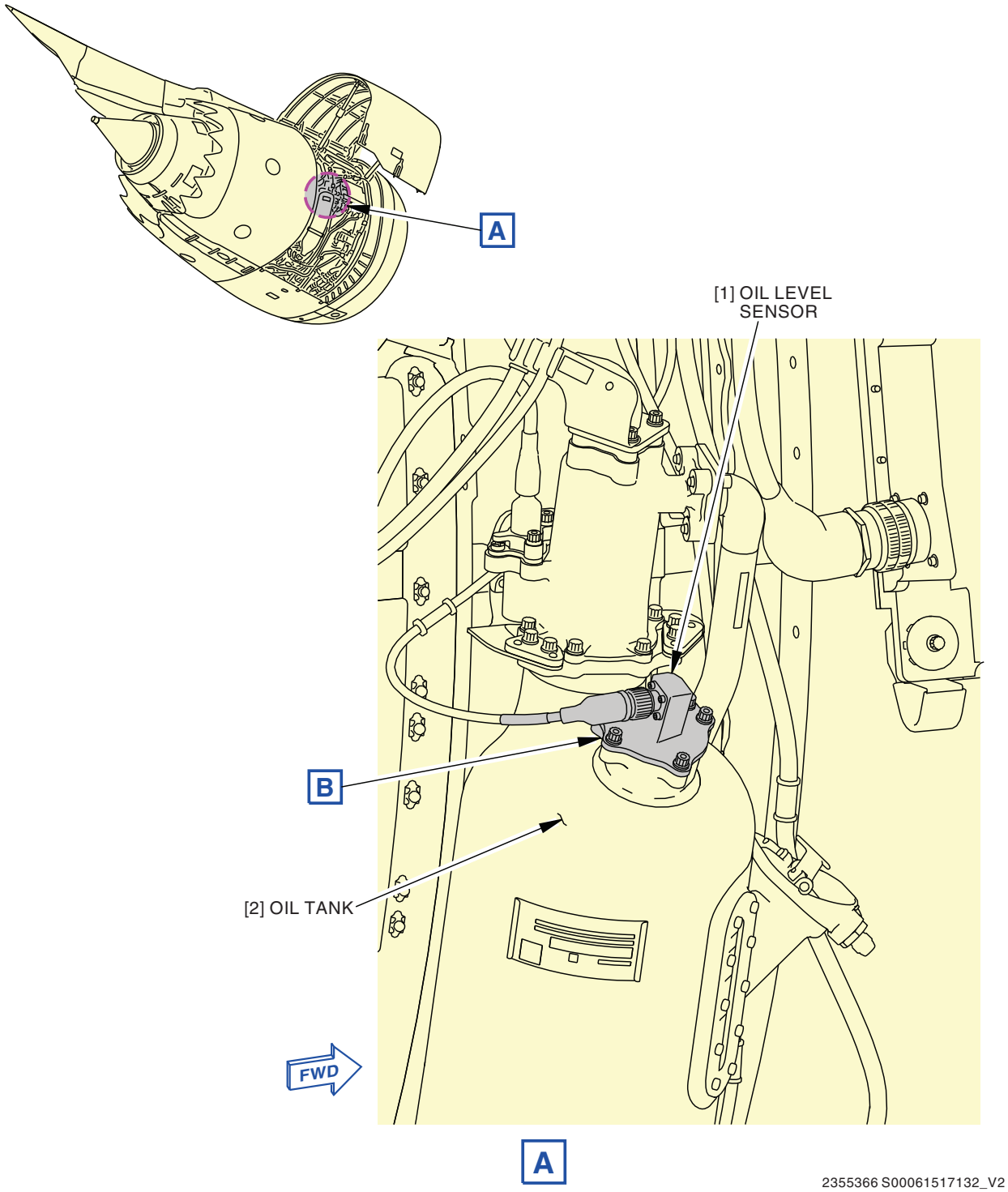
(f) Remove and discard the preformed packing [5] and the preformed packing [6] from the grooves of the oil level sensor [1].

(g) Remove the two Tool Set, SPL-14904 jackscrews from the extraction bolt holes of the oil level sensor [1].

(h) Install the protective cover on the open port on the oil tank [2].

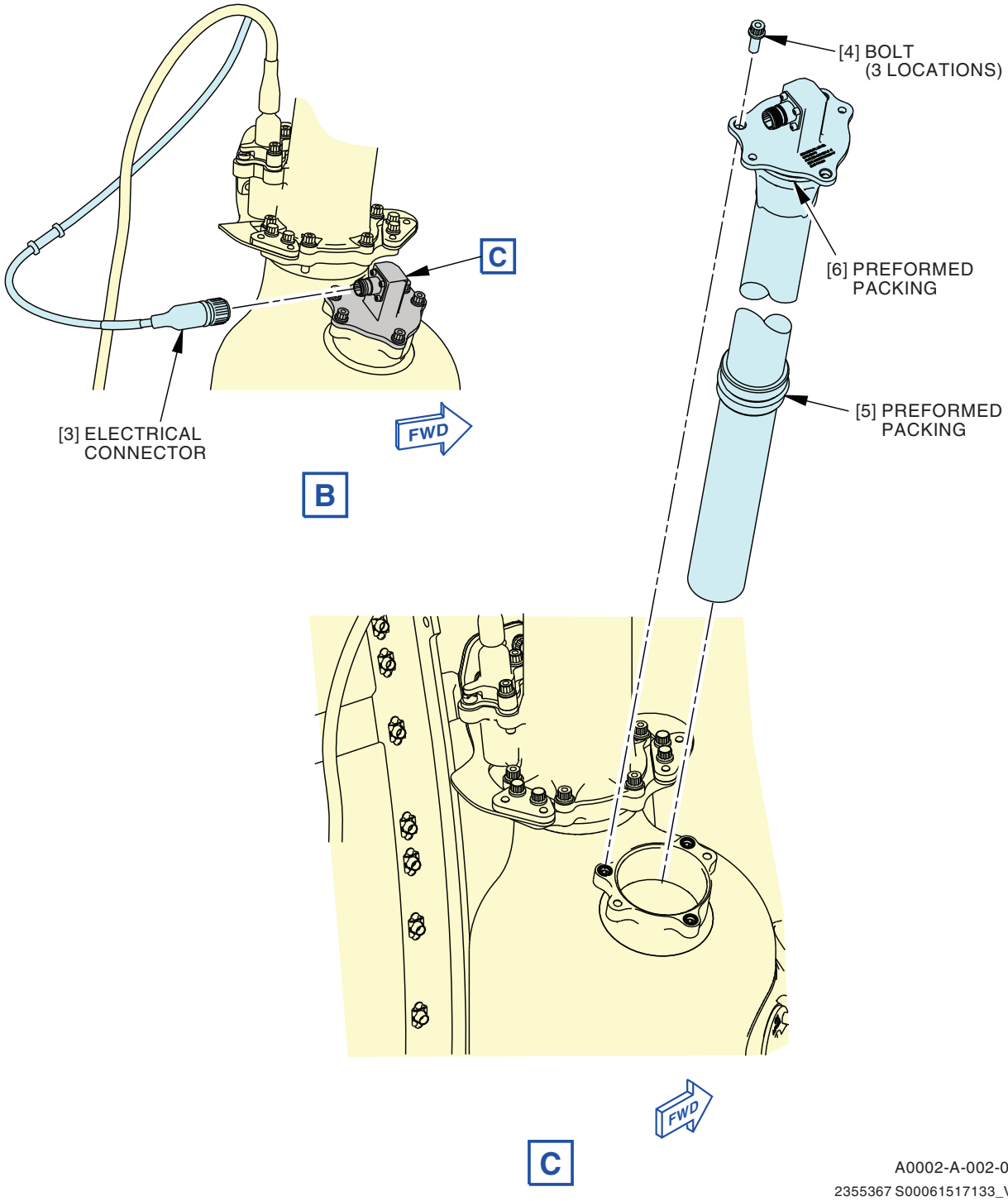
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Oil Level Sensor Installation
Figure 401/79-31-01-990-801-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Oil Level Sensor Installation
Figure 401/79-31-01-990-801-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 79-31-01-400-801-G00

3. Oil Level Sensor Installation

(Figure 401)

A. General

(1) This task provides the instructions on how to install the oil level sensor.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00601 [CP2101]	Vaseline - Graphite Mineral	
D00672 [CP5070]	Vaseline - Pure Mineral	V V-P-236

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Oil level sensor	79-31-01-01-020	PTF ALL
5	Preformed packing	79-31-01-01-040	PTF ALL
6	Preformed packing	79-31-01-01-030	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

H. Oil Level Sensor Installation

SUBTASK 79-31-01-420-001-G00

WARNING

DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION

DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Install the oil level sensor [1] as follows:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Remove the protective cover from the open port on the oil tank [2].

(b) Make sure that the mating faces are clean and clear of unwanted materials.

(c) Lubricate the new preformed packing [5] and the new preformed packing [6] with pure mineral vaseline, D00672 [CP5070].

(d) Install the new preformed packing [5], and the new preformed packing [6] in the grooves of the oil level sensor [1].

(e) Lubricate the threads of the three bolts [4] with graphite mineral vaseline, D00601 [CP2101].

(f) Install the oil level sensor [1] on the oil tank [2].

NOTE: If the oil level sensor is new, make sure to remove all protective covers before installation.

(g) Install the three bolts [4] that attach the oil level sensor [1] to the oil tank [2].

1) Tighten the three bolts [4] to 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m).

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(h) Use the teflon-jawed pliers, STD-664 to connect the electrical connector [3] to the oil level sensor [1]:

1) Remove the protective covers from the electrical connector [3] and the electrical receptacles on the oil level sensor [1].

2) Connect the electrical connector [3] to the oil level sensor [1] (TASK 70-00-01-910-803-G00).

(i) Clean the work area and clear the tools used to do the task.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 79-31-01-410-001-G00

(1) On the applicable engine, close the right fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

SUBTASK 79-31-01-860-002-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

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J. Oil Level Sensor Installation Test

SUBTASK 79-31-01-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

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OIL PRESSURE AND TEMPERATURE SENSOR - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Oil pressure and temperature sensor removal

(2) Oil pressure and temperature sensor installation.

TASK 79-32-01-000-801-G00

2. Oil Pressure and Temperature Sensor Removal

(Figure 401)

A. General

(1) This task provides the instructions on how to remove the oil pressure and temperature sensor.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-203	Container - Oil Resistant, 1 U.S Gal (3.8 L)
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

F. Prepare for the Removal

SUBTASK 79-32-01-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 79-32-01-010-001-G00

(2) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

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(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

G. Oil Pressure and Temperature Sensor Removal

SUBTASK 79-32-01-030-001-G00

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(1) Use the teflon-jawed pliers, STD-664 to disconnect the electrical connector HJ6A [3] and the electrical connector HJ6B [4] from the oil pressure and temperature sensor [1] as follows (TASK 70-00-01-910-803-G00):

(a) Disconnect the electrical connector HJ6A [3] and the electrical connector HJ6B [4] from the oil pressure and temperature sensor [1].

(b) Put the protective covers on the electrical connector HJ6A [3], the electrical connector HJ6B [4], and the electrical receptacles of the oil pressure and temperature sensor [1].

SUBTASK 79-32-01-020-001-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(2) Remove the oil pressure and temperature sensor [1] from the accessory gearbox [2] as follows:

(a) Put a oil resistant container, STD-203 below the oil pressure and temperature sensor [1].

(b) Remove the two bolts [5] that attach the oil pressure and temperature sensor [1] to the accessory gearbox [2].

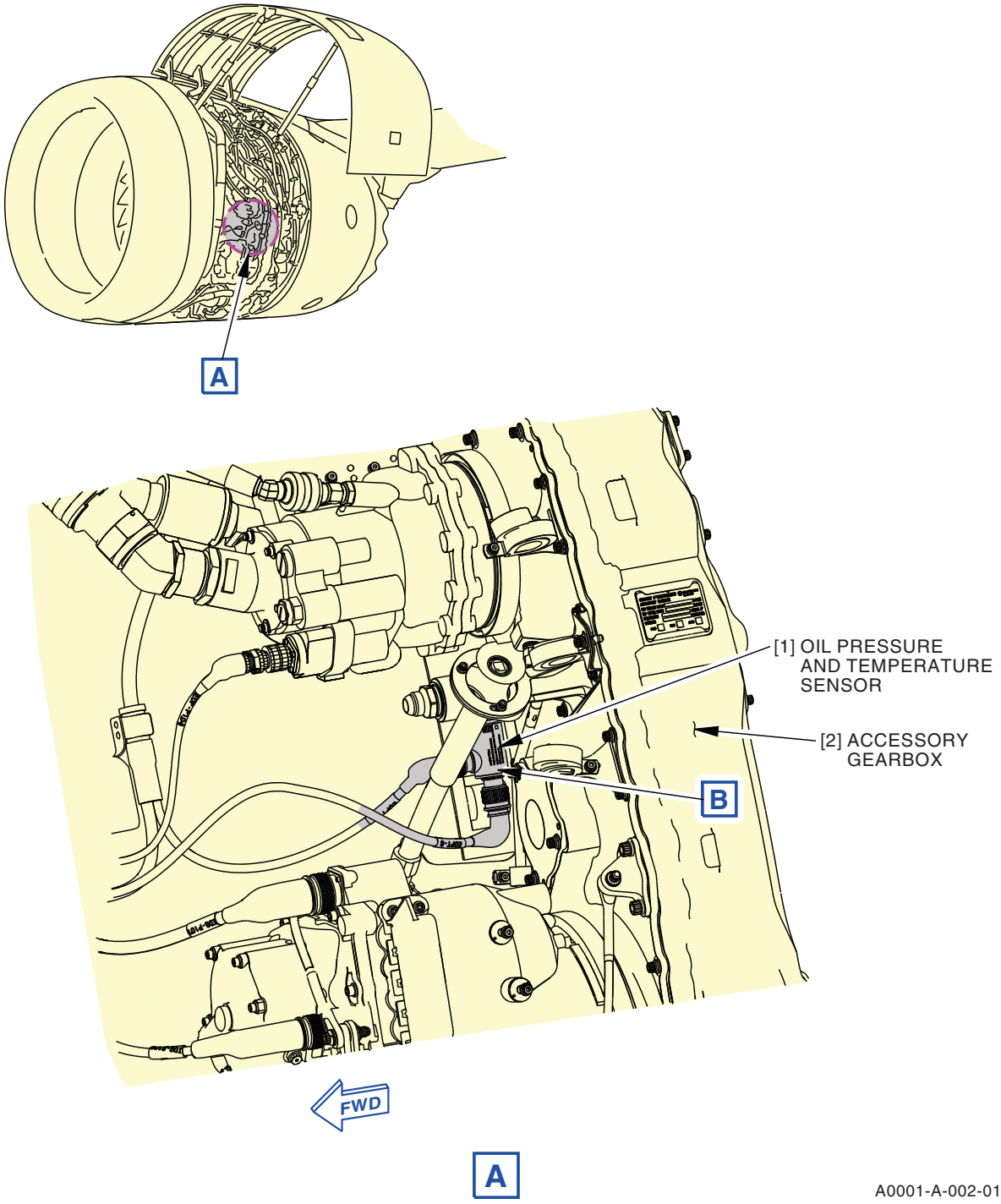
(c) Remove the oil pressure and temperature sensor [1] from the accessory gearbox [2].

(d) Remove and discard the preformed packing [6] and the preformed packing [7] from the oil pressure and temperature sensor [1].

(e) Install the protective covers on the open on the openings.

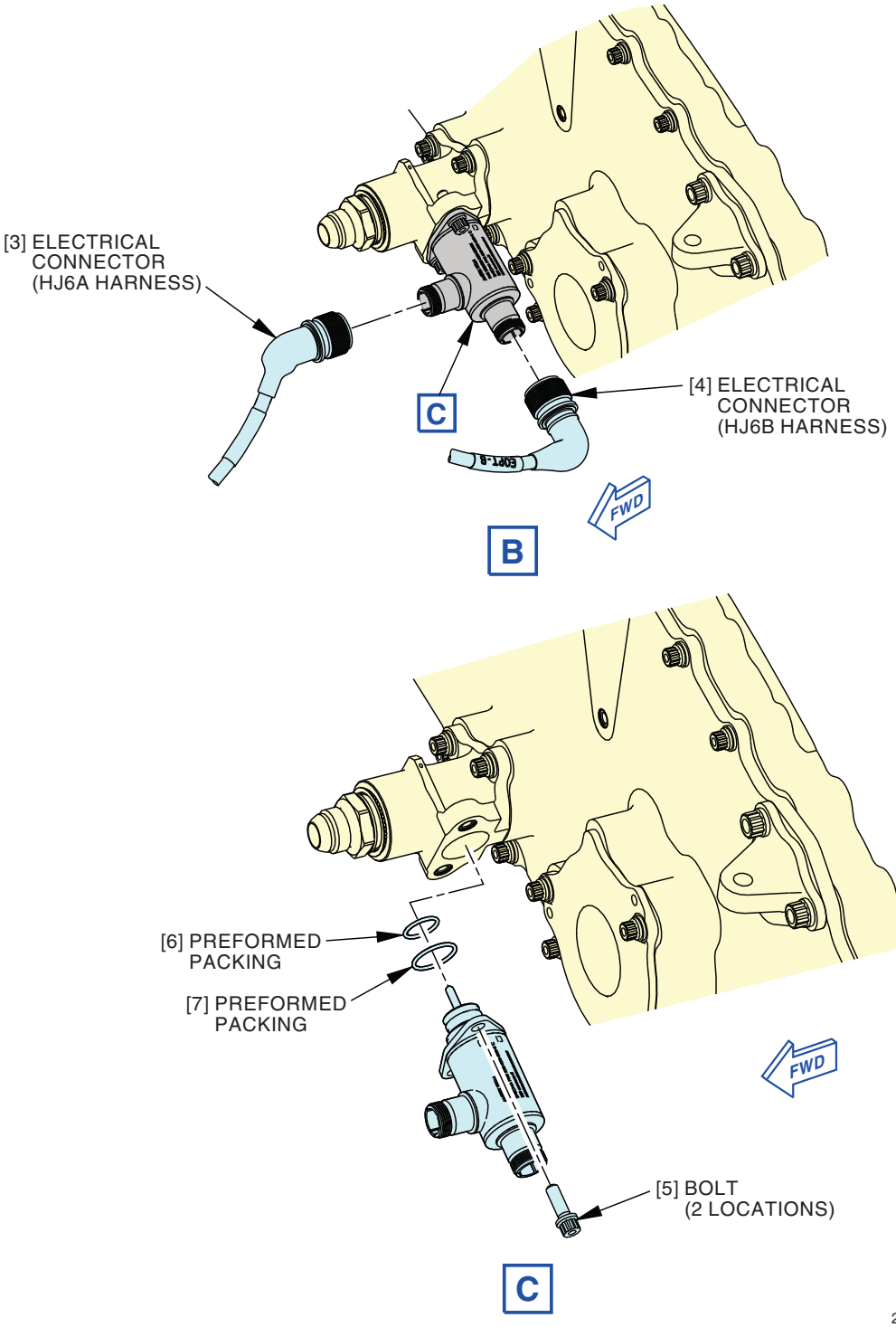
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Oil Pressure and Temperature Sensor Installation
Figure 401/79-32-01-990-801-G00 (Sheet 1 of 2)

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Oil Pressure and Temperature Sensor Installation
Figure 401/79-32-01-990-801-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 79-32-01-400-801-G00

3. Oil Pressure and Temperature Sensor Installation

(Figure 401)

A. General

(1) This task provides the instructions on how to install the oil pressure and temperature sensor.

(2) You must do the tests that are listed in the power plant test reference table after you install the oil pressure and temperature sensor.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00601 [CP2101]	Vaseline - Graphite Mineral	
D00672 [CP5070]	Vaseline - Pure Mineral	V V-P-236

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Oil pressure and temperature sensor	79-32-01-01-020	PTF ALL
6	Preformed packing	79-32-01-01-040	PTF ALL
7	Preformed packing	79-32-01-01-030	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

H. Oil Pressure and Temperature Sensor Installation

SUBTASK 79-32-01-420-001-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Install the oil pressure and temperature sensor [1] as follows:

(a) Remove the protective covers from the openings.

(b) Make sure that the mating faces are clean and clear of unwanted materials.

(c) Lubricate the new preformed packing [6] and the new preformed packing [7] with vaseline, D00672 [CP5070].

(d) Lubricate the threads of the two bolts [5] with graphite mineral vaseline, D00601 [CP2101].

(e) Install the new preformed packing [6] and the new preformed packing [7] onto the grooves of the oil pressure and temperature sensor [1].

(f) Install the oil pressure and temperature sensor [1] with the new preformed packing [6] and new preformed packing [7] on the accessory gearbox [2].

(g) Install the two bolts [5] that attach the oil pressure and temperature sensor [1] to the accessory gearbox [2].

1) Tighten the bolts [5] to 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m).

SUBTASK 79-32-01-430-001-G00

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

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(CAUTION PRECEDES)

CAUTION

DO NOT USE PLIERS THAT HAVE METAL JAWS TO TIGHTEN THE ELECTRICAL CONNECTOR. DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(2) Use the teflon-jawed pliers, STD-664 to connect the electrical connector HJ6A [3] and the electrical connector HJ6B [4] to the oil pressure and temperature sensor [1] as follows (TASK 70-00-01-910-803-G00).

(a) Remove the protective covers from the electrical connector HJ6A [3], the electrical connector HJ6B [4], and the electrical receptacles of the oil pressure and temperature sensor [1].

(b) Connect the electrical connector HJ6A [3] and the electrical connector HJ6B [4] to the oil pressure and temperature sensor [1].

(c) Tighten the electrical connector HJ6A [3] and the electrical connector HJ6B [4].

SUBTASK 79-32-01-860-002-G00

(3) Clean the work area and clear the tools and other items.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 79-32-01-860-003-G00

(1) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 79-32-01-410-001-G00

(2) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

J. Oil Pressure and Temperature Sensor Installation Test

SUBTASK 79-32-01-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OIL FILTER DIFFERENTIAL PRESSURE SENSOR - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Oil filter differential pressure sensor removal

(2) Oil filter differential pressure sensor installation.

TASK 79-35-01-000-801-G00

2. Oil Filter Differential Pressure Sensor Removal

(Figure 401)

A. General

(1) This task provides the instructions on how to remove the oil filter differential pressure sensor.

(2) The oil filter differential pressure sensor is referred to as the oil filter DP sensor.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-203	Container - Oil Resistant, 1 U.S Gal (3.8 L)
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

F. Prepare for the Removal

SUBTASK 79-35-01-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 79-35-01-010-001-G00

(2) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

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(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

G. Oil Filter Differential Pressure Sensor Removal

SUBTASK 79-35-01-030-001-G00

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(1) Use the teflon-jawed pliers, STD-664 to disconnect the electrical connector [3] from the oil filter DP sensor [1] (TASK 70-00-01-910-803-G00):

(a) Disconnect the electrical connector [3] from the oil filter DP sensor [1].

(b) Put the protective covers on the electrical connector [3] and the electrical receptacle of the oil filter DP sensor [1].

SUBTASK 79-35-01-020-001-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(2) Remove the oil filter DP sensor [1] as follows:

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(a) Put a oil resistant container, STD-203 below the oil filter DP sensor [1].

(b) Remove the two bolts [4] that attach the oil filter DP sensor [1] to the lubrication unit [2].

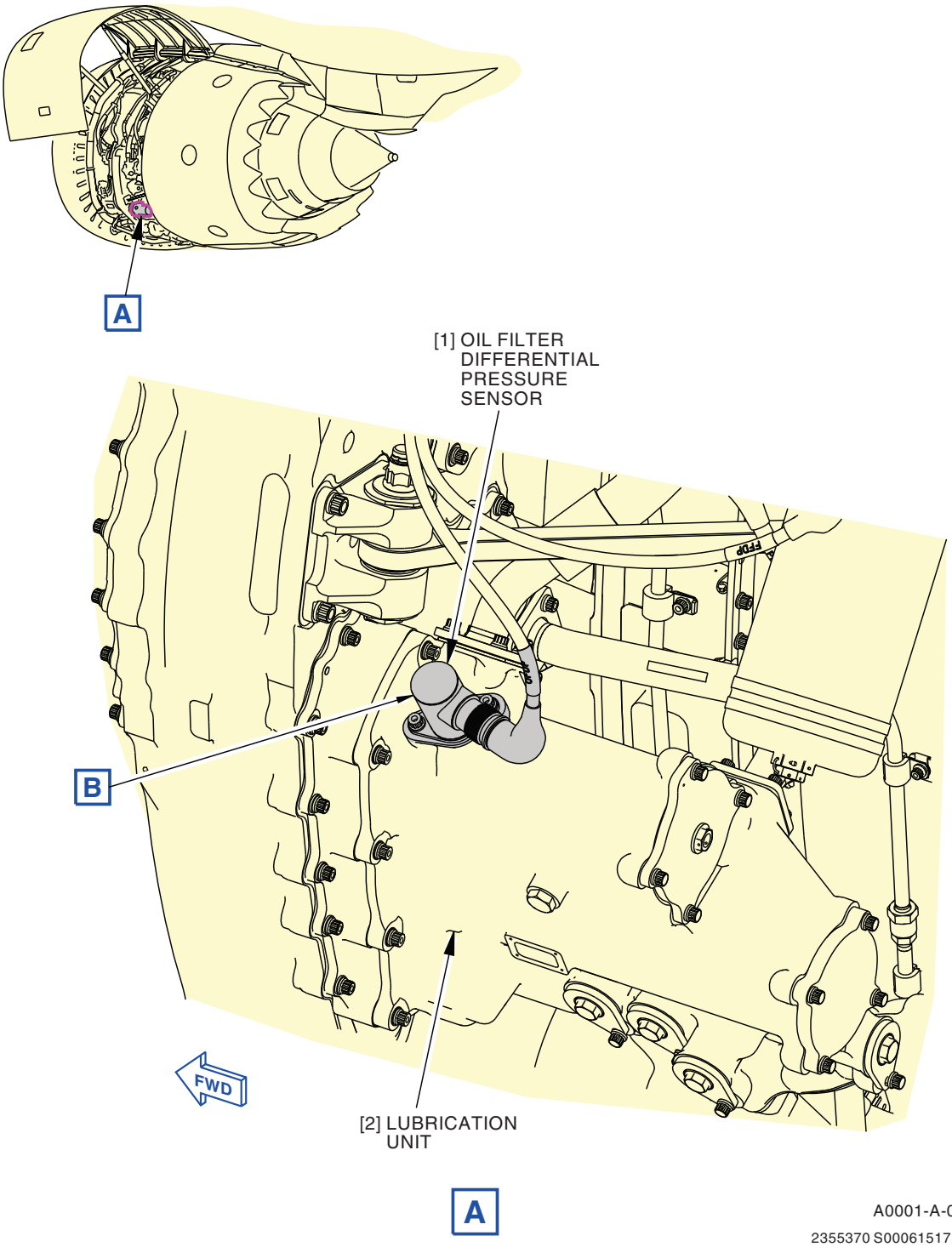
(c) Remove the oil filter DP sensor [1] from the lubrication unit [2].

(d) Remove and discard the preformed packing [5] and the preformed packing [6] from the oil filter DP sensor [1].

(e) Install the protective covers on the openings.

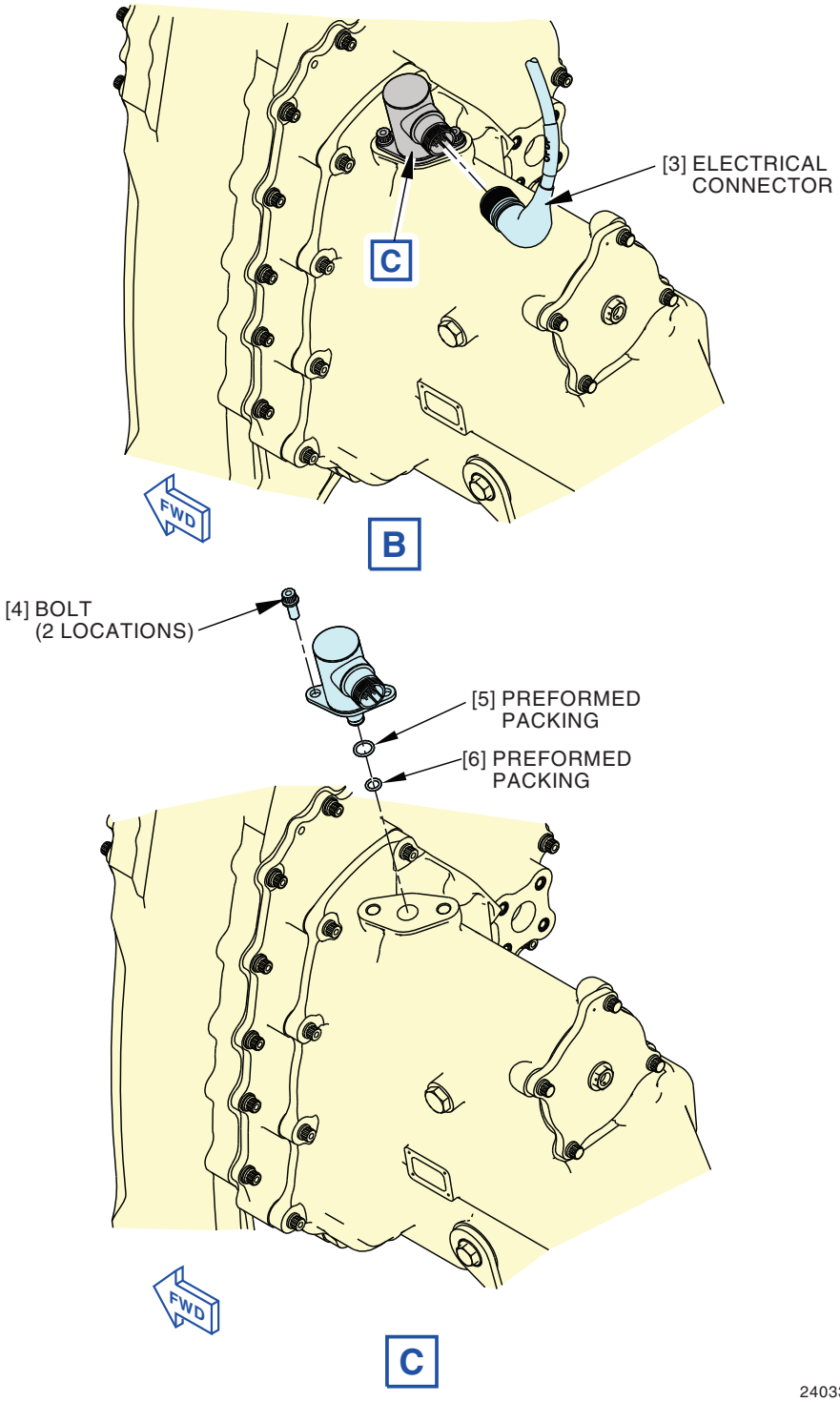
——— END OF TASK ———

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Oil Filter Differential Pressure Sensor Installation
Figure 401/79-35-01-990-801-G00 (Sheet 1 of 2)

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Oil Filter Differential Pressure Sensor Installation
Figure 401/79-35-01-990-801-G00 (Sheet 2 of 2)

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TASK 79-35-01-400-801-G00

3. Oil Filter Differential Pressure Sensor Installation

(Figure 401)

A. General

(1) This task provides the instructions on how to install the oil filter differential pressure sensor.

(2) The oil filter differential pressure sensor is referred to as the oil filter DP sensor [1].

(3) You must do the tests that are listed in the power plant test reference table after you install the oil filter DP sensor [1].

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00601 [CP2101]	Vaseline - Graphite Mineral	
D00672 [CP5070]	Vaseline - Pure Mineral	V V-P-236

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Oil filter DP sensor	79-35-01-01-020	PTF ALL
5	Preformed packing	79-35-01-01-040	PTF ALL
6	Preformed packing	79-35-01-01-030	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

H. Oil Filter Differential Pressure Sensor Installation

SUBTASK 79-35-01-420-001-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

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(WARNING PRECEDES)

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Install the oil filter DP sensor [1] as follows:

(a) Remove the protective covers from the openings.

(b) Make sure that the mating faces are clean and clear of unwanted materials.

(c) Lubricate the new preformed packing [5] and the new preformed packing [6] with vaseline, D00672 [CP5070].

(d) Lubricate the threads of the two bolts [4] with graphite mineral vaseline, D00601 [CP2101].

(e) Install the new preformed packing [5] and the new preformed packing [6] in the grooves of the oil filter DP sensor [1].

(f) Install the oil filter DP sensor [1] on the lubrication unit [2].

(g) Install the two bolts [4] that attach the oil filter DP sensor [1] to the lubrication unit [2].

(h) Torque the two bolts [4] to 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m).

SUBTASK 79-35-01-420-002-G00

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTOR AND RECEPTACLE ARE CLEAN WHEN YOU CONNECT THEM. DIRTY CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION DO NOT USE PLIERS THAT HAVE METAL JAWS TO TIGHTEN THE ELECTRICAL CONNECTOR. DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(2) Use the teflon-jawed pliers, STD-664 to connect the electrical connector [3] to the oil filter DP sensor [1] (TASK 70-00-01-910-803-G00):

(a) Remove the protective covers from the electrical connector [3] and the electrical receptacle of the oil filter DP sensor [1].

(b) Connect the electrical connector [3] to the oil filter DP sensor [1].

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(c) Tighten the electrical connector [3].

SUBTASK 79-35-01-860-002-G00

(3) Clean the work area and clear the tools and other items.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 79-35-01-860-003-G00

(1) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 79-35-01-410-001-G00

(2) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

J. Oil Filter Differential Pressure Sensor Installation Test

SUBTASK 79-35-01-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

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